Meter Reads — an as-built blueprint

Nineteen addresses a person can reach, what runs behind each of them, what the product says it cannot do, and what happened when somebody drove it. Two of the nineteen are built, translated into eleven languages, and nothing in the product points at either. **Since 2026-09-12 it also carries the rulings that answer that** — one tab retired, one renamed, two doorless screens folded into a cycle page that did not exist — drawn beside the as-built inside each screen's own entry, never in a second place.

READ AT	DATE OF THE READ	DESIGN SYSTEM FROM	IS THIS THE PRODUCT NOW?	ENTRIES	DRAWN
2.237.10 · 3706b459	2026-09-12	*tokens.css + AppShell · PageHeader · DataTable · KPIStrip · StateFrame · FormModal, and getComputedStyle on the running app*	*Could not check. `origin/HEAD` does not resolve and the remote carries two branches; HEAD is the tip of its own.*	21	17 of 21

DRAWINGS	PROCESSES	WIZARD STAGES	NAMING NO DOOR	JOURNEYS WALKED	JOURNEY ISSUES	RULE VIOLATIONS
26 · one per screen	3	5	2 of 19 as built · 0 revised	3 of 3	10	18

FIELDS SERVED	FIELDS NEVER TRACED	CROSS-MODULE FACTS	STATED ABSENCES	SPECIFICATIONS	OUT OF SCOPE
100 of 121	0	28	181	53	13

⚠ Before anybody builds from these drawings

The revised screens close every finding that is about a screen — and none of the nine that are about code. Two of those nine matter more than anything the revision fixes: **the reading detail cannot be opened at all today**, and **a tenant-scoping gate is reporting a clean zero over a file it never looked at**. Build these screens and both survive intact.

Each of the nine is specified to the point of being applicable, in the order it should be done, at §10 · the nine a wireframe cannot fix — with the patch for the crash, the trap in the obvious version of that patch, and the four screens it breaks rather than the one that was walked.

The three things a reader should not miss

1 · The reading detail page does not open. Clicking any meter on the reads list replaces the page with `TypeError: t.history.map is not a function`. The cause is one line: `getAssetReads()` is typed as an array over an endpoint that now answers a paged envelope. Every unit test of that screen supplies its own array fixture, so the fixture is right and the product is wrong. **Found by clicking, in one click.**

2 · Routes and Cycles have no door. Both are routed, built, tested and translated; the sidebar renders no submenus by two PO rulings, their route constants are imported by nothing, and no page links to either. They open only if you type the URL. The suite that looks like it would catch this asks whether a declared path renders a real screen — *not* whether a user can get there — and the gate that does ask the door question covers customer-portal screens only.

3 · A primary button renames itself for a door it will not open. On a billed reading the action becomes *Correct billed read* and is disabled with *“that path is not built”*. `POST`

`/reads/{id}/billed-correction` is served, the money screen exists, the server's own copy tells clerks to use it — and the front end has never called it.

How the flow was scoped, and what this document is not

Reading "Meter Reads" as the `metering` module's reading surface: the reads list and its two tabs, one reading and its three corrections, the five-stage upload, the single-entry and estimate dialogs, the bulk bar, the route sheet, and the route/cycle operational views with their chase list. **Stopping short of** premises, units and assets — which `metering` also owns, and which are three separate flows — and short of validation *settings*, which live on the Settings screen and are cited here rather than audited. Say so if that boundary is wrong.

This document is exhaustive over itself — every rule it states, figure it prints, screen it draws, decision it records. **It is not exhaustive over the product.** And it checks whether a number is served, reached and displayed with its unit; it does **not** check whether the number is right. That is B4/B5's 61 golden tests, of which 14 are engine-verified — see appendix F, which names it as the most expensive assumption here.

Description — grey rail. A statement about what the code does today, including an honest statement that something is missing. **It commits nobody to anything.**

Commitment — accent card. A named piece of work, what it waits on, the screens it changes, and what a person could do and see once it is done. **If a thing is not in a commitment card, nobody has undertaken to build it.**

Desktop only, for every audience

One drawing per screen, at a fixed **1440** viewport, scaled as a whole unit and never reflowed — a responsive drawing in a narrow window is a phone drawing nobody drew. **Every screen in this module is a staff screen**, so no audience question arises here; the absence of a phone drawing is a property of this document and not of the product.

01

The standards every screen here is drawn to

Nothing in this list was invented for this document. Each is an existing ratified rule of this repository, or a failure it already paid for, and every judgement in the sections below is made against one of them. Where a rule is **ratified but unenforced** the table says so, because in a repository where no gate reads rendered output a green suite is evidence of nothing.

The precedence, and it decides whether a finding is a finding

1 · `04-docs/02-decisions/14-ui-conventions.md` (B14) is the authority — 31 binding rules, numbered 1–21 and 31–40. Counted from B14's own status table on the read: 28 rows there (1–20, 32, 34–40) plus rules 21, 31 and 33, which are declared in their own sections further down the file. Rules 22–29 were never minted and 30 was declined.

2 · `05-prototype/r10-2026-08-16/UI-RULES.md` is B14's source, not its authority — twenty rules in five groups, Tables being 12–17.

3 · the build file, `.claude/skills/react-tailwind-css-ui-generator/SKILL.md`, is not a third rule set: it is how the two above are obeyed in this tree.

And two style worlds. `UI-RULES.md` names `.lt`, `.page-head`, `.date-input`, `.btn-sm` — classes the shipped app does not declare; they live in the prototype, whose runtime CLAUDE.md forbids porting. Every drawing below takes its **tokens from the shipped app** and its **layout from r10**.

Rule	What it says, and what it is doing here	Enforced?

B14 · 7	Date fields always use the date input — calendar affordance, keyed entry, Today shortcut. **Never a bare text input.**	PARTIAL min/max on four asset screens only; the Today half is unbuilt
B14 · 8	Dropdowns share one chevron style; **5+ options are searchable** (type-to-filter). The kit's threshold is `LONG_LIST_THRESHOLD = 10` and the control is `SearchSelectField`.	CHEVRON: gate, `nativeSelects: 0` · searchable half **unchecked**
B14 · 12	Table toolbar: `[Search…] [All | filter chips ×] [+ Filter ▾] … [▦ Columns]`. One row.	RATIFIED G43 covers clause (b) only
B14 · 15	Row actions are **icons with tooltips**, never text links. Max 2 visible + overflow.	G37 — ENFORCED
B14 · 16	A pagination footer on **every** list table. Wizard summary tables are exempt.	shared DataTable only
B14 · 21	New UI is written in Tailwind utilities and adds **no** class to a stylesheet — and a utility is emitted only for a file listed in `07-frontend/src/styles/route-b-migrated.txt`.	G31 ceiling + `routeBUtilities.test.ts`
B14 · 31	**Nothing internal is ever visible to a utility user** — not a primary key, not a rule number, not a path. Where the reference is what a user needs, it has a human-facing number and *that* is shown; the internal value may ride on `title` as evidence.	G28 over strings · a **uuid-shaped value** is enforced on S30 and S15 only, by render-check scripts
B14 · 32	A screen has a description budget — one sentence per section, no rule id or internal code in clerk-facing copy.	UNENFORCED
B14 · 33	On-screen text is capped by what it is *for*: label ≤ 3 words · fact ≤ 6 · instruction ≤ 6 per sentence · consequence ≤ 6 per sentence, at most two · **rationale is zero words on screen**.	G33 on refusals only, six words, baseline 606 may only shrink
B14 · 36	The captured packs give layout and detail; r10 gives the visual style; **the shared kit is how you comply**. Where the kit lacks a piece, add it to the kit.	UNENFORCED
B14 · 39	One spacing scale — 4 · 8 · 12 · 16 · 20 · 24 · 32, no ad-hoc 3/7/9/14 — and a field is **label → input → helper** at 5 : 6 : 20.	order asserted; the three figures have no gate
B14 · 40	**The page-head title is 32px on every screen**, set on the element rather than inherited from its tag. Ratified **2026-09-12 — the day of this read**.	ENFORCED by a component test *and* a render check against the built stylesheet
NFR-3	**Universal server pagination.** Load-bearing rather than tidy: it is the flexibility T4/T3 depend on.	no gate
NFR-5	**No hard caps anywhere.** No table, index or code path may enforce a maximum account count per tenant.	no gate
Hard rule 3	**Consumption always carries an explicit unit. Never assume.** A bare number is refused; a dash is the honest answer.	G44 display-truth ratchet
Hard rule 4	Financial immutability — no UPDATE or DELETE on issued invoices, lines, payments or journal lines. **Corrections are new rows.** A read follows the same shape by `BR-MET-024`: override **appends**.	DB + service
Hard rule	An address is stored on `premises` and nowhere else; every other entity walks to it.	G85

8		
Law 2	**Built is not wired** — and its mirror, a live screen behind a dead control. Nine defects in one day.	G13 one direction only
Law 6	**Diff the running app against the prototype.** No gate looks at rendered output.	a person with a walkthrough
Law 7	Prove a guard red before trusting it green — *and in the harness that will run it*; then keep reading its output.	habit
R14	New entries sort to the top. Default ordering on every list is `created_at` descending, not name ascending; a clerk's chosen sort still wins.	UNENFORCED
R15	Money never reaches JSX unformatted. Every amount through `formatMoney()`.	G44

What the drawings are made of

Palette, type scale, spacing and corner radius read from `07-frontend/src/styles/tokens.css`; component shapes from `AppShell.css` (240px inverted-ink rail, 56px topbar), `PageHeader.css`, `DataTable.css` (13.5px body, 12px uppercase header at .08em), `KPIStrip.css` (a band with 2px rules top and bottom — *not* cards), `StateFrame.css`, `FormModal.css` (560px, `.input` 36px min). Accent **#ec3013**, ground **#f3f2f2**, surface **#eae9e9**, ink **#201e1d**, **radius 0 at every size**, Archivo at heading weight 800.

The typeface is loaded, not merely named. These drawings link `fonts.googleapis.com/css2?family=Archivo:wght@400;600;800` — the same family at the same three weights `tokens.css` imports on line 22 — so the wireframes are set in the product's own type rather than in the reader's system font. A drawing uses no weight the product does not load.

The shell is one component, drawn one way. Rail 240px inverted ink, brand, tenant block, thirteen nav items, version, and the **identity in the footer** (initials · name · current-tenant role). Top bar 56px: **breadcrumb · global search · help ? · bell**. Both come from `AppShell.tsx`, and the Billing blueprint draws the same two because it is the same component.

One correction to the drawing rules this document was written under. They pin the page title at 40px, measured on the running app. **That is no longer the product.** B14 rule 40 was ratified on 2026-09-12 — the day of this read — and the title is **32px**; the component emits `text-h2` and two checks hold it there. The drawings use 32.

⚠ Where these rulings live, and the answer is mostly “only here”

Thirteen PO rulings of 12–13 September exist in this document and in no file of the repository

Every ruling quoted here was checked against `12-governance/` and `04-docs/`. **The split is clean and it is by date.**

Date	Ruling	Also in the repo?
2026-08-10	*“font in Prototype and font in application is different”*	decision log
2026-08-10	*“use 32 pixels — for every ordinary screen”*	decision log · CR-093
2026-08-27	*“(b) Accept and warn”*	decision log · open questions
2026-08-27	*“In plan Nav bar — no sub menu required”*	decision log · defect log
—	*“if you have to increase the length of search bar please increase it”*	decision log · EP-SHL-01
2026-09-12	*“Need Attention — does not add any value”*	NOWHERE
2026-09-12	*“we should say Read Cycles and within that we should show Routes and their completion”*	NOWHERE
2026-09-12	*“we will need a list view of read cycles” · “and a detail view”*	NOWHERE

2026-09-12	*“in detail section use Panel to information groups”*	NOWHERE
2026-09-12	*“on this screen can you show utility services chip in header”*	NOWHERE
2026-09-12	*“just keep the what is final screen”*	NOWHERE
2026-09-12	*“write it down in document and not in repo”*	NOWHERE
2026-09-13	*“a details view of Route … similar to details view of Cycle”*, option A	NOWHERE
2026-09-13	The walk-reorder question, raised and unanswered	NOWHERE

Every August ruling reached the repo. Not one from this session did. The cause is not carelessness — it is the last row of that table. *“Write it down in document and not in repo”* was the right call for a code patch, and it silently applied to the rulings as well.

Why it matters more than it looks. This document becomes the single point of failure for thirteen decisions. They **cannot be grepped**, so no lane building any other screen will meet them, and **nobody can count what depends on them** — which is the honest answer to “how many gaps are there?”: today, unknowable. A blueprint holding rulings that exist nowhere else is not a record, it is a hostage.

And a captured ruling is not a built one. Even written down, a ruling schedules nothing. Each of the thirteen has a commitment on the register — that is the undertaking — and **MR-53** is the one that gets the rulings themselves into greppable text.

Description, commitment, and the four chips

The chips on every screen heading are a **vocabulary, not decoration**, and this section states it because the document did not. Everything in here is one of two things.

Kind	What it is
Description	A statement about what the code does **today** — including an honest statement that something is missing. **It commits nobody to anything.** Every finding, every drawing of an as-built screen, every stated absence and every sentence in sections 05 to 08 is a description.
Commitment	A **change specification**: a named piece of work, what it waits on, the screens it changes, and what a person could do and see once it is done. Every one lives in §09, the specification register, and appears as a card in the section that owns it. **If a thing is not in a commitment card, nobody has undertaken to build it.**

Four state chips, and only four. A screen may carry more than one — most do. **Every chip on every heading is DERIVED from the register**, never typed beside it: the commitments that name a screen decide its states, so the two cannot drift apart and a check at assembly proves they have not. A screen no specification covers carries **No commitment**, and **that absence is the finding, not an oversight**.

Chip	What it asserts	Count
EXISTS BEHIND	The screen is routed and built **today**. It says nothing about whether a person can reach it — two of these have no door, and that is a description recorded in the entry.	19
NEW READ	The specification's change is to **show data the server already answers** — a column, a field, a figure. Nothing new is served; this is wiring.	7

NEW BUILD	The specification needs something **built, rewritten or removed** — an endpoint, a write, a screen, a control, a sentence.	17
BLOCKED	At least one commitment on this screen **waits on a question nobody has answered**. Four commitments are blocked — **MR-10** (a selection model), **MR-17** (an unruled tenant toggle), **MR-24** (`OQ-209`) and **MR-38** (the window disagreement) — and they reach seven screens between them.	7
NO COMMITMENT	No specification covers this screen. **The absence is the finding.** After the sweep of 2026-09-12 exactly **one** screen carries it — entry 19, which is a second address rather than a screen.	1

This vocabulary drifted during the revision pass, and the drift is worth recording

Between 2026-09-12's first and ninth versions the headings grew **nine chips that are not in the vocabulary** — *Revised*, *New screen*, *Drawn from code*, *Not walked*, *Throws on open*, *Retired*, *Retired as a screen*, *Behind a doorless screen*, *Second address*. Every one was true, and **every one was a DESCRIPTION wearing a commitment chip's clothes** — which is precisely the confusion the two-word distinction above exists to prevent. They are gone; each fact now sits in the entry's own text, where a description belongs. **Found by the PO reading the headings, by no check** — nothing in this document validated its own vocabulary, and now the count column above does.

Five screens were also carrying a revision with **no commitment card behind it**, which by the rule above means nobody had undertaken to build them. `MR-26` to `MR-30` close that.

And a second pass found the larger half. PO instruction: *"commitment chip — not just on screens but all changes that we need to ask Claude Code to do."* Seven pieces of work were described in full and undertaken by nobody — the copy budget across eleven screens, the Panel ruling, three false comments, a duplicated line in the migration list, four seam reads, and the **window disagreement that three drawn figures compute from**. `MR-31` to `MR-38` close those. **38 commitments, eight of which change no screen at all, and every chip on every heading is derived from the register rather than typed beside it.**

How a form is built — spacing, pickers and dates

B14 rule 38 settles this: where a drawing and our UI rules disagree about how a form is BUILT, the UI rules win. What a form CONTAINS is still the drawing's to decide. So every field below is drawn to the rules, not to the prototype, and where the running product departs from them it is a row in section 07 rather than a licence here.

What	The rule	How the drawings obey it
Spacing	One scale, no ad-hoc values: 4 · 8 · 12 · 16 · 20 · 24 · 32 · 40.	Every gap in every drawing is a step on it. A dialog is 24px inside; fields are 20px apart; a label sits 6px above its input and its help 4px below.
Label order	Label above the input, help under it — never beside, never after.	Drawn that way on every field, including the two the product hand-rolls.
Two per row	At most two fields on a line, and a field that needs the width takes the row.	The estimate and add-reading dialogs pair register with date; every long field is full width.
Dropdowns	The shared `SelectField`. **Past ten options it must be `SearchSelectField`** — `BR-SHL-033`, `LONG_LIST_THRESHOLD = 10`; B14 rule 8 asks for type-to-filter from five.	Drawn 36px high with a single ▾, one border, the product's focus ring. **Six of the eight pickers in this module are under ten options today and over it at the top of the ICP** — the table below says which.
Dates	The product's own date field, with its calendar affordance — not a bare `<input type="date">`, whose look is the browser's and changes per machine.	Drawn with the 📅 affordance and the same 36px box as every other input.

The two containers, side by side

Both are real and both are in the product now. **The PO has ruled that the detail screens**

use the left one, so every drawing here does — but a reader comparing a drawing with the running app needs to know which is which, and that the read detail does not look like this yet.

PANEL · COMPONENTS/PANEL.TSX · USED 149×

AMOUNT

Previous balance	$0.00
Current charges	$251.45
Total due	**$251.45**

BREAKDOWN

Water $132.02 · Sewer $94.50 · Trash $15.00

A box. 1px divider border on all four sides · surface fill · 16px padding · 16px below · heading 16px inside it.
Where: the bill detail, the upload stamp, the cycle page — anywhere a group is an *object* you could lift off the page.

SECTION · READDETAILPAGE.TSX:139

CALCULATION

(4,320 – 4,110) × 1 = **210 kWh**

FLAGS

High usage · 154% of a 5.2 kWh average

Not a box. A **rule across the top** and 14px of air under it · no border at the sides or bottom · no fill.
Where it is TODAY: the read detail. **Where it is in these drawings: nowhere.**

COMPONENT SPECIMEN · both drawn to their component's own declarations, not to a screen — **The PO ruled between them on 2026-09-12, pointing at the bill detail:** *"use Panel … in detail section use Panel to information groups"*. **Every information group on every detail screen in this document is now a Panel** — the read detail's five, the upload stamp's five, the cycle page's four. **That is a CHANGE, not a record of the product**, and it is written down as one: the read detail today renders the right-hand pattern (`ReadDetailPage.tsx:139`, `"border-t border-divider pt-3.5"`), and these drawings render the left. **The heading follows the ruling too** — `Panel.tsx` declares its heading at 16px, and the screen the ruling names shows the red small-caps label, so that is what is drawn. A tone on a Panel stays a **border, never a fill**: *"one component, one visual decision"*, settled after the customer portal shipped two `Panels` under one name that disagreed about exactly that.

Every picker and date in the module, and where its options come from

One place, because the same four lists are drawn on five screens. A control's source is a fact about the screen, so it is stated once here and cited from each entry.

Control	Screen	Where the options come from	How many	Built with
Cycle	07	`getCycles()` → `listAll("/cycles")` — walks *every* page at 1000 a page, cached five minutes, `retry:false` deliberately (*"a clerk waiting on three retries for a picker is worse than a picker that says it is unavailable"*).	Every cycle the tenant has	`SelectField`
Route	07	`getRoutes()` → `listAll("/routes")`, the same shape. **Not narrowed by the cycle chosen above it** — the list is every route in the tenant, and the help line asks the clerk to leave it empty for a whole cycle.	Every route the tenant has	`SelectField`
Read method	07	`getReadMethods()` — the tenant's own reference list (`BR-MET-123`: the source is a TAG chosen at upload). Says *unavailable* only once the answer is in and empty, never while it is still loading.	A tenant lookup	`SelectField`
Utility services	07	`getTenantServices()` on the **shared** `queryKeys.tenantServices` — the same list the settings screen reads, on purpose, so two caches cannot disagree about what a tenant sells. *Could not be read* and *sells nothing* are told apart.	Up to eleven	multi-select
Register	08,	`READ_CHANNELS` — a client constant at `api/endpoints.ts:11657` mirroring the server's `models.CHANNELS`. **Seven and fixed:** total · tou_on	Seven	A raw

	12, 18	· tou_mid · tou_off · export · generation · demand. Adding a register is a string in two places, by design.		**A raw `<select>` on 08**; `SelectField` on 12 and 18
Route (sheet)	17	`getRoutes()` again, and a cycle picker beside it behind a segmented scope.	Every route	`SelectField`
Reading month	07	Typed. Any day in the month; the batch stores the first of it.	—	date field
Read date	08, 12, 18	Typed.	—	**A raw `<input type="date">` on 08**; the product's own on 12 and 18

> **Two of these do not obey the rules above, and both are on one screen**
>
> Stage 2's manual grid hand-rolls its register dropdown and its date box — `<select className="input">` and `<input type="date">` — while stage 1, three panels earlier, uses the shared `SelectField` for the same kind of choice. **Rows 7-05 and 7-06 in section 07.**
>
> **And none of the eight is type-to-filter.** `BR-SHL-033` puts the threshold at ten options and B14 rule 8 at five. Six cycles and six routes pass today; a utility at the top of this product's stated range has neither. **Row 7-07.**

02

Reading the meters

Seven entries — the list a billing clerk lives in, one reading answered in full, and the three corrections that hang off it. Every screen here carries **what it is today and what it becomes**, in that order, so there is one place per screen and no second register of revised drawings to keep in step.

> **Ruled 2026-09-12 — and it reverses a signed rule**
>
> **“Need Attention — does not add any value.”** The tab goes. That supersedes `BR-MET-060`, which says Reads *opens* on the exceptions, and retires `US-MET-019`, the story that built the three counts. **Stated rather than assumed**, because a ruling that collides with a signed rule has to name what it displaces.
>
> **Two of the three counts survive the move and one does not.** *Flagged* and *Disputed* are already declared filters on All reads, so they become a filter click. ***Missing* has no filter and never has** — the server cannot slice the reads list that way. It moves to Read cycles instead, which is the only place the number means anything: a cycle has a window and an expected set, and *“short by five”* is a sentence about a cycle.

---

01 ~~**Reads — Needs attention**~~ EXISTS BEHIND · NEW READ · NEW BUILD · MR-02 · MR-03 · MR-26

~~`/:tenant/reads?tab=attention`~~ — the tab and its three tiles are withdrawn

~~WAS: the landing tab. Three tiles — Missing 18 · Flagged 24 · Disputed 0 — with a drill link on two of them, and an all-clear line when every count is zero.~~

NOW: RETIRED by PO ruling of 2026-09-12. Kept as an entry rather than deleted, so the next reader finds out what happened to it instead of finding nothing.

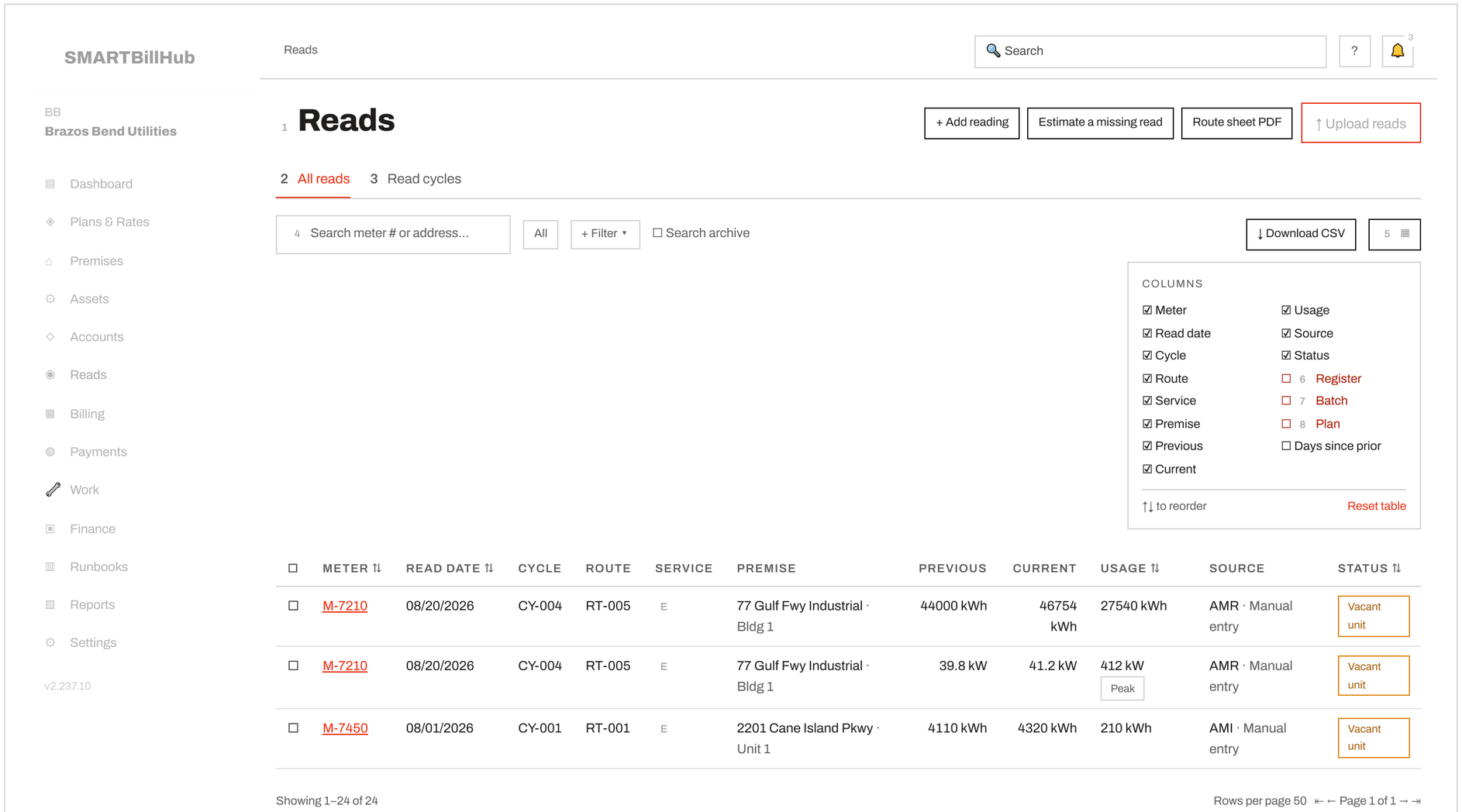
Where its content went	**Flagged** → the `status` filter on entry 02, which counts with the same predicate the tile did. **Disputed** → the `disputed` filter, likewise. **Missing** → entry 20, the cycle detail, where it is stated per cycle against that cycle's expected register set, and summarised per cycle on entry 15's *Registers collected* column.
What is genuinely lost	**The all-clear.** There is no longer one place that says “nothing needs attention” across the whole tenant. Per cycle, yes; across the tenant, no. Recorded as a cost of the ruling rather than argued against it.
What it costs to build	Front end only. The endpoint behind it (`GET /reads/needs-attention`) keeps its other caller on the Dashboard.

02 Reads

EXISTS BEHIND · NEW READ · NEW BUILD · MR-04 · MR-26 · MR-27 · MR-31

`/:tenant/reads · now the landing tab · every control writes itself into the URL`

The question: *“Show me the reading for this meter so I can check it against the bill.”*



Showing the last two years. Older readings need the archive search.

DW Dana Whitfield
System Admin

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — 1440, with the column picker open. Twelve of fifteen columns are visible, which is what a clerk actually sees** — *Register*, *Batch*, *Plan* and *Days since prior* are `defaultVisible: false` and live behind ▦. The picker is drawn open because the three changes on this screen are all inside it, and because a control this document keeps citing is worth showing once. The first row is shaded to show the whole row is now the target, not the meter cell alone. **An earlier version drew all fifteen at once and overflowed the window by 64px**, clipping the primary action and the pager — recorded in §10.

1 Page header · unchanged · B14 2, 40 · **UNCHANGED**
2 All reads · **now the landing tab** — `?tab` absent lands here · **CHANGED**
3 Read cycles · replaces the two doorless screens · r10 RDS-02R/02C · **NEW**
4 Search · posts `?q=`; matches the meter number **and the premise address** · UI-RULES 12 · **NEW READ** — needs the server
5 Column picker · show/hide, ↑↓ reorder, Reset · B14 14 · **UNCHANGED**
6 **Register**, was **Channel** · `meter_reads.channel` · **RELABELLED** — one word for one thing, and *Register* is the clerk's
7 Batch · `read_import_batches.batch_no`, **not the database id** · B14 31 · **NEW READ** — needs the server

As-built element	Disposition	Why
Header, kicker, four header actions	**UNCHANGED**	—
Needs attention tab	**REMOVED**	**PO ruling, 2026-09-12:** *"Need Attention — does not add any value."* Its three counts are accounted for in the callout above
All reads tab	**MOVED**	Becomes the landing tab
Search box	**RELABELLED**	*"Search meter #…"* → *"Search meter # or address…"*, because the read behind it widens. A clerk on a phone call has an address
7 filters · archive toggle · CSV · column picker · pager	**UNCHANGED**	—
13 of the 15 columns	**UNCHANGED**	—
Channel column	**RELABELLED**	→ **Register**. A user who knows the old word will not find it, which is why this is its own disposition
Batch column	**UNCHANGED in place, its VALUE changes**	uuid → `RB-2608-0044`
Row affordance	**MOVED**	The identity cell keeps working; the whole row becomes the target

```
As-built fields   19
  unchanged       15
  moved            2   (All reads tab, row affordance)
  relabelled       2   (Channel → Register, the search placeholder)
  removed          0   ← the retired TAB is entry 01, not a field of this screen
Remedy fields     19   = 19 − 0 removed + 0 new
```

Nothing on this screen is lost. The two server changes — the wider search and the batch

number — change what a field *contains*, not whether it is there.

What runs behind it

```
meter_reads (lineage heads only, superseded_by_id IS NULL)
  ├→ assets.Asset.asset_no ──────────────── the Meter cell            ⇄ metering→assets
  ├→ metering.ServicePoint → premises ──── the Premise cell          ⇄ walked to premises
  ├→ quantity_display.quantity_with_unit — Previous · Current · Usage
  └→ selectors.read_dict / read_list      selectors.py:5292-5335
      └→ GET /api/{tenant}/reads?q&sort&dir&page&per&f.*&archive   api.py:1395
          └→ useQuery(readsQueryKey) → <DataTable config={readsTableConfig}/>
                                                                   WHOLE
```

#	Field	Comes from	Owned by	Health
①	Select checkbox	*nothing — page state; `bulkSelectable(row)` decides enabled*	—	SERVED
②	Meter	`assets.assets.asset_no` via `read_dict.meter`	metering ⇄ assets	SERVED
③	Read date	`meter_reads.read_at` — a **calendar date**, stamped 00:00Z of the read day	metering	SERVED
④	Cycle	`metering.cycles.code` joined server-side	metering	SERVED
⑤	Route	`metering.routes.code`	metering	SERVED
⑥	Service	`tenancy.services.code` through the service point	metering ⇄ tenancy	SERVED
⑦	Premise · unit	`premises.address_line1` + `units.label`, walked	metering (premises is metering's own table)	SERVED
⑧	Previous	`prior_value_display` — the server composes value+unit	metering	SERVED
⑨	Current	`current_value_display`	metering	SERVED
⑩	Usage	`usage_display`	metering	SERVED
⑪	Source	`read_method` + `source`, two catalogue lookups	metering	SERVED
⑫	Status	`meter_reads.status`	metering	SERVED
⑬	Channel hidden	`meter_reads.channel`	metering	SERVED
⑭	Days since prior hidden	`meter_reads.days_since_prior`	metering	SERVED

	⑮ Batch hidden	`meter_reads.import_batch_id` — **the raw uuid**. `read_dict` serves the id and not `batch_no`.	metering	SERVED, BUT RULE 31
	⑯ Search box	posts `?q=`; matched in SQL on the meter number only	metering	SERVED, NARROWER THAN r10 DRAWS
	⑰ Route / Cycle filter options	`GET /routes`, `GET /cycles` through `listAll()` — every page, fetched whole	metering	SERVED, UNBOUNDED
	⑱ Service filter options	`GET /settings/services`, the tenant's enabled set	metering ⇄ tenancy	SERVED
	⑲ Total row count	`count` on the paged envelope	metering	SERVED

Screen health: RULE 31 on ⑮ — 19 fields, 16 SERVED, 2 served but narrower or unbounded than the screen implies, 1 a rule-31 breach. 0 ungraded. Flow WHOLE.

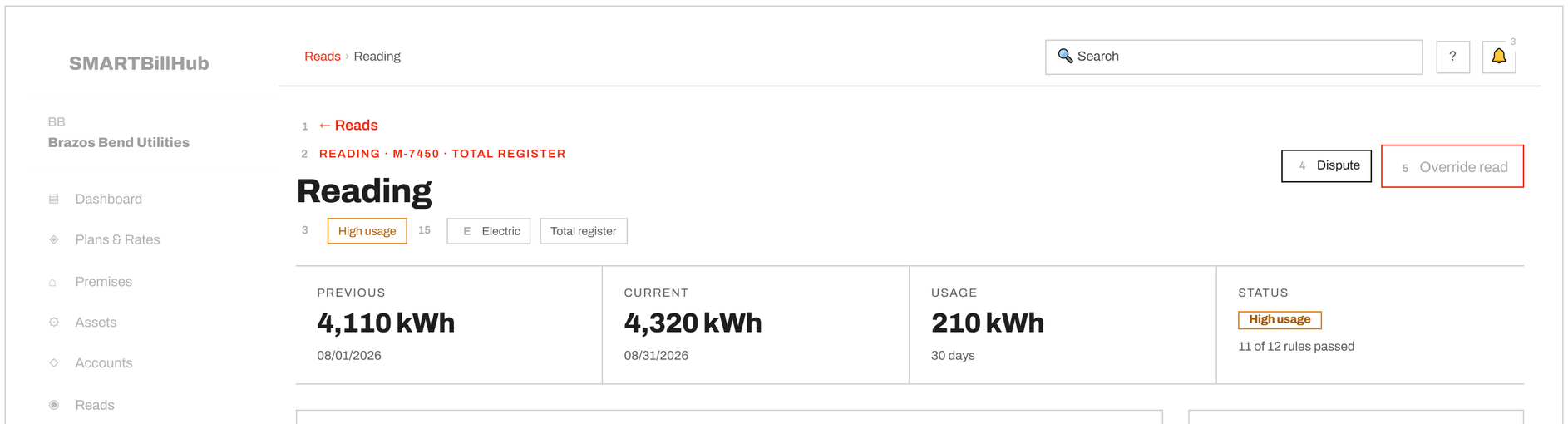
03 One reading

EXISTS BEHIND · NEW READ · NEW BUILD · MR-01 · MR-05 · MR-15 · MR-32 · MR-39 · MR-45 · MR-46 · MR-47 · MR-48 · MR-52

`/:tenant/reads/:readId` · second address: `/:tenant/reads?read=<id>` (entry 19)

The question: *"Why is this reading flagged, and can I prove to a regulator where the number came from?"*

Ten sections become six, and the page splits into two columns — the left answers *is this number right?*, the right answers *where did it come from?* Nothing is dropped; the merge is proved by the arithmetic below.



6 CALCULATION

(4,320 – 4,110) × 1 = **210 kWh**

13 **Registers this plan expects** · Time of use · On-peak ✓ · Off-peak ✓ · Mid-peak missing · Total ✓ 3 of 4 · see the meter →

7 FLAGS

High usage

154% of a 5.2 kWh average · threshold 150% · 3 readings in the average

Could not run · Master vs sub-meters — no master meter on this premises

14 **On a demand register this line reads differently** · the billable figure is the **peak itself in kW**, never a difference of two peaks

8 HISTORY

v1 · current version · no corrections

READ DATE	CURRENT	USAGE	SOURCE
08/01/2026	8,634 kWh	424 kWh	File import
07/01/2026	8,210 kWh	—	File import

9 Showing 1–2 of 2 Rows 50 ⇤ ← 1/1 → ⇥

10 LOCATION

CY-001 | RT-001 | Total register | File import | Walk-by

2201 Cane Island Pkwy · P-00002
Unit 1
11 ACC-2042

12 EVIDENCE

Batch	RB-2609-0006
File	reads.csv
Hash	808e48f7…e804
By	Dana Whitfield
At	09/12/2026 08:31
Session	sess-4417 · 10.2.0.14
Certified	Yes
Approvals	None required

RA Rosa Alvarez
Billing Clerk

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Revised · 1440.** The four figures stay the `KPIStrip` band — 2px rules top and bottom, not four cards. Below it the page is **two columns**: everything a clerk reads to judge the number on the left, everything they read to prove where it came from on the right. At narrow widths the right column drops beneath the left; nothing reflows inside a section.

1 Exit · ← `Reads` · B14 1 — an exit names where it goes · **UNCHANGED**
2 Kicker · meter · register · **UNCHANGED**
3 Status pill · `meter_reads.status` · B14 2 keeps state in the head · **UNCHANGED**
4 Dispute · outlined · `POST /reads/{id}/dispute` · **UNCHANGED**
5 Override read · filled accent; on a billed reading it becomes **Correct billed read** and is **LIVE** · `POST /reads/{id}/billed-correction` · **CHANGED**
6 **Calculation** · assembled from stored operands · BR-MET-056 · **RELABELLED** from *How this usage was worked out*
7 **Flags** · the stored outcome bag, thresholds included, and the rules that could not run · **RELABELLED** from *Why this reading is flagged*
8 **History** · versions + corrections + this meter's past reads, **merged into one** · **MOVED**
9 Pager · the envelope's own `count`, already served · B14 16 · **NEW**
10 **Location** · the six tags and the premise walk, **merged into one** · **MOVED**
11 Account · `meter_reads.account_id` — a real column served to nobody today · **NEW READ** — needs the server
12 **Evidence** · the compliance stamp · BR-MET-017 · **RELABELLED** from *Compliance metadata*
13 **Registers this plan expects** · **NEW READ** · **R1** — the set from the stored `plan_registers` outcome, ticked off. **Today this page says nothing at all about the other registers**, so a clerk looking at one TOU window cannot see that a sibling never arrived
14 **A demand register is a MAXIMUM** · **NEW READ** · **R4** — `BR-MET-051`. Running `BR-MET-019`'s delta over two peaks is `DEF-797`: **412 kW billed as 14 kWh, $154.00 against $4,532.00**
15 **Service and register, in the header** · **NEW READ** · PO ruling 2026-09-12. Both are served on the row today and both were buried: the service sat in *Location*, four groups down, and the register only in the kicker. **They belong with the identity because they decide how the number is read** — the unit, the plan's expected set and the rate all follow from them, and a clerk scanning a flagged reading should not have to scroll to learn it is the demand register on an electric meter. The *Location* group keeps the cycle, route, source and method; the service leaves it rather than appearing twice

Two decisions inside this redraw, both worth disagreeing with before it is built

The Estimate button is gone. It could never be enabled on this screen — estimation fills a *missing* reading and this reading exists, so its reason was permanent rather than a state. A control that can never become live is not a disabled control, it is a word in the wrong place. **Estimating is still reachable**, from the Reads header and from the shortfall on a cycle.

The account is drawn as a link, and it is not served today. `meter_reads.account_id` is a real foreign key in the reviewed cross-module set, and the screen currently prints *"the account is not served on a reading; open it through the unit"* over the top of it. **A stored column and a printed absence for the same fact is the two-homes problem waiting to happen.** Drawn served; if the ruling goes the other way the column is retired instead and this line reverts to the sentence.

As-built section	Disposition	Why
The four figures	**UNCHANGED**	—
Dispute banner · net-meter pair	**UNCHANGED**	Both conditional, both still above the fold when they apply
How this usage was worked out	**RELABELLED**	→ **Calculation**
Why this reading is flagged	**RELABELLED**	→ **Flags**; the "rules that could not run" list folds in under it rather than sitting apart
Tags	**MOVED**	Into **Location**, which is what the six tags are: which service, which cycle, which route, which register, how it arrived
Premise / unit / account	**MOVED**	Into **Location**, beneath the tags
Compliance metadata	**RELABELLED**	→ **Evidence**. Same eight rows, same order
Supersession history	**MOVED**	Into **History**
Billed corrections	**MOVED**	Into **History**, still conditional
Read history (this meter)	**MOVED**	Into **History**, and gains a pager
Estimate button	**REMOVED**	**PO ruling, 2026-09-12** — approved as change 2.3. It is a permanent refusal rather than a state, and estimating stays reachable from two other places

```
As-built fields   17
  unchanged        9
  moved            5   (tags · premise-unit-account · chain · corrections · read history)
  relabelled       3   (calculation · flags · evidence)
  removed          0   ← the Estimate BUTTON is an action, not a field; its row is above
Remedy fields     19   = 17 − 0 removed + 2 new (the history pager, the account link)

Sections        10 → 6   (band · calculation · flags · history · location · evidence)
```

The field census: 41 served, 32 shown, 9 nowhere

Counted from the payload, not from the drawing. `read_dict` serves **41** fields; **32** reach this screen. **Nine are served to nobody:**

Field	What is lost with it

`dispute_cleared_at` `dispute_cleared_by`	**The sharp one.** Entry 06 restricts clearing a dispute to a System Admin and records who and when — **and this page shows no trace it ever happened**, so the next person reopens a question that was answered
`imported_by` · `disposition` · `disposition_reason`	How this reading got in and what the clerk decided about it. **Evidence** shows the batch's stamp and not the row's
`event_context` · `trues_up_read_ids`	Why the read happened, and which readings it trues up
`read_type` · `period_end`	The consumption/export label a pinned contract keeps, and the closing end of the period whose start is shown

Why this count and not a count of drawn rows. A drawing can be missing a row nobody ever served, or show one nobody asked for; **served against shown is checkable against the code**, and it is the only field number in this document that can be re-derived by somebody who does not trust it. MR-52.

What runs behind it

```
meter_reads (one head) + rule_outcomes JSON + read_corrections + the chain walk
  ├→ selectors.read_dict + read_chain      selectors.py:5234
  │    └→ GET /api/{tenant}/reads/{id}     api.py:1447       → sections ①–⑩, ⑫–⑮
  ├→ read_import_batches (compliance)      GET /read-batches/{id}   api.py:1228
  │    └→ audit actor_names seam ————————— "Uploaded by" as a NAME  ⇄ metering→audit
  └→ assets read seam                      GET /assets/{id}/reads
       └→ the history table                                    BREAKS AT PAGINATION
```

#		Field	Comes from	Owned by	Health
	①	Kicker — meter · channel	`read_dict.meter` + `.channel`	metering ⇄ assets	SERVED
	②	Previous / Current	`prior_value_display` / `current_value_display`, composed with the unit server-side	metering	SERVED
	③	Usage · days	`usage_display`, `days_since_prior`	metering	SERVED
	④	Peak demand (a demand register draws this instead of Usage)	`consumption` + `unit`, the register's own value at the cycle read	metering	SERVED
	⑤	Rules passed / evaluated	`rule_outcomes` JSON, tallied in the browser from stored outcomes	metering	SERVED
	⑥	Math line	assembled from `prior_value`, `current_value`, `multiplier` and the server's own `usage_display`	metering	SERVED
	⑦	Flag reason + thresholds	the stored outcome bag — nothing is compared again here	metering	SERVED
	⑧	Tags: service · cycle · route · channel · transport · method	`read_dict` since DEF-045	metering ⇄ tenancy (service name)	SERVED

⑨	Premise · premise no · unit	walked to `premises`; links to the premises detail	metering	SERVED
⑩	**Account**	*nothing. A reading does not carry its account, and the screen refuses to join it in the browser*	—	NOT SERVED — stated
⑪	Batch number	`read_import_batches.batch_no`, falling back to the id when absent	metering	SERVED
⑫	File hash	`compliance.file_hash` (SHA-256)	metering	SERVED
⑬	Uploaded by	`compliance.uploaded_by_name`; the key rides on `title` as evidence	metering ⇄ audit	SERVED
⑭	Supersession chain	the server's walk of `superseded_by_id`	metering	SERVED
⑮	Billed corrections	`read_corrections` — references and a reason; **no money figure**, the credit note is billing's	metering ⇄ billing	SERVED
⑯	Read history rows	`GET /assets/{id}/reads`, the assets read seam	metering ⇄ assets	SERVED, UNPAGINATED
⑰	Net meter pair	`net_pair`, served on the read to save a round trip; renders for a net meter and nothing else, and **never nets the two**	metering	SERVED

Screen health: NOT SERVED on ⑩ — 17 fields, 15 SERVED, 1 unpaginated, 1 stated absence. 0 ungraded. Flow BREAKS AT THE HISTORY FETCH — and it takes the whole screen with it.

04 Correct the reading

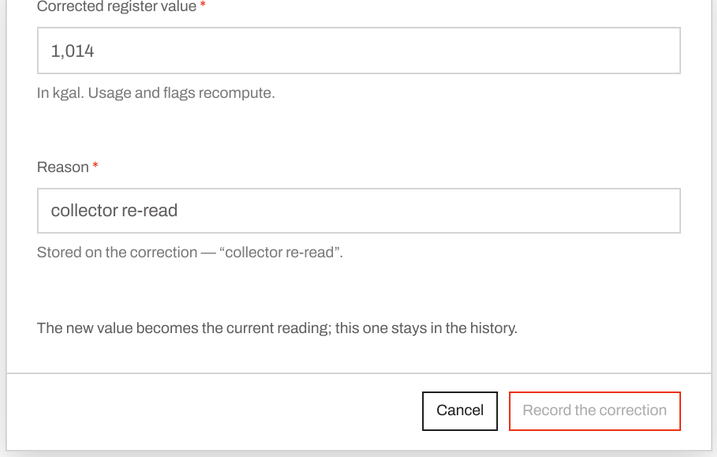
EXISTS BEHIND · NEW BUILD · BLOCKED · MR-24 · MR-31

`a dialog on entry 03 — no address of its own`

The question: *"The collector re-read it. Put the right number in without destroying the old one."*

READING · MTR-0452

Correct the reading for MTR-0452



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **As built, 1440.** `FormModal` at its declared 560px, ink scrim at 45%, 2px rules under the header and over the footer, square corners. The helper sits **under** its input — B14 rule 39's order, built and measured in the staff kit on 2026-09-12.

For	A clerk who has a better number for a reading that is not yet billed.
How you get here	Entry 03's primary action. **No other door**, and none is wanted: a correction needs the evidence on the page behind it.
Primary action	**Record the correction** — `POST /reads/{id}/supersede`.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — and it is an **append**: a new `meter_reads` row, the old head closed with `superseded_by_id` and `superseded_at`, monthly usage refreshed outside the transaction, one audit row. Re-superseding an already-superseded head is refused by name; a **billed** read is refused here and routed to the other door.
Walked	**Not walked.** Entry 03 throws before its header renders, so none of its three dialogs can be reached in the running app. Drawn from source, and graded UNVERIFIED for that reason.
Cognitive load	The helper states the unit, so the clerk types a register value and not a usage.
States	**Loading** the submit button shows working. **Empty** n/a. **Failed** the server's refusal renders under the field that caused it. **Domain state:** the button is disabled with its reason when the read is not the head, is billed, or the role is read-only — never missing.
Words	*Correction*, never *edit*. Nothing here is edited.
Components	`FormModal` · `TextField` · `GuardedButton`
Rules	BR-MET-024 · hard rule 4's shape · B14 10 (the asterisk), 11 (the consequence line), 39
Audience	Staff.

```
the form → POST /api/{tenant}/reads/{id}/supersede      api.py:1467 → 201 + Location
  ↳ services.supersede_read          new meter_reads row · old head closed · audit_log
      ↳ _refresh_monthly_usage       meter_read_monthly_usage  (outside the transaction)
          ↳ navigate(/reads/{new_id})  the page follows the new head        WHOLE
```

#	Field	Comes from	Owned by	Health
①	Corrected register value	*typed. The unit in the label is* `read.unit`	metering	SERVED
②	Reason	*typed; required; stored on the correction*	metering	SERVED
③	Consequence line	*a catalogue string*	—	SERVED

Screen health: SERVED — 3 of 3 on a read. Flow UNVERIFIED — not walked, because entry 03 throws first.

“What if the wrong REGISTER was read?” — this screen does not help, and nothing else does either

`supersede_read` accepts **two fields: `value` and `reason`** (`services.py:8398–8406`). It cannot move a reading to a different register, and neither can entry 06's billed correction. **That is not an oversight — it is enforced in the database.** Metering migration `0007`'s trigger leaves the marker columns free and **FREEZES THE MEASUREMENT**, in the code's own list: *asset · read_at · **channel** · prior/current value · multiplier · consumption · unit*. The only line in the module that ever sets a channel is the import parser.

So a reading filed on the wrong register is permanent, and hard rule 4 means there is no delete to fall back on. Two things then happen at once, and only one of them is visible:

What happens	What the product says
The **right** register is still missing	**Visible.** `plan_registers` warns, the meter joins the chase list, and somebody estimates it — an estimate standing in for a reading that *already arrived*, filed one column over
The **wrong** register now holds a reading	**Informational only.** It rides along as `unexpected` in the rule detail and **is never a severity** — the rule *“judges only the SHORTFALL”*, in its own docstring

On a demand or TOU plan this is a money path, and this repository has already paid for it once. A max-demand figure filed on `total` is priced as energy: that is `DEF-797` exactly — **412 kW billed as 14 kWh, $154.00 against $4,532.00**. The registers' units differ (`kW` against `kWh`, `BR-MET-143`), so the wrong column is not merely the wrong label.

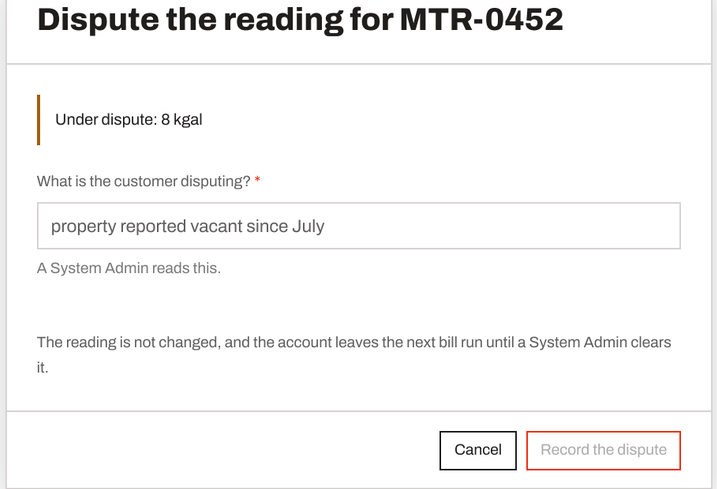
This is raised, not decided — the freeze is `OQ-209`, reported and unresolved, and widening it collides with a signed rule. **MR-24** puts the four shapes on the table: *(a)* let a correction carry a channel, which reopens OQ-209 · *(b)* a re-register path that supersedes the wrong row and creates the right one, two rows and no update, which is the only shape that leaves hard rule 4 intact · *(c)* leave it and say so on screen, which today no screen does · *(d)* **stop it at the door** — refuse at validate when a row's register is not in that meter's expected set, where today it is a silent `unexpected`. **(d) is the preventive half and costs nothing the other three cost**; it is the only one of the four this document recommends without a ruling.

05 Dispute the reading

EXISTS BEHIND · NEW BUILD · MR-31

a dialog on entry 03

The question: *“The customer says this is wrong. Stop the bill until somebody looks.”*



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **As built, 1440.** The consequence is stated *before* the commit, which is B14 rule 11 — and it is a real consequence: the account leaves the bill run from this moment. A **billed** reading is disputable on purpose; the issued bill stands (BR-MET-133) and the answer says so.

For	A clerk taking a customer's word for it, without changing a number.
How you get here	Entry 03's secondary action, which flips to *Clear the dispute* (entry 06) once one is open.
Primary action	**Record the dispute** — `POST /reads/{id}/dispute`.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — three marker columns (`disputed_at`, `dispute_reason`, `disputed_by`) plus an audit row. **It changes no value, by design**; the route exists because DEF-046 found those columns written by nothing.
Walked	**Not walked** — unreachable behind entry 03's crash. Drawn from source.
Cognitive load	It removes the question “will this stop the bill?” by answering it on the form.
States	**Loading** working state on submit. **Empty** n/a. **Failed** under the field. **Domain state:** already disputed — the button is disabled with *“This reading already carries an open dispute.”*
Words	The helper quotes an example sentence rather than describing one, because a System Admin will decide from exactly that text.
Components	`FormModal` · `TextField` · warning rail
Rules	BR-MET-024 · BR-MET-133 · B14 11, 33 (the consequence cap), 39
Audience	Staff.

```
the form → POST /api/{tenant}/reads/{id}/dispute        api.py:1496
  ↳ services.dispute_read   meter_reads.disputed_at / dispute_reason / disputed_by + audit_log
     ↳ the account is held out of the bill run from this moment  ⇄ metering→billing
```

↳ billed? the answer carries `issued_bill_stands_br_met_133` WHOLE

#	Field	Comes from	Owned by	Health
①	Under dispute: figure	`usage_display` from the read	metering	SERVED
②	What is being disputed	*typed; required; stored on the marker*	metering	SERVED
③	Consequence line	*catalogue string*	—	SERVED

Screen health: SERVED — 3 of 3 on a read. Flow UNVERIFIED — not walked, because entry 03 throws first.

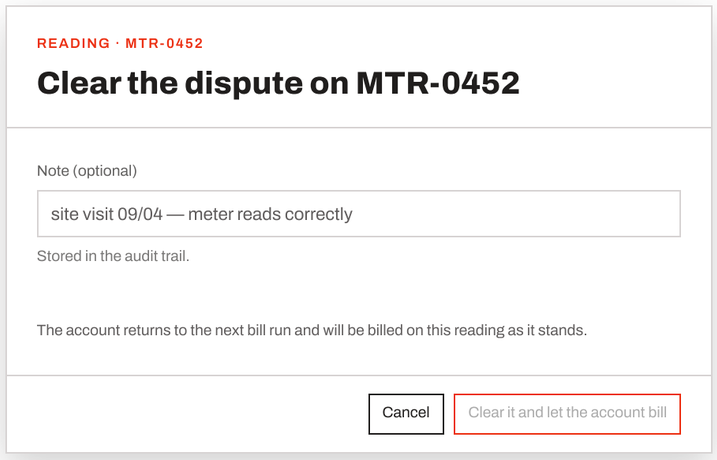
> **And the same question of a dispute: it marks, it does not fix**
>
> A dispute writes **a marker and a note**. It changes no value, no register and no consumption — the same migration `0007` freeze applies, and the docstring calls a dispute *"an ORTHOGONAL marker"* in the rule's own words. So the answer to *"the register is wrong"* is not here either: raising a dispute records that somebody objects and leaves the reading exactly where it is.
>
> **What it does settle is the bill**, and that one IS ruled: on a reading already billed, `BR-MET-133` says the issued bill **stands** — the code is `issued_bill_stands_br_met_133`, and it was renamed from a value that said *unruled* after the ruling landed. So a dispute is the right door for *"the customer says this is wrong"* and the wrong door for *"we filed it wrong"*.

06 Clear the dispute

EXISTS BEHIND · NEW BUILD · MR-31

a dialog on entry 03 · **System Admin only**

The question: *"I have looked. Let the account bill."*



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — As built, 1440.** The submit label is a sentence rather than a verb, because what it does is not *clear* — it is *let the account bill*. There is **no cleared-reason column**: the note goes to the audit trail only, and the helper says so rather than implying it lands on the reading.

For	A System Admin releasing an account back into the bill run.
How you get here	Entry 03's secondary action, which becomes this one whenever `read.disputed` is true. A Billing Clerk sees it **disabled with its reason**: *"Only a System Admin can clear a dispute and let the account bill again."*
Primary action	**Clear it and let the account bill** — `POST /reads/{id}/dispute/clear`.
Tabs	None.
Table controls	None.
Write surfaces	**Update** — `dispute_cleared_at`, `dispute_cleared_by`, plus audit. The optional note is audit-only.
Walked	**Not walked** — unreachable behind entry 03's crash. Drawn from source.
Cognitive load	The button says the consequence, so the admin does not have to hold "what happens to the bill" in their head.
States	**Loading** working. **Empty** n/a. **Failed** under the field — *"There is no open dispute on this reading to clear."* **Domain state:** not an admin.
Words	*Clear* is the verb the register uses; the button spells out what clearing does.
Components	`FormModal` · `TextField` · `GuardedButton`
Rules	BR-MET-024 · BR-SHL-025 (disabled with its reason, at the route *and* in the service) · B14 11
Audience	Staff — System Admin.

```
the form → POST /api/{tenant}/reads/{id}/dispute/clear   api.py:1529  [SystemAdminOnly]
  ↳ services.clear_dispute   dispute_cleared_at / dispute_cleared_by + audit_log
      ↳ the account rejoins the next bill run                      WHOLE
```

#	Field	Comes from	Owned by	Health
①	Note	*typed, optional; lands in `audit_log` and nowhere else*	metering ⇄ audit	SERVED
②	Consequence line	*catalogue string*	—	SERVED

Screen health: SERVED — 2 of 2 on a read. Flow UNVERIFIED — not walked, because entry 03 throws first.

03

Getting the readings in

Seven entries — a five-stage wizard, the counter case beside it, and the bulk bar that clears the flags it produces. This is the first step of a meter-to-cash system, and for five days in August it could not be performed by a human at all: the whole import pipeline shipped served, unit-tested and with no screen, so `meter_reads` held zero rows on a fully seeded

stack and no bill this product had ever produced came from a meter reading.

> **Five stages, five drawings — R6 — and the rail's names are not the code's**
>
> **This is a correction made by walking the app, and it is the reason Step 5 exists.** The wizard's own step ids are `["source","upload","validate","review","post"]` (`UploadReadsWizard.tsx:221`), and a first draft of this document drew and named the five screens from them. **The running app draws something else.** S28 mounts the **shared** `WorkflowRunner` (`UploadReadsWizard.tsx:3118`, `WORKFLOW_KINDS.reads` at `kinds.ts:548`), and the rail a clerk actually reads is **Extract · Transform · Validate · Load** WRITES **· Result**.
>
> **They do not line up one to one**, which is the part no static read would have produced: the `validate` and `review` panels **share stage 3**, `post` is stage 4, and stage 5 (*Result*) has no entry in `UPLOAD_STEP_IDS` at all. Every drawing below is the running app, measured at 1440 in Chromium against the deployed join on 2026-09-12.

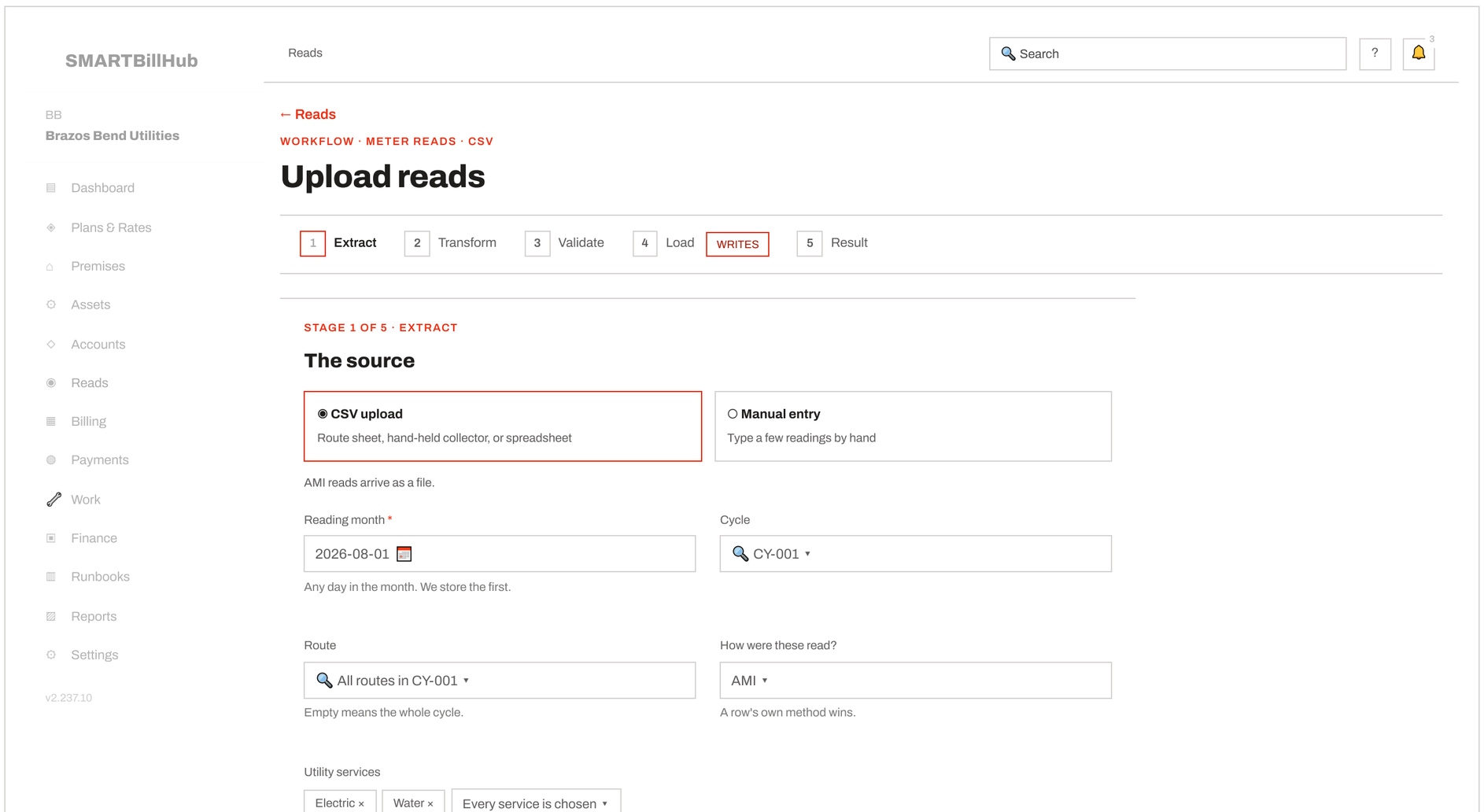
---

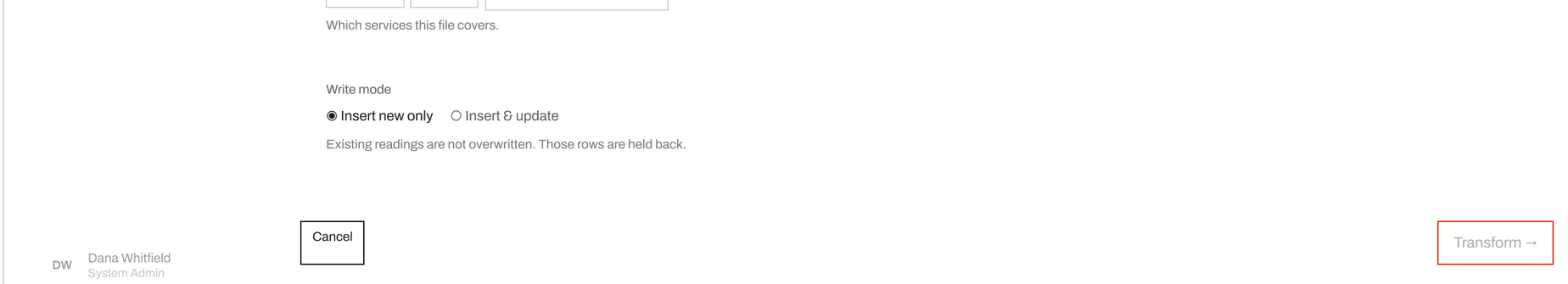
07 Upload reads · stage 1 · Extract

EXISTS BEHIND · NEW BUILD · MR-06 · MR-31

`/:tenant/reads/upload · stage 1 of 5 · panel source`

The question: *"Before I hand over the file — what is this file of?"*





REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — 1440 — measured in Chromium against the deployed join, then trimmed to the copy budget.** Six controls, and the write mode sits last on purpose: the five above it are facts *about* the file and this is the one question about what the file *means*. **There is no third transport card.** AMI replay is refused by the server and is not offered — a transport card with nothing behind it is not a card — and one short line says where AMI actually lands instead.

For	A clerk with a file, tagging it before anything is staged.
How you get here	**Two named doors:** entry 02's primary ↑ *Upload reads* (`ReadsListPage.tsx:838`) and the Dashboard's *Import reads* (`DashboardPage.tsx:1198`) — the latter **hidden, not disabled,** for a role without `import_reads`. Both walked.
Primary action	**Transform** →, navigation, per B14 rule 4's grammar. Nothing is written on this stage.
Tabs	None — a five-stage rail, B14 rule 3. Completed stages are clickable; stages ahead are disabled, which is right.
Table controls	None.
Write surfaces	**NOTHING.** The tags are page state until stage 2 stages the batch.
Walked	**Yes, 2026-09-12.** Signed in as the seeded owner, opened the wizard, set the reading month and cycle, pressed *Transform* →, and it advanced. No error.
Cognitive load	Six decisions taken once, before the file, rather than six times per row afterwards.
States	**Loading** the cycle, route and read-method lists arrive independently. **Empty** a tenant with no read methods gets a stated sentence instead of an empty dropdown. **Failed** the rail's refusal strip. **Domain state:** manual entry, which changes stage 2 entirely.
Words	*Reading month*, not *period*. The six legacy “read source” values are no longer offered — one of three competing vocabularies collapsed by CR-038.
Components	`WorkflowRunner` + `StageBar` · `SourceCards` · `DatePickerField` · `SelectField` ×3 · `ChipMultiSelect` · `WriteModeField`
Rules	BR-MET-014 · BR-MET-123 · BR-MET-012 · B14 3, 4, 8, 10, 39
Audience	Staff.

```
GET /cycles              → Cycle options           served PAGED; the client fetches EVERY page
GET /routes              → Route options           served PAGED; the client fetches EVERY page
GET /reads/read-methods  → How were these read?    unpaginated (exists because /lookups is admin-only)
GET /settings/services   → the service chips       ⇄ metering→tenancy
  page state only — nothing is posted until stage 2                    WHOLE
```

#	Field	Comes from	Owned by	Health

	① Transport	*a literal pair, `SOURCE_KINDS`; the third value exists only to name the server's refusal*	—	SERVED
	② Reading month	*a native `<input type="date">`; truncated to the first of the month server-side*	metering	SERVED, BROWSER-FORMATTED
	③ Cycle — options	`GET /cycles` through `listAll()`, which walks **every page** at 1000 a time	metering	SERVED, FETCH-EVERYTHING
	④ Route — options	`GET /routes`, same helper	metering	SERVED, FETCH-EVERYTHING
	⑤ Read method — options	`GET /reads/read-methods`, the **tenant's** list, not a constant	metering	SERVED
	⑥ Service chips	`GET /settings/services`, the enabled set	metering ⇄ tenancy	SERVED
	⑦ Write mode	*a two-value choice; posted with the batch*	metering	SERVED

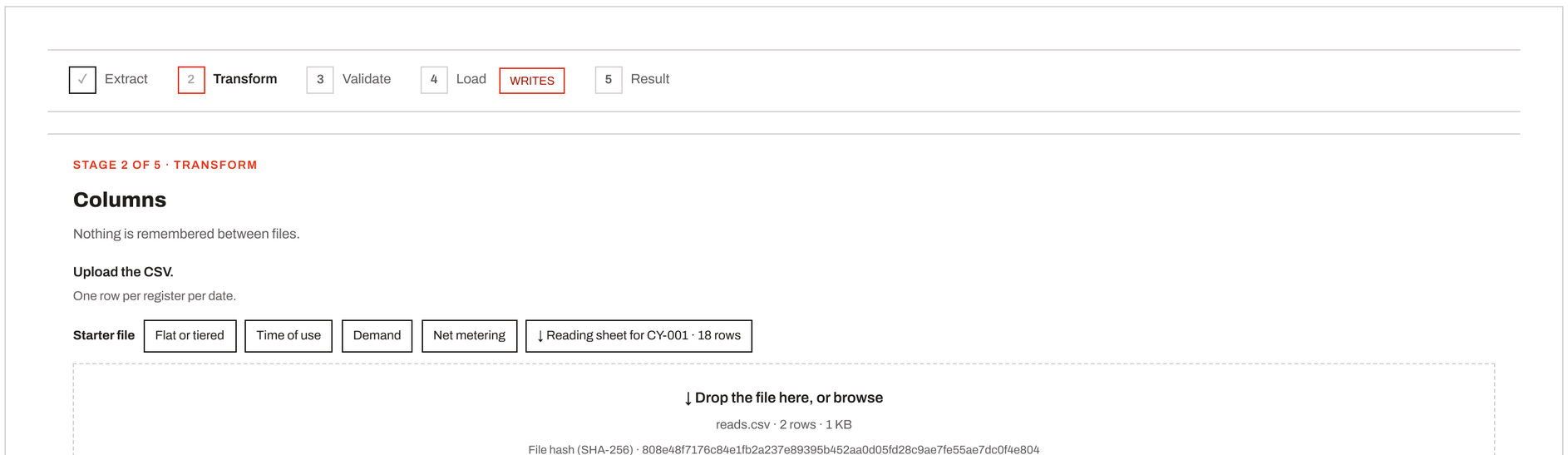
Screen health: SERVED, three qualified — 7 fields, 4 SERVED, 3 qualified. 0 ungraded. Flow WHOLE.

08 Upload reads · stage 2 · Transform

EXISTS BEHIND · NEW BUILD · BLOCKED · MR-08 · MR-24 · MR-31 · MR-40

`stage 2 of 5 · panel upload — two different screens behind one stage, chosen on stage 1`

The question: *“Here is the file. Did you get what I meant to send?”*



...or paste the rows

```
meter_no,read_date,channel,reading
```

Stage the batch

Batch RB-2609-0005 · 2 rows staged.

ROWS IN BATCH	METERS	SERVICES	READ DATE	TOTAL REGISTER
2	**2**	**—**	**2026-08-31**	**2**

SOURCE COLUMN	SAMPLE VALUE	MAPS TO	TRANSFORM	STATUS
meter_no	M-7450	meter_id	—	Matched
read_date	2026-08-31	read_date	—	Matched
channel	total	channel	—	Matched
reading	8700	reading	—	Matched

Not in your file: row_key, service, read_method, notes, demand_reset.

Only stage 4 creates readings.

← Back

Validate →

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **As built, 1440, after a real 2-row paste and a real stage.** Every figure in this drawing is what the app printed on 2026-09-12, including the batch number and the hash. **Four starters, and the shape is the file layout rather than the tariff** — there is deliberately no *seasonal* button, because a season is a property of the price list, so a seasonal read is a flat read on a date and a fifth button would send a clerk looking for a column no parser reads. The drop target **states what it received** — name, row count and the SHA-256 the receipt promises at stage 4.

Enter the readings

One row per register per date.

METER	DATE	REGISTER	READING	NOTES
M-7450	2026-08-31	Total register	8700	

+ Add a row

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **The manual branch of the same stage, drawn to the kit. As built, two of these five cells are hand-rolled** — the date is a raw `<input type="date">` and the register a raw `<select>`, both carrying `className="input"` rather than the kit's date field and `SelectField` (`UploadReadsWizard.tsx` ~1478–1515), while stage 1 three panels earlier uses the kit for the same kind of choice. **Drawn here as the kit renders them**, because that is what this screen should be; the departure is section 07's rows 7-05 and 7-06. The native date box is not a cosmetic difference — its calendar, its parsing and its keyboard are the browser's, so the control a clerk meets changes with the machine.

For	Handing the readings over — as a file, as pasted text, or typed.
How you get here	Stage 1's *Transform* →, or the rail once complete.
Primary action	**Stage the batch** — the first write in the flow, and the only one before stage 4.
Tabs	None.
Table controls	The mapping table and the manual grid have **no search, filter, columns, sort or pager** — correct: both are wizard tables, which B14 rule 16 exempts.
Write surfaces	**Add** *+ Add a row* on the manual branch. **Update** `POST /read-batches` → `read_import_batches` + `read_import_rows`. **Nothing lands in** `meter_reads`. **Delete** *Discard this batch* marks it discarded, never destroys it.
Walked	**Yes.** Pasted two rows, pressed *Stage the batch*, and the batch came back as **RB-2609-0005 · 2 rows · hash 808e48f7…** with all four columns matched and five template columns correctly reported absent. No error.
Cognitive load	The starters remove six unstated decisions; the pre-filled reading sheet removes the previous reading from the clerk's head entirely; the “not in your file” line removes the question *did it silently drop something?*
States	**Loading** *Staging….* **Empty** *“No file chosen yet.”* and *“Nothing has entered Reads yet.”* **Failed** a refusal strip naming the column or row. **Domain state:** a cycle with no read window — the sheet button explains itself rather than failing.
Words	*Register*, not *channel*, on the clerk-facing grid.
Components	`DropZone` · `MappingTable` · `ReconcileStrip` · `TextField multiline` · `PlainTable` · **raw `input` and `select`** in the manual grid
Rules	BR-MET-012 · BR-MET-032 (exactly one published template, no mapping memory) · BR-MET-054 · BR-MET-059 · B14 4, 7, 8, 16, 36
Audience	Staff.

```
the file bytes → POST /api/{tenant}/read-batches     api.py:1217 → 201 + Location
  ↳ services.stage_read_batch    read_import_batches + read_import_rows
      ├→ 🔒 file SHA-256 + tag fingerprint   read_import.py:741
      │    ↳ DB unique uq_read_import_batches_idempotency — it BLOCKS AT POST, not at stage
      └→ GET /read-batches/{id}    api.py:1228   EVERY staged row, no pager, cap 50,000
                                                 BREAKS AT PAGINATION
```

#		Field	Comes from	Owned by	Health
	①	Four starter buttons	`GET /reads/template?shape=`, validated against `READ_SHAPES`	metering	SERVED
	②	Reading sheet · row count	`GET /cycles/{id}/reading-sheet/summary` — the button names its own number	metering ⇄ rating	SERVED
	③	File name · rows · size · SHA-256	the staged batch's own compliance block	metering	SERVED
	④	Batch number	`read_import_batches.batch_no` — **a human-facing number, not the key**	metering	SERVED

	⑤ The five “what is in this file” figures	the staged batch's counts	metering	SERVED
	⑥ Column mapping + “not in your file”	`read_import.detect_mapping`, re-detected on every upload; nothing is stored	metering	SERVED
	⑦ Manual grid — Register options	`READ_CHANNELS`, *a literal array at `api/endpoints.ts:11657`*	—	HARDCODED
	⑧ Staged rows behind the panel	`GET /read-batches/{id}` — the whole batch, to a 50,000-row cap	metering	SERVED, UNPAGINATED

Screen health: HARDCODED on ⑦ — 8 fields, 6 SERVED, 1 unpaginated, 1 hardcoded. 0 ungraded. Flow BREAKS AT PAGINATION.

09 Upload reads · stage 3 · Validate

EXISTS BEHIND · NEW READ · NEW BUILD · MR-08 · MR-20 · MR-31 · MR-40 · MR-49

`stage 3 of 5 · panels validate and review — two of the code's five steps share this one stage`

The question: *“What in this file is wrong, what will not post, and what do I do about each one?”*

✓ Extract · ✓ Transform · 3 **Validate** · 4 Load WRITES · 5 Result

STAGE 3 OF 5 · VALIDATE

The rules

Errors block the load. Warnings do not.

Run the rules

ROWS IN BATCH	PASSED	WARNINGS	ERRORS	MISSING RECORDS
2	0	1	1	2

What the rules found

RULE	SEVERITY	HITS	ACTION
Negative usage	Error	1	Must be resolved
Low usage	Warning	1	Accept with a reason
Outside the read window	Warning	2	Accept with a reason
Registers the plan expects	Warning	2	Accept with a reason
Vacant unit	Warning	2	Accept with a reason

Meter factor	No signed threshold exists for this rule, so it reports instead of firing. **Cannot run**

Registers expected

PLAN	METERS	EXPECTS	ARRIVED	SHORT	DECISION
Time-of-use	2	8	2	6	On-peak · Mid-peak · Off-peak Estimate all Send someone out

Flagged rows

Give each row a decision. An undecided error will not post.

All flagged (2) | Errors (1) | Warnings (1) | Accept all warnings | Note demand reset (0)

This batch has no demand rows. **1 row will post. 1 will not.**

← Back | Discard this batch | Load →

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **As built, 1440, after a real validation run.** Every figure is the app's own from 2026-09-12. Three things are worth reading twice. **“Meter factor — cannot run”** is a rule declaring that it did not fire and why, which is hard-won law 1 obeyed rather than quoted. The five figures reconcile: passed 0 + warnings 1 + errors 1 = 2 rows. And **“1 rows will post”** is a plural bug — see section 07, row 7-09.

For	A clerk deciding whether this file is fit to post, and disposing of what is not.
How you get here	Stage 2's *Validate* →, or the rail.
Primary action	**Run the rules** — and it **writes**, which is the thing most easily missed about this stage: it stamps `severity` and `rule_outcomes` onto every staged row. A “check” that writes is still a write.
Tabs	None. Four **segments** over the flagged rows — All flagged · Errors · Warnings · Note demand reset — which are a segmented control, not tabs, and not in the URL.
Table controls	**Search NONE · Filter** the four segments, client-side over the rows already fetched · **Columns NONE · Sort NONE · Pagination NONE.** Exempt as a wizard table — but the rows behind it are the whole batch, which is the finding.
Write surfaces	**Update** `POST /read-batches/{id}/validate`; per-row `PATCH …/rows/{row_no}` (by **row number**); `DELETE …/rows/{row_id}/delete` (by **uuid** — the two row routes are keyed differently, worth knowing before a third is written); bulk demand-reset; and the two shortfall actions. **Add** NOTHING.
Walked	**Yes.** Pressed *Run the rules* on a real 2-row batch. Came back with 6 named rules, one of them declaring it could not run, a plan-shortfall table, and a split statement. No error.
Cognitive load	It puts the rule, its severity, its hit count and the action on one line, so the decision never needs a second screen.
States	**Loading** *Running….* **Empty** the strip is absent until a run has happened, rather than five zeros that would read as a clean file. **Failed** a refusal strip. **Domain state:** does-not-reconcile, which gets its own alert.
Words	*Errors* and *Warnings* are the server's own buckets, not a second vocabulary.
Components	`ReconcileStrip` · `RuleResultsTable` · `RegisterCompleteness` · `RowHandlingOptions` · `PostSplitStatement` · `RefusalStrip`
Rules	BR-MET-020 (the engine reads the settings row, never a constant) · BR-MET-051 · BR-MET-130 · BR-MET-940 · B14 4, 12, 16 (exempt), 17
Audience	Staff.

```
POST /api/{tenant}/read-batches/{id}/validate     api.py:1249
  ↳ read_validation_settings + read_validation_service_settings   (the S64 row, never a constant)
      ↳ read_import_rows.severity / rule_outcomes  +  batch counts
          ↳ GET /read-batches/{id}?severity=  → the rows the segments filter, in the browser
                                                BREAKS AT PAGINATION
```

#	Field	Comes from	Owned by	Health
①–⑤	Rows · Passed · Warnings · Errors · Missing records	the validate answer's own counts	metering	SERVED
⑥	Rule · severity · hits · action	`rule_outcomes` rolled up per rule	metering	SERVED
⑦	“Cannot run” + its reason	the outcome's own `reason` — here, *no signed threshold exists*	metering	SERVED
⑧	Registers each plan expects	`completeness` against the plan-required set	metering ⇄ rating	SERVED
⑨	Estimate all / Send someone out	`POST /reads/bulk-estimate` and the work-order path	metering ⇄ work	SERVED
⑩	Row disposition — options	*a literal disposition list in `workflowKit`*	—	HARDCODED
⑪	What will post / will not	`postSplit()` — the same computation stage 4 names on its button	metering	SERVED

Screen health: HARDCODED on ⑩ — 11 fields, 10 SERVED, 1 hardcoded. 0 ungraded.
Flow BREAKS AT PAGINATION.

10 Upload reads · stage 4 · Load

EXISTS BEHIND · NEW READ · NEW BUILD · BLOCKED · MR-09 · MR-17 · MR-20 · MR-31 · MR-49

`stage 4 of 5 · panel post — the only stage that writes`

The question: *“What exactly am I signing, and what will and will not go in?”*

Batch	RB-2609-0006
Source	CSV upload · reads.csv
File hash (SHA-256)	808e48f7176c84e1fb2a237e89395b452aa0d05fd28c9ae7fe55ae7dc0f4e804
Uploaded by	Dana Whitfield
Uploaded at	09/12/2026 08:31
Session · IP	2 Not captured

3 COMPOSITION

WILL POST	HELD BACK	NO DECISION	REJECTED
1	1	1	0

Held rows come back as a file. A warning still posts.

4 COVERAGE

3 of 5 meters covered. 2 meters still unread.

5 Short registers · **tou_mid** on MTR-0471 · **demand** on MTR-0488

Unread meters join the chase list.

6 CO-SIGN

Co-sign this batch — Optional. It does not hold the post.

7 CERTIFY

☐ I certify these readings are accurate to the best of my knowledge

This creates 1 reading. A posted reading cannot be un-created.

← Back — Discard this batch

8 1 reading will post — 9 Post readings

Tick the certification to post.

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Revised · 1440.** The stamp becomes **Evidence**, the strip becomes **Composition**, and the two sentences under them become five words each. The commit is **disabled until the tick**, with the reason under it and the consequence above it — so the refusal arrives *before* the click rather than after it. The button carries no number; the number is a fact on the line beside it, where it can be a sentence.

1 **Evidence** · the six stamp rows, now a named group · **RELABELLED** — matches entry 03's own Evidence group

2 Session · IP · **"Not captured"** in the refusal colour, never blank · **CHANGED** · finding 4.3

3 **Composition** · four counts, labels cut to two words · **RELABELLED** · B14 33

4 **Coverage** · unchanged in substance, promoted out of a trailing paragraph · **MOVED**

5 **Short registers, named** · **NEW READ** — from the `plan_registers` outcome the batch already stores. A TOU meter short of one window is invisible today until stage 3 is re-read

6 **Co-sign** · unchanged; its 12-word rationale becomes 6 · B14 33

7 **Certify** · the tick, then the consequence: *"This creates 1 reading."* + *"A posted reading cannot be un-created."* — 4 and 6 words, two sentences, which is exactly what B14 11 and 33 ask of a consequence before an irreversible act · **NEW READ**

8 The count, as a fact beside the button · **MOVED** off the label · finding 4.2

9 **Post readings** · two words, no count, **disabled until the tick**, reason underneath · **CHANGED** · findings 4.1 and 4.2

As-built element	Disposition	Why
All six stamp rows · four composition counts · the certification · co-sign · coverage	**UNCHANGED**	Every fact survives
"Session · IP — · —"	**RELABELLED**	An em dash reads as *nothing to say*; the audit row's whole job is to say whether there is
"This is the only stage that writes. Everything before it was rehearsal."	**REMOVED**	11 words of rationale. The WRITES chip on the rail says it without a sentence
"The batch is stamped with everything below…" (19 words)	**REMOVED**	Rationale. The group heading **Evidence** is the label
"A row that cannot be filed is held back with its reason and comes back as a file you can correct and upload again. A reading with a warning on it is still filed." (32 words)	**RELABELLED**	→ *"Held rows come back as a file."* + *"A warning still posts."* — 7 and 4
"Commit 1 rows ✓"	**RELABELLED**	→ **Post readings**. A label is ≤ 3 words and carries no sentence; the count was what broke the grammar
The refusal strip	**MOVED**	From *after* the press to *under* the disabled button. The strip existed because the press was accepted

```
As-built fields   16
  unchanged       13
  relabelled       3   (Session · IP, the held-rows line, the commit label)
  moved            1   (the refusal, from after to before)
  removed          0   fields — 62 WORDS of rationale
Remedy fields     18   = 16 + 2 new (short registers by name, the consequence line)
```

For	The clerk about to create real readings, and the auditor who will ask about them in eighteen months.
How you get here	Stage 3's *Load* →, which is disabled until the rules have run — walked and confirmed.
Primary action	**Commit N rows ✓** — a commit verb with its count. Not the danger tier, and that is right: posting is additive, and B14 rule 6 reserves danger for destructive commits.
Tabs	None.
Table controls	None — a stamp, a composition strip and two controls.
Write surfaces	**Update** `POST /read-batches/{id}/post`, the widest write in the module: `meter_reads` per row, `read_import_rows`, the batch, `premises_submeter_reconciliations`, `meter_read_monthly_usage` and asset monthly usage, plus audit. Also **co-sign**, System Admin or Executive, enforced in the *service* rather than at the route.
Walked	**Yes, and the refusal is real.** Pressed *Commit 1 rows ✓* with the certification unticked: the button was enabled, the click was accepted, and the screen answered *"Tick the certification before posting the batch."* Nothing posted.
Cognitive load	It removes the two questions that otherwise arrive after the fact: *how many* and *what did not go in*.
States	**Loading** the post is **synchronous** — BR-MET-055 asks for a background job and this deployment registers no job store, so a 202 with nothing behind it was refused and the

	wrapper is reported instead. **Empty** *"Nothing is staged yet."* blocks the stage. **Failed** refusal strip. **Domain state:** already posted — the idempotency guard names who posted it, when, and which batch.
Words	*Certify* is the clerk's own statement (BR-MET-017), recorded verbatim on the batch. *Commit* rather than *Post* on the button, which is the shared runner's word, not this screen's.
Components	`ReconcileStrip` · `CosignPanel` · `CycleShortfall` · `WriteModeStated` · `RefusalStrip`
Rules	BR-MET-017 · BR-MET-018 / OQ-044 (idempotency) · BR-MET-130 · BR-MET-940 · BR-MET-163 · hard rule 6 · hard rule 7 · B14 4, 6, 11
Audience	Staff.

```
POST /api/{tenant}/read-batches/{id}/post        api.py:1289   [synchronous]
  ├─🔒 guard 1  status == "posted"           → BIZ_IDEMPOTENT
  ├─🔒 guard 2  _idempotency_block(batch)    → refuses an identical file-hash + tag re-submission,
  │                                            NAMING who posted it, when, and which batch
  ├→ meter_reads                    (one row per posting staged row)
  ├→ read_import_rows               (rule_outcomes, meter_read_id)
  ├→ premises_submeter_reconciliations
  ├→ meter_read_monthly_usage  +  asset monthly usage
  └→ audit_log                                                  WHOLE
```

#	Field	Comes from	Owned by	Health
①	Batch number	`batch_no` — human-facing	metering	SERVED
②	Source · file name	`source_kind` + `compliance.file_name`	metering	SERVED
③	File hash	`compliance.file_hash` — the same hash stage 2 showed	metering	SERVED
④	Uploaded by	`compliance.uploaded_by_name`. This printed a raw primary key to a clerk for sixteen days and then *explained* it.	metering ⇄ audit	SERVED
⑤	Uploaded at	`compliance.uploaded_at`, rendered in the tenant's zone — measured 09/12/2026 08:31 against 13:31 UTC, which is correct for America/Chicago	metering	SERVED
⑥	Session · IP	`session_ref`, `ip` — **both empty on the walked post**	metering	SERVED, EMPTY IN PRACTICE
⑦	The seven composition figures	`postSplit()`	metering	SERVED
⑧	Cycle shortfall	`cycle_coverage`, or its absent-reason when the batch carries no cycle	metering ⇄ rating	SERVED
⑨	Certify checkbox	posted as `certified`; stored as	metering	SERVED

		`certified_by_uploader`		
	⑩ Co-sign	`POST /read-batches/{id}/cosign` — and the screen says plainly that it **does not hold the post**	metering	SERVED, HALF THE RULE

Screen health: SERVED, two qualified — 10 fields, 8 SERVED, 2 qualified. 0 ungraded. Flow WHOLE.

11 Upload reads · stage 5 · Result

EXISTS BEHIND · NEW BUILD · MR-41

`stage 5 of 5 · no entry in UPLOAD_STEP_IDS — the shared runner's receipt`

The question: *"It is done. What did I just do, and what do I do with the rows that did not go in?"*

> **Drawn from the source, and labelled as such — because the walk never rendered it**
>
> **The walk did not reach stage 5**, and the reason is worth keeping: the commit refuses until the certification is ticked, and the tick could not be driven from the script. The stage was reached in the rail and named, and the post was refused with its own sentence — but the receipt itself was never rendered in front of a person.
>
> **So this drawing is composed from `Receipt`, `RejectsDownload`, `SupersededSummary` and `RefusedBilledRows`, and it is the ONE drawing in this document that is a picture of the source rather than of the product.** Every other figure here was measured in a browser. Read it as a specification of what those four components compose, not as evidence of what a clerk saw. **The fix is a test that can tick the certification**, after which this becomes a walked drawing like the rest.

✓ Extract ✓ Transform ✓ Validate ✓ Load 5 **Result**

STAGE 5 OF 5 · RESULT

Posted

COMPOSITION

POSTED	SUPERSEDED	HELD BACK	REFUSED
1	0	1	0

HELD BACK

1 row could not be filed.

↓ Download the rejects file

Correct it and upload again.

BATCH

Batch RR-2609-0006



AS-BUILT, FROM CODE · composed from `Receipt`, `RejectsDownload`, `SupersededSummary`, `RefusedBilledRows`; the walk never rendered it — **1440 · composed from the source, not walked.** Three `Panels`, because a receipt is three separate objects — what happened, what did not, and the batch itself. **The rollback row says *Not available* rather than offering one**: the shared runner's rollback sentence is, for reads, a **refusal** — a posted read batch is immutable and a committed reading can only be superseded. Printing the runner's generic *"rolling back reverses every reading this run created"* here would be one target's sentence repeated where it is false.

For	The clerk immediately after the post.
How you get here	Stage 4's commit, once certified.
Primary action	*Not walked.* The runner's own `onDone` returns to `/reads` (`UploadReadsWizard.tsx:2998`).
Tabs	None.
Table controls	*Not walked.*
Write surfaces	**NOTHING** — the write happened at stage 4. The rejects file is a read (`GET /read-batches/{id}/results.csv?kind=failures`, streamed).
Walked	**No.** Reached in the rail; the receipt was not rendered. The drawing above is composed from the source and says so on its face — the only one in this document that is.
Cognitive load	*Not assessed.*
States	*Not walked.* The components it composes are `Receipt`, `RejectsDownload`, `SupersededSummary` and `RefusedBilledRows`.
Words	The runner's rollback sentence is, for reads, a **refusal**: a posted read batch is immutable and a committed reading can only be superseded, never un-created. Saying "rolling back reverses every reading this run created" here would be one target's sentence repeated where it is false.
Components	`Receipt` · `RejectsDownload` · `SupersededSummary` · `RefusedBilledRows`
Rules	BR-MET-017 · hard rule 4 · B14 17
Audience	Staff.

Screen health: UNGRADED — not walked. Flow UNVERIFIED.

12 Add a single reading

EXISTS BEHIND · NEW BUILD · MR-25 · MR-31

`a dialog on entry 02 — no address of its own`

The question: *"Somebody rang in one reading. I am not walking a five-step wizard for it."*

ONE METER, ONE DATE, ONE VALUE

Add a single reading

Meter number *

MTR-0451

We look it up.

MTR-0451 · 1408 Elm · 4B · Total register · ×1
Cycle CY-002 · period August 2026

This meter is on no route. The reading still posts, in August 2026.

Read date *

08/31/2026

Reading · kWh *

45910

The dial, not the usage.

ON A MULTI-REGISTER METER

On-peak · kWh *

12,480

Off-peak · kWh *

31,220

Mid-peak · kWh *

8,010

Total register · kWh *

51,710

Its plan expects four. All four post together.

Note

Cancel

Post reading ✓

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — 1440, with MR-25 drawn in.** The context panel is fetched **on blur, not per keystroke** — a per-character request asks the server about "M", "M-", "M-4" and every one is a miss. The amber strip is the **off-cycle warning** and it does *not* disable the post: the PO ruled *"(b) Accept and warn"*, because nothing is wrong with the reading. **The lower block is the change.** As built, a meter with more than one register renders *no value fields at all* and hands off to the five-stage wizard; here it grows the rows it has already fetched — by **name**, not by code, which is the other half of the fix, since this screen today prints `tou_on` where the reads list prints *On-peak*.

For	The counter case — one meter, one date, one value, one register.
How you get here	Entry 02's *+ Add reading* secondary action (`ReadsListPage.tsx:812`), which is unconditionally enabled.
Primary action	**Post reading ✓** — and it stages, validates and posts through the **same three endpoints** as a file. There is no second pipeline.
Tabs	None.
Table controls	None.
Write surfaces	**Add** — the whole dialog. **Update** — three calls in sequence: `POST /read-batches` → `/validate` → `/post`. The certification is **implicit**: pressing *Post reading* on a four-field form filled from the meter in front of you *is* the BR-MET-017 statement, and the stamp is recorded identically either way.
Walked	**Yes, 2026-09-12.** Opened from entry 02's *+ Add reading*, which is enabled. The dialog renders with the meter field and its helper *"Type the number on the meter, then check it."* The value fields stay hidden until a meter resolves, which is the `simple` gate working. Not posted.
Cognitive load	It removes the wizard. The context panel removes the second question — *is this the right meter?* — before the value is typed.
States	**Loading** the context fetch has its own state. **Empty** — a meter that resolves to nothing says so. **Failed** the refusal renders under the field. **Domain state:** a meter with more than one register — *"This meter has 3 registers. Use Upload reads to enter all of them."* and the value fields do not render.
Words	*Reading*, with the unit in the label, and the helper says *the number on the dial, not the usage* — the one mistake this form exists to prevent.
Components	`FormModal` · `TextField` ×3 · `DatePickerField` · warning rail
Rules	BR-MET-012 · BR-MET-017 · BR-MET-129 (a warned reading posts) · BR-MET-140 (the off-cycle warning) · OQ-422 closed *"(b) Accept and warn"* · B14 7, 10, 39
Audience	Staff.

```
on blur → GET /api/{tenant}/reads/quick-context?meter_no=   api.py:2155  (writes NOTHING)
  └→ submit → POST /read-batches → POST /read-batches/{id}/validate → POST /read-batches/{id}/post
      └→ the same pipeline a 412-row file walks, with one row              WHOLE
```

#	Field	Comes from	Owned by	Health
①	Meter number	*typed; resolved by exact lookup on blur*	metering ⇄ assets	SERVED
②	Context: address · unit · register · multiplier	`quick_read.quick_entry_context`	metering	SERVED
③	Derived cycle and period	the same payload; absent when the meter is on no route	metering	SERVED
④	Read date	*picked*	metering	SERVED

	⑤ Reading, with its unit in the label	*typed*; the unit is `context.meter.unit`	metering	SERVED
	⑥ Note	*typed, optional*	metering	SERVED
	⑦ Register name	the served **code** — no signed vocabulary maps a channel to a display word on this surface	metering	SERVED AS A CODE

Screen health: SERVED, one code where a word belongs — 7 fields, 6 SERVED, 1 a raw code. 0 ungraded. Flow WHOLE.

"Can it handle every kind of register?" — no, and the refusal is the good part

The modal asks the server what registers the meter has the moment you leave the meter box, and then **branches** (`QuickReadModal.tsx:400–418`):

The meter	What the modal does	Rate types
One register	Shows the reading field and posts with `channel: default_channel` — the meter's *own* register, not a hardcoded `total`. Four fields, one value, done.	**Flat · tiered · seasonal · capacity**
More than one	**It does not render the reading fields at all.** An amber-ruled panel says *"This meter has 3 registers."* + *"Use Upload reads to enter all of them."* and hands off.	**Time of use · peak/off-peak · demand · net-metered**
Registers unknown	Same hand-off, different sentence — it does not guess a register.	—

Refusing is right, and the reason is three entries back. A modal with one value box on a TOU meter would file one window and leave two missing — and because the register of a filed reading **can never be changed** (entry 04), a half-filed TOU meter is a permanent one. The boundary is deliberate: *"a multi-register meter is detected on SELECTION and the modal hands off — it does not grow rows inside itself."*

What is wrong is where it hands off TO. A clerk standing at a demand meter with two numbers on a card is sent into a **five-stage file wizard**. The wizard's manual grid does take them — one row per register — so the path exists; it is five stages for two numbers.

MR-25 · let the modal grow the rows it already knows about. It has already fetched `channels` — the exact list, by name. Rendering three labelled number boxes instead of one is the same form, the same endpoint and the same pipeline, and it deletes the detour. **The hand-off stays for `channel_count === null`**, which is the only case where the modal genuinely does not know what to ask for.

And one copy fault while we are here: this screen renders the register as the served **code** — `tou_on`, not *On-peak* — because, in its own words, *"no signed vocabulary maps a channel to a display word."* The reads list *does* translate it (`metering.reads.channel.*`), so the same register reads two ways on two screens. *"Use Upload reads to enter all of them."* is also 7 words against rule 33's 6 — *"Upload reads takes all three."* is 5.

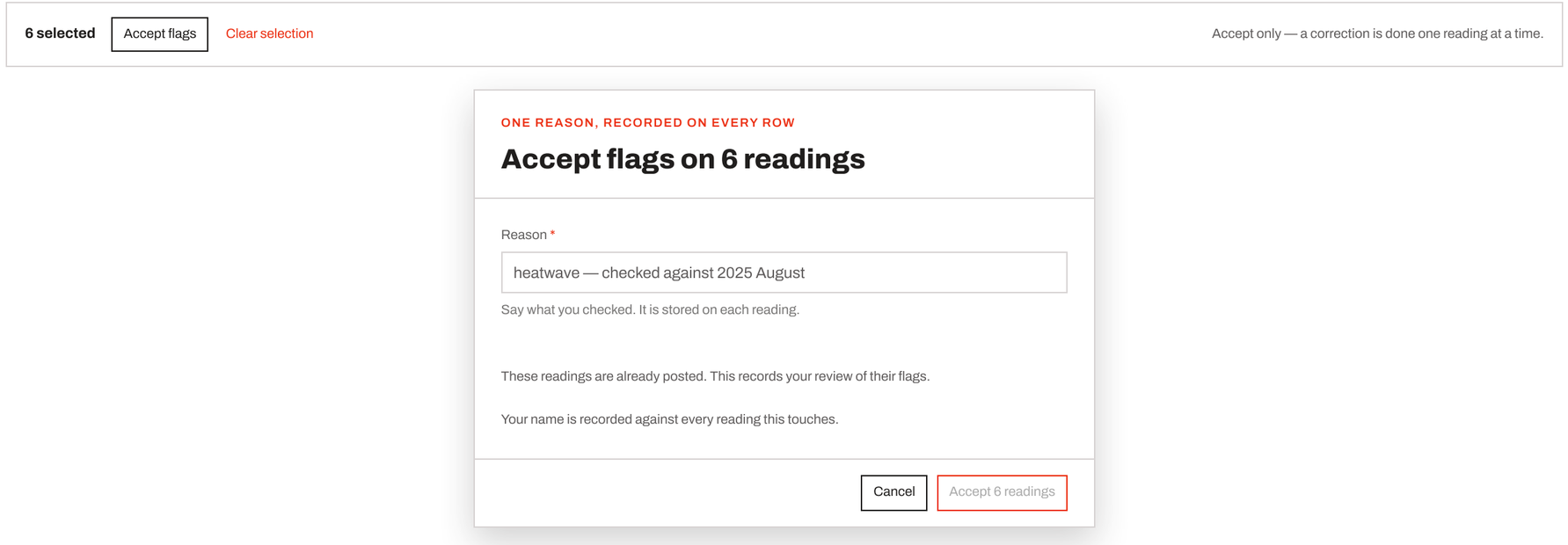
---

13 Accept flags in bulk

EXISTS BEHIND · BLOCKED · MR-10

a bar and a dialog on entry 02 — the bar renders only once a row is ticked

The question: *"I have checked forty high-usage flags and they are all the heatwave. Clear them in one go, and put my name on it."*



AS-BUILT · measured in Chromium on 2026-09-12 at `3706b459` — **As built, 1440.** The bar renders **nothing** until something is selected — a permanent empty toolbar over every list is chrome nobody reads. A row that cannot be accepted keeps its checkbox and has it **disabled with the reason on hover**, so a clerk learns why rather than wondering where the control went.

For	A clerk clearing a batch of flags they have genuinely checked.
How you get here	Ticking a row checkbox on entry 02. No other door, and none is possible — the selection *is* the entry.
Primary action	**Accept N readings** — `POST /reads/bulk-fix`, capped at **500 rows** server-side.
Tabs	None.
Table controls	The bar's own refusal list is a plain list, no controls.
Write surfaces	**Update** — three marker columns per read (`fix_disposition`, `fix_reason`, `fixed_by`) and one audit row each. **It cannot clear an error and cannot hold a read out of billing. Add / delete** — NOTHING.
Walked	**Yes.** Ticked four row checkboxes on entry 02; the bar appeared reading *"4 selected · Accept flags · Clear selection · Accept only — a correction is done one reading at a time."* Not submitted.
Cognitive load	One reason typed once, about rows the clerk can see. Changing a filter or a page **clears the selection** deliberately — accepting forty rows you can no longer see, with a reason about the ones you could, is the failure this prevents.
States	**Loading** working on the submit. **Empty** — the bar is absent. **Failed** the answer is **200 with per-row outcomes**, never a 4xx for a partly-refused selection, and every refusal names its meter and its reason. **Domain state:** a selection larger than 500.
Words	*Accept*, and the bar says so: the other BR-MET-023 verbs each need a per-row value or a per-row reason, and the screen states that rather than leaving a clerk hunting for Override.
Components	`ReadsBulkBar` · `FormModal` · `TextField` · refusal list

Rules	BR-MET-023 · BR-MET-129 (a warned reading posts; a flag is not a failure) · BR-SHL-025 · DEF-046 finding 2 (the three columns written by nothing) · hard rule 7
Audience	Staff.

```
row checkboxes (page state) → POST /api/{tenant}/reads/bulk-fix   api.py:2112
  ↳ services._stamp_fix    fix_disposition / fix_reason / fixed_by  + one audit_log row per read
      ↳ 200 with applied[] and refused[]      cap 500 rows          WHOLE
```

#	Field	Comes from	Owned by	Health
①	Selected count	*page state*	—	SERVED
②	Reason	*typed; required; stored on every row*	metering	SERVED
③	Result — applied of selected	the endpoint's own answer	metering	SERVED
④	Per-row refusals	`refused[]` with a named reason each	metering	SERVED
⑤	Why a row cannot be selected	`bulkSelectable(row)` in the browser, from the row the list already holds	—	SERVED

Screen health: SERVED — 5 of 5. Flow WHOLE.

04

Read cycles

Six entries, and the group that changes most. Two screens that nothing in the product pointed at become **one tab with a list and a detail behind it** — and the reading work a route does gets read where it means something, against the cycle whose window it belongs to.

Ruled 2026-09-12

“Cycle and Route — we should say Read Cycles and within that we should show Routes and their completion.” And: **“we will need a list view of read cycles and we will need a detail view of read cycles.”**

Routes stops being a screen. It becomes a section of a cycle, which is what it always was in substance: a route's collection progress is measured against a cycle's window, and on its own it had no window to be measured against. **The two doorless addresses are closed by the same change that fixes what they were saying** — the three denominators, the clock in the date column, the countdown that could not say “closed”.

On the numbers in the margin

Entry numbers are **kept as they were** rather than resequenced, because sections 05 to 09 and the whole of appendix C refer to screens by number — renumbering would silently invalidate every one of those references. So the new cycle detail is **entry 20** and is read here, in its place, rather than at the end.

14 ~~Routes~~ EXISTS BEHIND NEW READ NEW BUILD MR-02 · MR-11 · MR-12 · MR-13 · MR-16 · MR-19 · MR-21 · MR-28

~~`/:tenant/reads/routes`~~ — the address redirects; the content moves into entry 20

~~WAS: a top-level list of every route with its collection progress — and nothing in the product linked to it. Nine columns, no toolbar, client-side paging, a date column printing a clock a day early, and a Collected column counting readings beside a Meters column counting meters.~~

NOW: RETIRED as a screen by PO ruling of 2026-09-12. Every column survives, inside the cycle it belongs to — see entry 20.

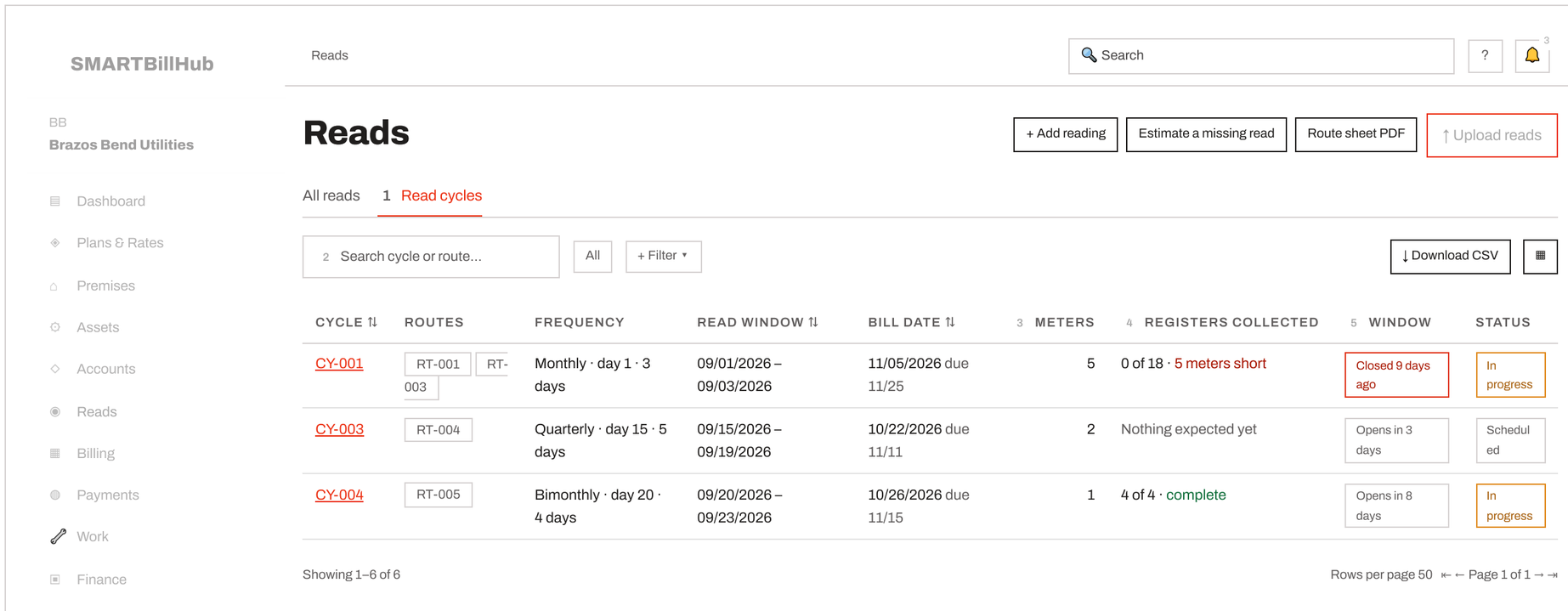
Not drawn. A retired screen has no final state, and this document now carries one drawing per screen — the one that will exist. What was on it, and where each column went, is the table below.

Where its content went	All nine columns move into entry 20's **Routes** section, under the cycle whose window they are measured against. *Cycle* stops being a column there, because the cycle is the page.
What is genuinely lost	**The all-routes view.** A supervisor who wants "every route in the utility, one list" no longer has one — they read routes cycle by cycle. Recorded as a cost of the ruling rather than argued against it; if it bites, the answer is a Routes tab, not the old orphan screen.
The address	**Kept and redirected** to the Read cycles tab, the way the retired drawer's `?read=` links are. There are tickets with these URLs in them.

15 Read cycles — the list EXISTS BEHIND NEW READ NEW BUILD BLOCKED MR-02 · MR-07 · MR-11 · MR-12 · MR-13 · MR-16 · MR-21 · MR-29 · MR-38

`/:tenant/reads?tab=cycles` · was ~~`/:tenant/reads/cycles`~~, which had no door

The question: *"Which cycles can I bill on Friday, and which are short?"*





REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Revised · 1440.** Ten columns become nine: *Start day* and *Days to complete* fold into **Frequency**, which is where a person reads them anyway (*"Monthly · day 1 · 3 days"*), and the countdown becomes a **Window** pill that can say the thing the old one could not — **that the window has closed**. The row is the target; it opens entry 20.

1. Read cycles tab · replaces both doorless addresses · **NEW**
2. Search · cycle code and route code · UI-RULES 12 · **NEW READ**
3. Meters · `cycle.expected` · **UNCHANGED** — and now the only meter count on the row
4. **Registers collected** · the chase read · BR-MET-057 · **RELABELLED** from *Completeness*, because it counts registers and the column beside it counts meters
5. **Window** · derived from the window dates against the tenant's today · **NEW** — replaces *"0 days left"*, which said that nine days after the window shut

> ⛔ **Two of these depend on a disagreement nobody has settled**
>
> The **Window** pill and the **Registers collected** figure are both computed from the cycle's window — and **two derivations of "the current window" give different answers**. Measured on one day: the cycle row printed August while the completeness engine answered September. The screen works around it today by passing the row's own window into the chase call, so both halves of one line describe one period.
>
> **Build these two on the fork and you get a screen that disagrees with itself.** It is a filed backend disagreement, not a design question, and it needs settling before ⑤ and ④ are trustworthy. Everything else on this drawing is safe to build now.

As-built element	Disposition	Why
Cycle · Routes · Bill date · Meters · Status	**UNCHANGED**	—
Read window	**UNCHANGED**	—
Start day · Days to complete	**MOVED**	Into the Frequency cell they describe. Both still on screen, in the same row
Completeness	**RELABELLED**	→ **Registers collected**. Someone searching for the old word will not find it, which is why this is its own disposition
"N days left"	**RELABELLED**	→ the **Window** pill, which can also say *Closed 9 days ago*
Chase link	**MOVED**	Into entry 20, where the meters it names are listed
Manage in Settings →	**MOVED**	Into entry 20's header, beside the cycle it configures

```
As-built fields   11
  unchanged        6
  moved            3   (start day, days to complete, the chase link)
  relabelled       2   (completeness, days-left)
```

```
  removed          0
Remedy fields     13   = 11 − 0 removed + 2 new (search, the Window pill)
```

For	The person deciding whether a cycle can be billed.
How you get here	**The Read cycles tab** — which is the whole point of the change. Today: **NOTHING**.
Primary action	None on the list. `BR-MET-011` puts every create, edit and delete in Settings, and this file imports no mutation wrapper at all.
Table controls	**Search · filter · columns · sort · server pager** — all four, where today there are none and the pager runs in the browser over a fetch-everything read.
Write surfaces	**Detail** the row opens entry 20. **Add / edit / update NOTHING**, by rule.
Walked	**The as-built was**, by typing the URL — six cycles, and CY-001 reading *5 · 0 of 18 · 0% · 0 days left · chase 5 meters* on one line. The revision is drawn, not walked.
States	**Loading** table skeleton. **Empty** *"No cycles are configured yet."* **plus *Add a cycle in Settings*** →, which it lacks today. **Failed** reason + Retry. **Domain state:** before the window opens.
Rules	BR-MET-010 · 011 · 057 · 078 · CR-029 D2, D7 · B14 12, 13, 14, 16, 17 · NFR-3
Audience	Staff. Drawn desktop.

20 A cycle — the detail

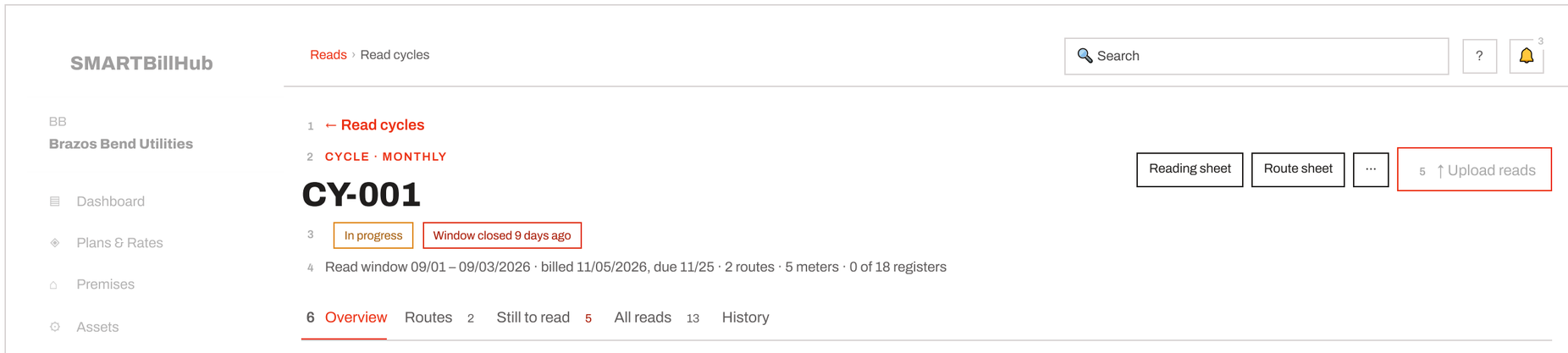
NEW READ · NEW BUILD · BLOCKED · MR-10 · MR-28 · MR-30 · MR-38 · MR-43 · MR-44 · MR-49 · MR-50 · MR-51

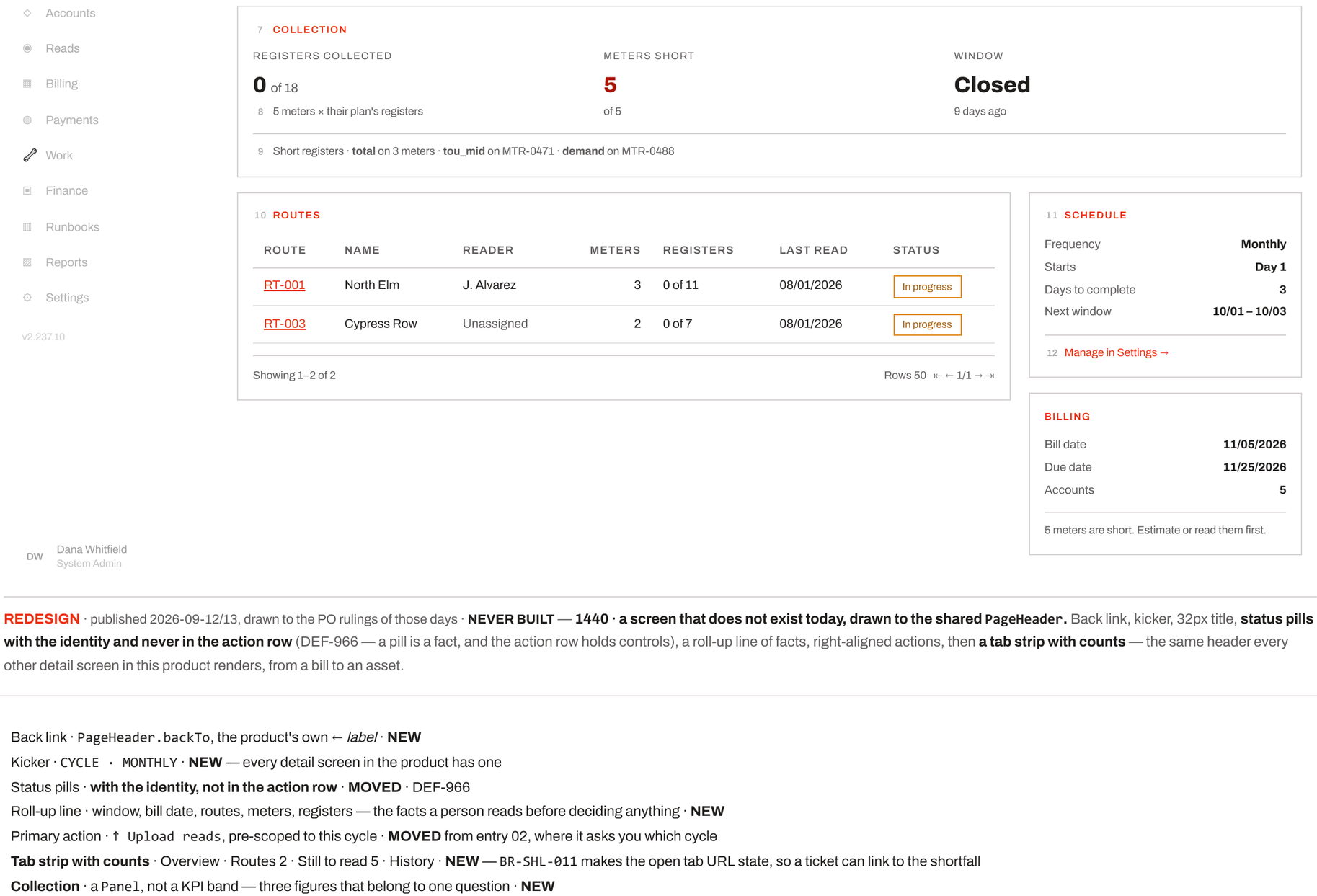
`/:tenant/reads/cycles/:cycleId` · opened by a row on entry 15

The question: *"CY-001 is short. Which routes, which meters, and what do I do about it before Friday?"*

Why this screen exists

It is the second half of the ruling — **"within that we should show Routes and their completion"**. Retiring entry 14 moved nine columns with nowhere to go; this is where they went. Everything drawn here is **served today** by an endpoint the product already answers: the cycle, its routes, its chase list, its two sheets and its estimate dialog. **No new endpoint is proposed.**



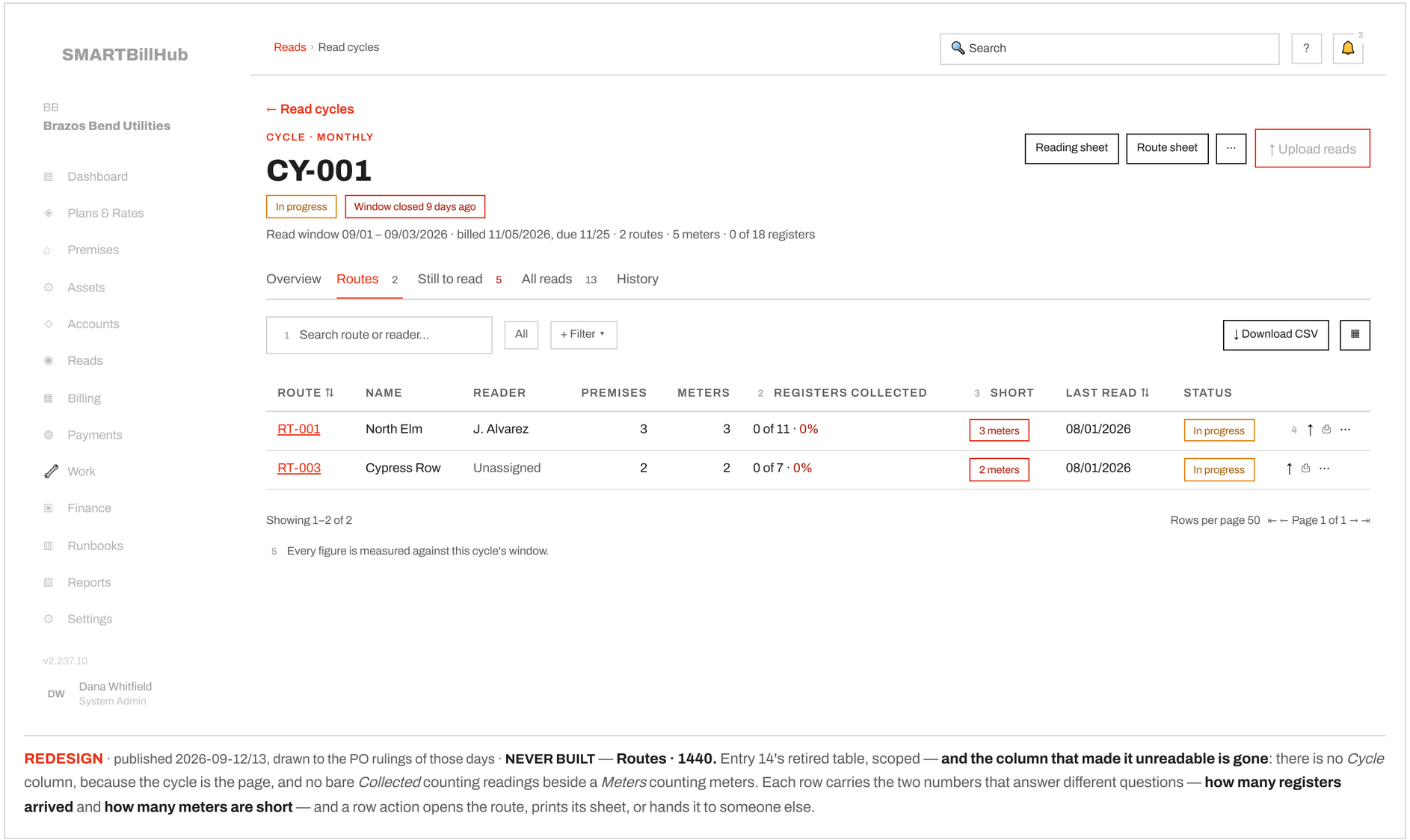


REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **1440 · a screen that does not exist today, drawn to the shared `PageHeader`.** Back link, kicker, 32px title, **status pills with the identity and never in the action row** (DEF-966 — a pill is a fact, and the action row holds controls), a roll-up line of facts, right-aligned actions, then **a tab strip with counts** — the same header every other detail screen in this product renders, from a bill to an asset.

1 Back link · `PageHeader.backTo`, the product's own ← *label* · **NEW**
2 Kicker · `CYCLE · MONTHLY` · **NEW** — every detail screen in the product has one
3 Status pills · **with the identity, not in the action row** · **MOVED** · DEF-966
4 Roll-up line · window, bill date, routes, meters, registers — the facts a person reads before deciding anything · **NEW**
5 Primary action · ↑ `Upload reads`, pre-scoped to this cycle · **MOVED** from entry 02, where it asks you which cycle
6 **Tab strip with counts** · Overview · Routes 2 · Still to read 5 · History · **NEW** — `BR-SHL-011` makes the open tab URL state, so a ticket can link to the shortfall
7 **Collection** · a `Panel`, not a KPI band — three figures that belong to one question · **NEW**
8 **The denominator, explained** · *"5 meters × their plan's registers"* · **NEW READ** — **R5**. Today 18 is a number with no stated basis
9 **Short registers, by name** · **NEW READ** — from the same resolver stage 3 already uses. This is the rate-type answer on this screen: a TOU meter short of one window says *which* window
10 **Routes** · entry 14's table, scoped, one column lighter — *Cycle* is the page · **MOVED**
11 **Schedule** · the three configuration facts, out of the list's columns · **MOVED**
12 **Manage in Settings** → · the config escape, on the panel it configures · **MOVED**

> **Four tabs, four screens**
>
> `BR-SHL-011` makes the open tab **URL state**, so each of these is separately addressable and separately linkable — a ticket that says *"CY-001 is five short"* points at the third one, not at the page. The header, the roll-up line and the actions are the same on all four; only the body below the strip changes. **The drawing above is the Overview tab.**

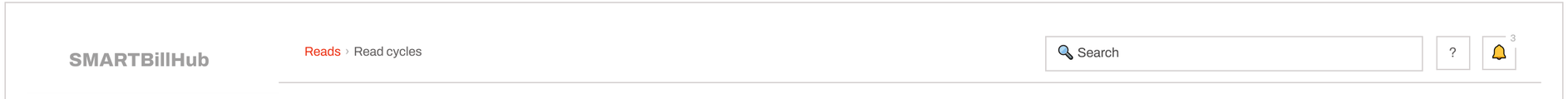
Routes — the tab



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Routes · 1440.** Entry 14's retired table, scoped — **and the column that made it unreadable is gone**: there is no *Cycle* column, because the cycle is the page, and no bare *Collected* counting readings beside a *Meters* counting meters. Each row carries the two numbers that answer different questions — **how many registers arrived** and **how many meters are short** — and a row action opens the route, prints its sheet, or hands it to someone else.

1. Search and filter · B14 12 · **NEW** — entry 14 had no toolbar at all
2. **Registers collected** · the plan-driven denominator, per route · **RELABELLED** from *Collected*, which counted readings beside a column counting meters
3. **Short** · meters, not registers — the two questions kept apart · **NEW READ**
4. Row actions · upload for this route · print its sheet · reassign · **NEW** · B14 15
5. One floor note, 8 words · B14 33

Still to read — the tab



BB
Brazos Bend Utilities

- Dashboard
- Plans & Rates
- Premises
- Assets
- Accounts
- Reads
- Billing
- Payments
- Work
- Finance
- Runbooks
- Reports
- Settings

v2.237.10

← Read cycles

CYCLE · MONTHLY

CY-001

Reading sheet | ··· | ↑ Upload reads

In progress | Window closed 9 days ago

Read window 09/01 – 09/03/2026 · billed 11/05/2026, due 11/25 · 2 routes · 5 meters · 0 of 18 registers

Overview | Routes 2 | Still to read 5 | All reads 13 | History

Search meter or address… | 1 All 5 | Total register (3) | On/off-peak (1) | Demand (1) | ↓ Download CSV

2 2 selected | Estimate these | Send someone out | Clear

☐	METER	PREMISES	UNIT	3 REGISTERS STILL NEEDED	PLAN	ROUTE	4 DAYS LEFT	LAST ACTUAL
☑	MTR-0463	22 Cypress Row	—	Total register	Tiered	RT-003	−9	08/01/2026
☑	MTR-0471	1408 Elm	2C	On-peak, **Mid-peak**, Off-peak	Time of use	RT-001	−9	08/01/2026
☐	MTR-0488	9 Pecan Way	—	**Demand**	Demand	RT-001	−9	08/01/2026
☐	MTR-0492	9 Pecan Way	—	Total register	Tiered	RT-001	−9	07/01/2026
☐	MTR-0501	31 Cypress Row	—	Total register	Tiered	RT-003	−9	None on file

5 Showing 1–5 of 5 — Rows per page 50 ⇤ ← Page 1 of 1 → ⇥

6 MTR-0501 has no actual reading. It cannot be estimated.

DW Dana Whitfield
System Admin

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Still to read · 1440.** Entry 16's panel, out from behind a table cell and made actionable. **The register column is the rate-type answer on this screen** — a TOU meter names the *window* it is short of, a demand meter names **Demand**, and the filter chips above count by register so a clerk can work one kind at a time. The dark bar is the product's bulk bar, appearing on selection. **The last row is the case that cannot be fixed here**: no actual reading has ever been taken, so `BR-MET-026`'s trailing average has nothing to average and the estimate refuses by name — the floor note says so before the clerk ticks it.

1. **Filter by register** · counts by what is missing · **NEW** — this is the rate-type filter, and it is what lets a clerk estimate all the demand meters in one pass
2. **The bulk bar** · `POST /reads/bulk-estimate`, **served since CR-038 and called by nothing**. The code deferred it to "W1's completeness surface"; this is that surface
3. **Registers still needed** · named per meter, the missing one in bold · **UNCHANGED** from entry 16, which had it right
4. **Days left, negative once the window shuts** · **CHANGED** — as built it printed 0 nine days after close
5. Server pager · NFR-3 · **NEW** — the as-built list is unpaginated and sorted in Python
6. A meter with no actual reading **cannot be estimated** · **NEW READ** · BR-MET-026 refuses it by name, and saying so before the tick is cheaper than saying it after

History — the tab

SMARTBillHub

Reads › Read cycles

Search | ? | 3

BB
Brazos Bend Utilities

← Read cycles

CYCLE · MONTHLY

··· | ↑ Upload reads

Dashboard
Plans & Rates
Premises
Assets
Accounts
Reads
Billing
Payments
Work
Finance
Runbooks
Reports
Settings
v2.237.10
DW Dana Whitfield System Admin

CY-001

In progress | Window closed 9 days ago

Read window 09/01 – 09/03/2026 · billed 11/05/2026, due 11/25 · 2 routes · 5 meters · 0 of 18 registers

Overview | Routes 2 | Still to read 5 | All reads 13 | History

1 **NOTHING SERVES THIS YET**

This tab needs one endpoint that does not exist. Everything else on this page is served today.

PERIOD ⇅	READ WINDOW	2 REGISTERS COLLECTED	METERS SHORT	3 ESTIMATED	BILL DATE	STATUS
09/2026	09/01 – 09/03	0 of 18 · 0%	5	0	11/05/2026	In progress
08/2026	08/01 – 08/03	18 of 18 · 100%	0	2	10/05/2026	Billed
07/2026	07/01 – 07/03	17 of 18 · 94%	1	1	09/05/2026	Billed
06/2026	06/01 – 06/03	18 of 18 · 100%	0	0	08/05/2026	Billed

Showing 1–4 of 12 — Rows per page 50 ⇤ ← Page 1 of 1 → ⇥

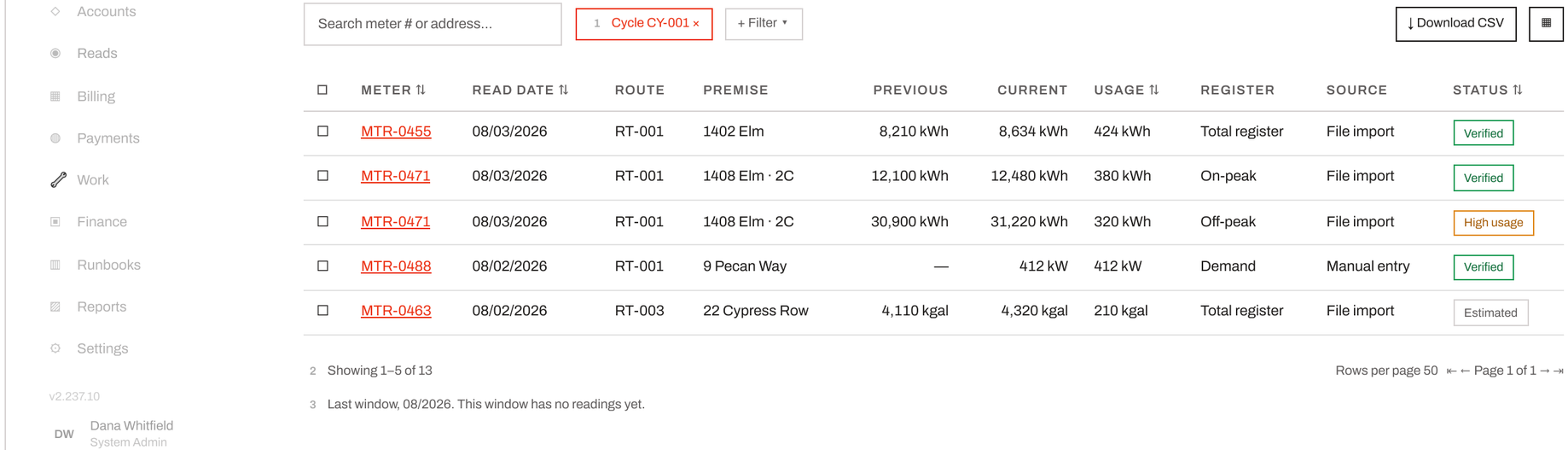
4 A period is dropped once a bill run consumes it.

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — History · 1440 · the one tab that needs building.** Twelve windows of this cycle, one row each. **The Estimated column is why this tab is worth having** — a cycle reading 100% on estimates is not the same as a cycle read by a person, and no other screen in the module puts those two numbers side by side. **Completeness answers one window at a time today**, so this is the only figure on entry 20 with no endpoint behind it; it is drawn in its own tab precisely so it can be dropped from wave one without moving anything else.

1 **The honest absence** · drawn as a warning Panel, not hidden · BR-SHL-025
2 Registers collected, per window · **NOT SERVED** — completeness answers one window at a time
3 **Estimated** · how much of that 100% was a person and how much was arithmetic · **NOT SERVED**, and the reason this tab earns its place
4 A period is dropped once a bill run consumes it · BR-MET-132

All reads — the tab

SMARTBillHub
Reads › Read cycles › CY-001
Search ? 3
BB Brazos Bend Utilities
Dashboard
Plans & Rates
Premises
Assets
← Read cycles
CYCLE · MONTHLY
CY-001
Reading sheet | Route sheet | ··· | ↑ Upload reads
In progress | Window closed 9 days ago
Read window 09/01 – 09/03/2026 · billed 11/05/2026, due 11/25 · 2 routes · 5 meters · 0 of 18 registers
Overview | Routes 2 | Still to read 5 | All reads 13 | History



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **All reads · 1440.** The reads list of entry 02, **pre-filtered to this cycle and nothing else** — same table, same toolbar, same columns, with the cycle chip already in the filter bar and removable. **No new endpoint:** `f.cycle` is one of the six filters `GET /reads` already declares. It answers the question the other four tabs cannot — *"what DID arrive, and does it look right?"* — and the floor note is the one thing it must say out loud, because a tab showing five readings for a cycle that has collected nothing this window would otherwise read as progress.

1 The cycle chip, already applied and removable · **NEW** — `f.cycle` is one of the six filters `GET /reads` declares, so this tab needs **no new endpoint**

2 Server pager · the same one entry 02 uses · **NEW**

3 **Which window these readings belong to** · **NEW READ** — a tab showing five readings for a cycle that has collected nothing this window would otherwise read as progress

What a person can DO to a cycle — the four actions, and why there are only four

Every create, edit and delete of a cycle lives in Settings (`BR-MET-011`). That is not a gap to be filled here: a cycle's frequency, its start day and its window length are configuration, and putting an edit on this page would put the same field in two places. So this screen acts on **the reading work**, never on the cycle's definition.

Action	Tier	What it does	Served today by
↑ Upload reads	Primary	Opens the wizard with this cycle already chosen — the single most common answer to a short cycle	Entry 07, which asks for the cycle on stage 1
Reading sheet	Secondary	The blank sheet to walk with, scoped to this cycle	Entry 18's dialog, whose cycle half was added 2026-08-27
Route sheet	Secondary	The printed walk, scoped to this cycle	Entry 17's dialog, cycle scope
Estimate these	On the shortfall	Estimates the ticked meters — the answer when nobody is going back out	Entry 19's dialog, and stage 3 of the wizard has the same control for the same shortfall
Send someone out	On the shortfall	Raises a visit for the ticked meters	The **work** module — **and this is the one action that crosses a module boundary**, so it is drawn and flagged, not assumed
Manage in Settings →	Escape	Edit the cycle itself — frequency, day, window,	Settings, per BR-MET-011

routes

What is deliberately NOT here: no *Close the window*, no *Reopen*, no *Bill this cycle*. The window is derived from dates and cannot be closed by hand; billing is the billing module's run and putting a trigger here would be a second way to start it.

For	The person who has to answer for a cycle on bill day.
How you get here	A row on entry 15. Also a direct address, because a ticket that says *"CY-001 is short"* needs a link to paste.
Primary action	↑ **Upload reads**, pre-scoped.
Tabs	**Four** — Overview · Routes 2 · Still to read 5 · History, each with its own address (`BR-SHL-011`). Overview carries the figures a person needs *together*; the other three are the work, and each is a list that wants its own toolbar and its own pager. **The counts ride on the strip**, so where the work is can be read without opening anything.
Table controls	**Server pager on both tables.** No search and no filter: both are bounded by the cycle, and a cycle with 2 routes does not need a search box. **At the top of the ICP a cycle can hold thousands of meters**, so the chase table's pager is not decoration.
Write surfaces	**Detail** route → entry 14's content in place; meter → the asset. **Update** Estimate these, Send someone out — both on the shortfall table, both through dialogs that already exist. **Add / edit NOTHING**, by BR-MET-011.
Walked	**No — it does not exist.** Every figure on it was read from an endpoint that answered during the walk, except History (see below).
States	**Loading** band skeleton, then each section. **Empty** a cycle with no routes says so and offers *Add a route in Settings* →. **Failed** per section, so a broken chase list does not take the page down with it. **Domain states:** before the window opens · window open · window closed and short · complete.
Rules	BR-MET-010 · 011 · 049 · 057 · 132 · B14 17, 33, 40 · NFR-3 · NFR-5
Audience	Staff. Drawn desktop.

#	Group / field	Comes from	Health
①	Band · read window, registers collected, still to read, bill date	`GET /cycles/{id}` + `completeness`	SERVED
②	Routes · seven columns	`GET /routes?cycle={id}`	SERVED
③	Still to read · six columns	`GET /cycles/{id}/chase-list`	SERVED · NEEDS A PAGER
④	Schedule · frequency, start day, days, next window	`GET /cycles/{id}`	SERVED
⑤	Billing · bill date, due date, accounts	`GET /cycles/{id}`; account count walks membership	SERVED
⑥	History · last three windows	**NOTHING SERVES THIS** — completeness answers one window at a time	NOT SERVED

Screen health: SERVED — 5 of 6. **History is the one thing on this page that would need building**, and it is drawn in the right-hand column precisely so it can be dropped from wave one without moving anything else.

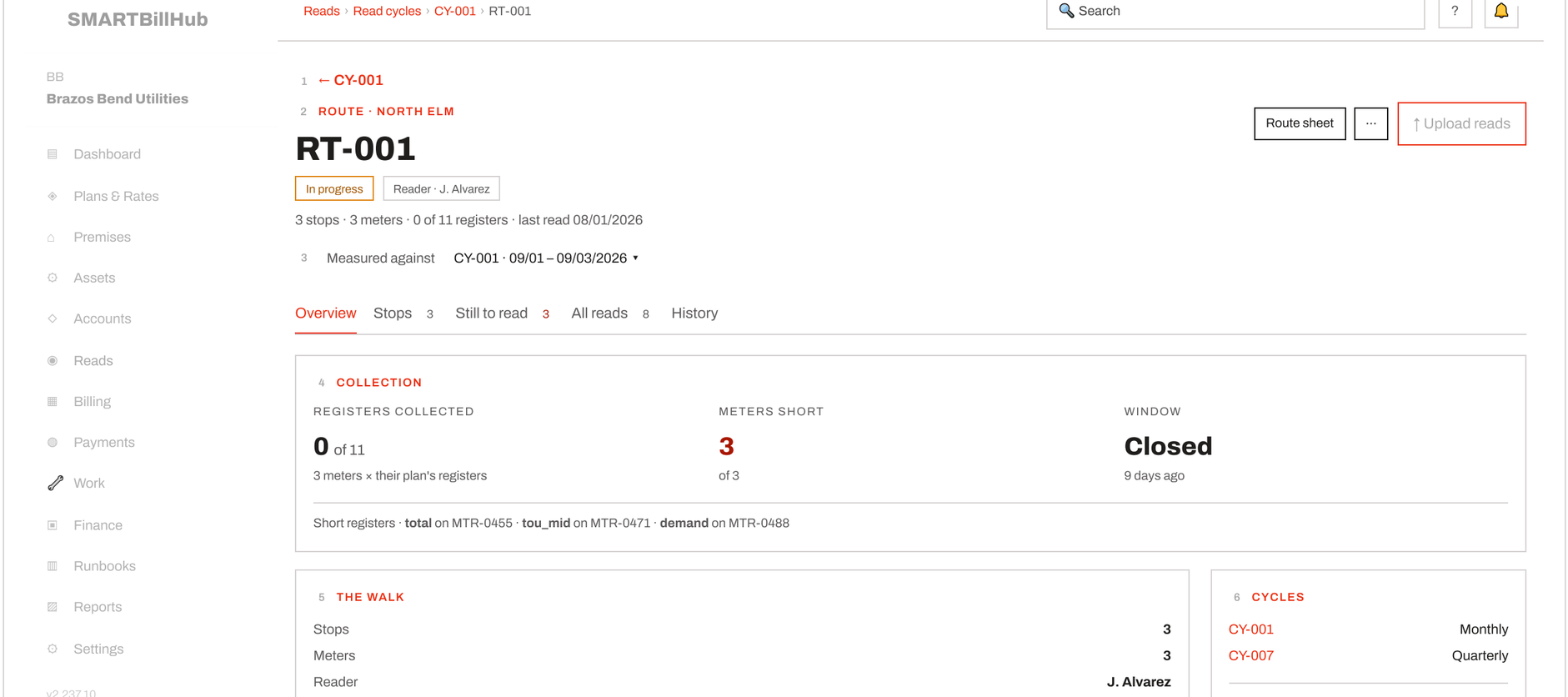
21 **A route — the detail** NEW READ NEW BUILD MR-44 · MR-49 · MR-50 · MR-51

`/:tenant/reads/routes/:routeId?window=2026-09` · opened by a route row on entry 20

The question: *"RT-001 is short. Which stops, in what order, and did the reader actually walk it?"*

> **Why a route needs a page, and why it needs a window with it**
>
> PO ruling 2026-09-13: *"what if User click on Route link — we will have to give a details view of Route. this view should be similar to details view of Cycle."* **Five tabs, the same header, the same Panels** — it answers the cycle's question one level down.
>
> **The one thing a cycle page never needs:** a route can be walked on **more than one cycle**, and this one is on two. *"0 of 11 collected"* has no denominator until a window is named, so the header carries the window this page is measured against — **PO ruling: option A, its own address with a selector**, defaulting to the cycle you arrived from. A ticket can link straight at `?window=2026-09`; the old Routes screen had no address anybody could use and that is how it became an orphan.
>
> **Almost all of it is served.** `GET /routes/{id}` answers `membership` and `premises_order`; `GET /routes/{id}/read-windows` answers the history and is **wrapped and called by nothing**; the reads tab is `f.route`, already declared. **No new endpoint is proposed.**

Overview



Excluded premises	**1**

The stop order is ruled and printed on the sheet. See **Stops**.

DW Dana Whitfield
System Admin

7 This route is walked on two cycles. Every figure above is measured against the one chosen in the header.

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Overview · 1440 · a screen that does not exist today.** The same header, the same three-figure `Panel` and the same two-column body as the cycle page, because it answers the same question one level down. **The one control the cycle page does not need is ④'s companion in the header: the window this route is measured against.** A route can be walked on more than one cycle — this one is on two — so *"0 of 11"* is meaningless until a window is named, and the selector defaults to the cycle you arrived from.

1 Back to the cycle you came from · **NEW**
2 Kicker · `ROUTE · NORTH ELM` · **NEW**
3 **The window this page is measured against** · **NEW** — the one control the cycle page does not need, and the reason this screen has an address of its own
4 **Collection** · the same three figures as the cycle, scoped to the route · **NEW READ**
5 **The walk** · stops, meters, reader, exclusions · **NEW READ** — from `premises_order` and `membership`
6 **Cycles** · every cycle this route is walked on · **NEW READ**
7 Why the selector exists, said once where the two cycles are listed

Stops

SMARTBillHub

Reads › Read cycles › CY-001 › RT-001

Search ? 3

BB
Brazos Bend Utilities

- Dashboard
- Plans & Rates
- Premises
- Assets
- Accounts
- Reads
- Billing
- Payments
- Work
- Finance
- Runbooks
- Reports
- Settings

v2.237.10

← CY-001

ROUTE · NORTH ELM

RT-001

In progress | Reader · J. Alvarez

3 stops · 3 meters · 0 of 11 registers · last read 08/01/2026

Route sheet ··· ↑ Upload reads

Measured against CY-001 · 09/01 – 09/03/2026 ▾

Overview | Stops 3 | Still to read 3 | All reads 8 | History

Search premises...

Route sheet ↓ Download CSV

1 STOP	PREMISES	UNIT	METER	SERVICE	2 REGISTERS EXPECTED	LAST READ	STATUS
1	1402 Elm	—	MTR-0455	E	Total register	08/01/2026	Short
2	1408 Elm	2C	MTR-0471	E	On-peak, Mid-peak, Off-peak, Total	08/01/2026	Short
3	9 Pecan Way	—	MTR-0488	E	Total register, Demand	08/01/2026	Short
—	17 Pecan Way	—	—	—	3 Excluded · vacant since 06/2026		Not walked

Showing 1–3 of 3 stops · 1 excluded

Rows per page 50 ⇤ ← Page 1 of 1 → ⇥

4 THE ORDER IS RULED, AND THIS SCREEN DOES NOT SET IT

The walk order is stored and printed on the route sheet. `PUT /routes/{id}/premises-order` is served and writes it. **Whether a clerk may reorder a walk from Reads, or only from Settings, is not ruled** — `BR-MET-011` puts route configuration in Settings, and `SET-06D` numbers these stops there. Drawn read-only until somebody rules. Manage in Settings →

DW Dana Whitfield
System Admin

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Stops · 1440.** `premises_order` read back — **one row per PREMISES, not per meter**, which is what CR-029 D11 ruled and what the route sheet prints. A premises the route skips keeps its row and says why, because a stop that vanishes from a walk is a stop a reader will look for. **Registers expected** is the plan-driven set again, so a TOU stop reads as four and a demand stop as two before anybody leaves the office.

1. **Stop number** · the ruled walk order, CR-029 D11 · **NEW READ**
2. **Registers expected** · the plan-driven set per stop, so a TOU stop reads as four before anybody leaves · **NEW READ**
3. **An excluded premises keeps its row** · **NEW READ** — a stop that vanishes is a stop a reader will look for
4. **Drawn read-only, and it names the question** · `PUT /routes/{id}/premises-order` is served and writes the order; whether Reads may call it or only Settings is **not ruled**

Still to read

SMARTBillHub

Reads › Read cycles › CY-001 › RT-001

Search ? 3

BB
Brazos Bend Utilities

- Dashboard
- Plans & Rates
- Premises
- Assets
- Accounts
- Reads
- Billing
- Payments
- Work
- Finance
- Runbooks
- Reports
- Settings

v2.237.10

DW Dana Whitfield
System Admin

← CY-001

ROUTE · NORTH ELM

RT-001

In progress | Reader · J. Alvarez

3 stops · 3 meters · 0 of 11 registers · last read 08/01/2026

Measured against CY-001 · 09/01 – 09/03/2026 ▾

Route sheet | ··· | ↑ Upload reads

Overview | Stops 3 | Still to read 3 | All reads 8 | History

Search meter or address... | All 3 | Total register (1) | On/off-peak (1) | Demand (1) | ↓ Download CSV

1 3 selected — Estimate these | Send someone out | Clear

☐	STOP	METER	PREMISES	UNIT	REGISTERS STILL NEEDED	PLAN	DAYS LEFT	LAST ACTUAL
☑	1	MTR-0455	1402 Elm	—	Total register	Tiered	−9	08/01/2026
☑	2	MTR-0471	1408 Elm	2C	**Mid-peak**	Time of use	−9	08/01/2026
☑	3	MTR-0488	9 Pecan Way	—	**Demand**	Demand	−9	08/01/2026

Showing 1–3 of 3 — Rows per page 50 ⇤ ← Page 1 of 1 → ⇥

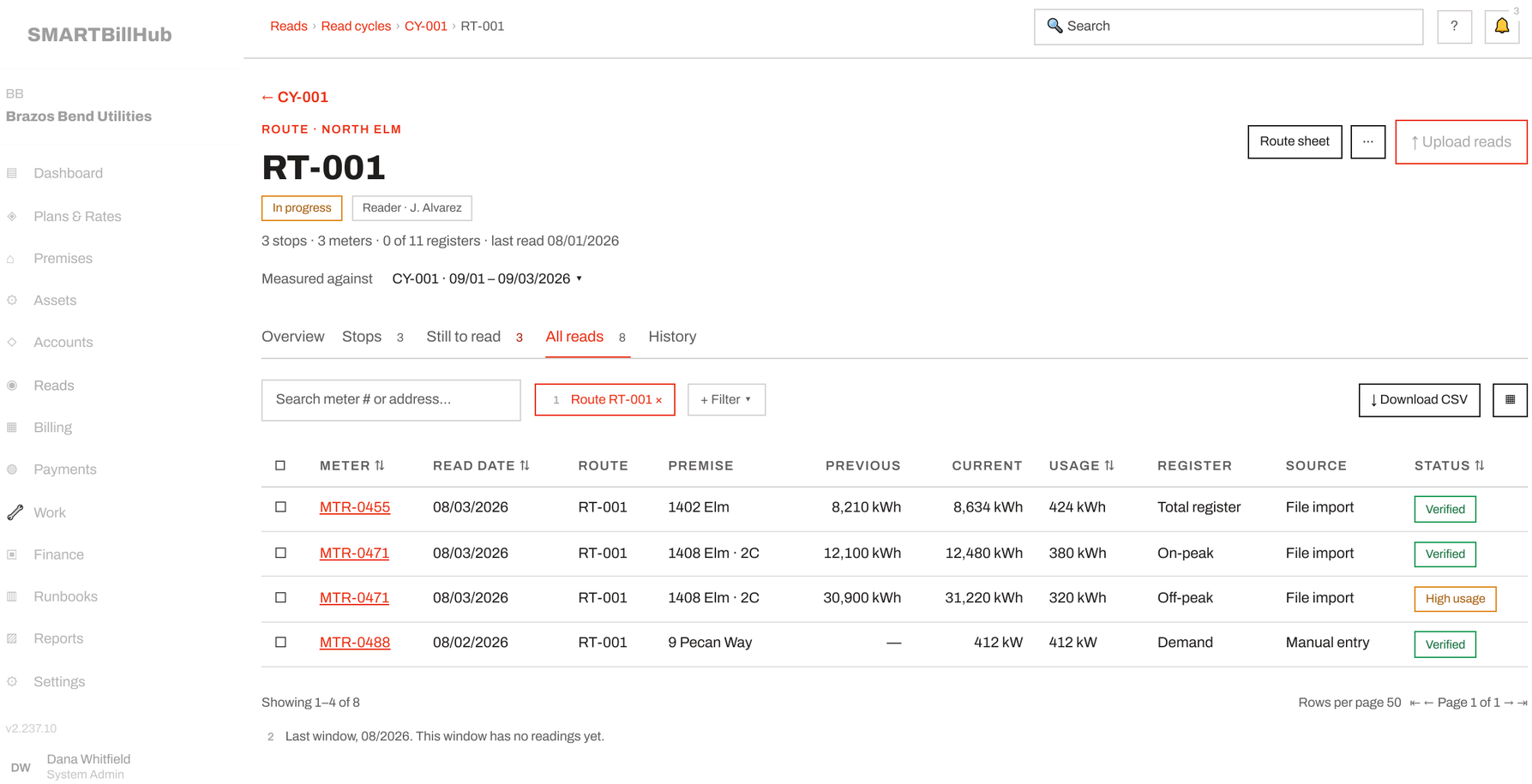
2 A demand register is never estimated. Send someone out.

REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **Still to read · 1440.** The cycle's chase list narrowed to this route, and **carrying the stop number** — which is the one column the cycle-level list cannot have and the only one that tells a reader the order to walk them in. **The floor note is a refusal the product already makes and no screen states:** a demand peak is a maximum,

not an average, so `BR-MET-026` will not estimate one. Ticking all three and pressing *Estimate these* would succeed for two and refuse for one — saying so first is cheaper than saying it after.

1 The bulk bar · `POST /reads/bulk-estimate` · the same control as entry 20
2 **A demand register is never estimated** · **NEW READ** — `BR-MET-026` refuses it, and ticking all three would succeed twice and refuse once

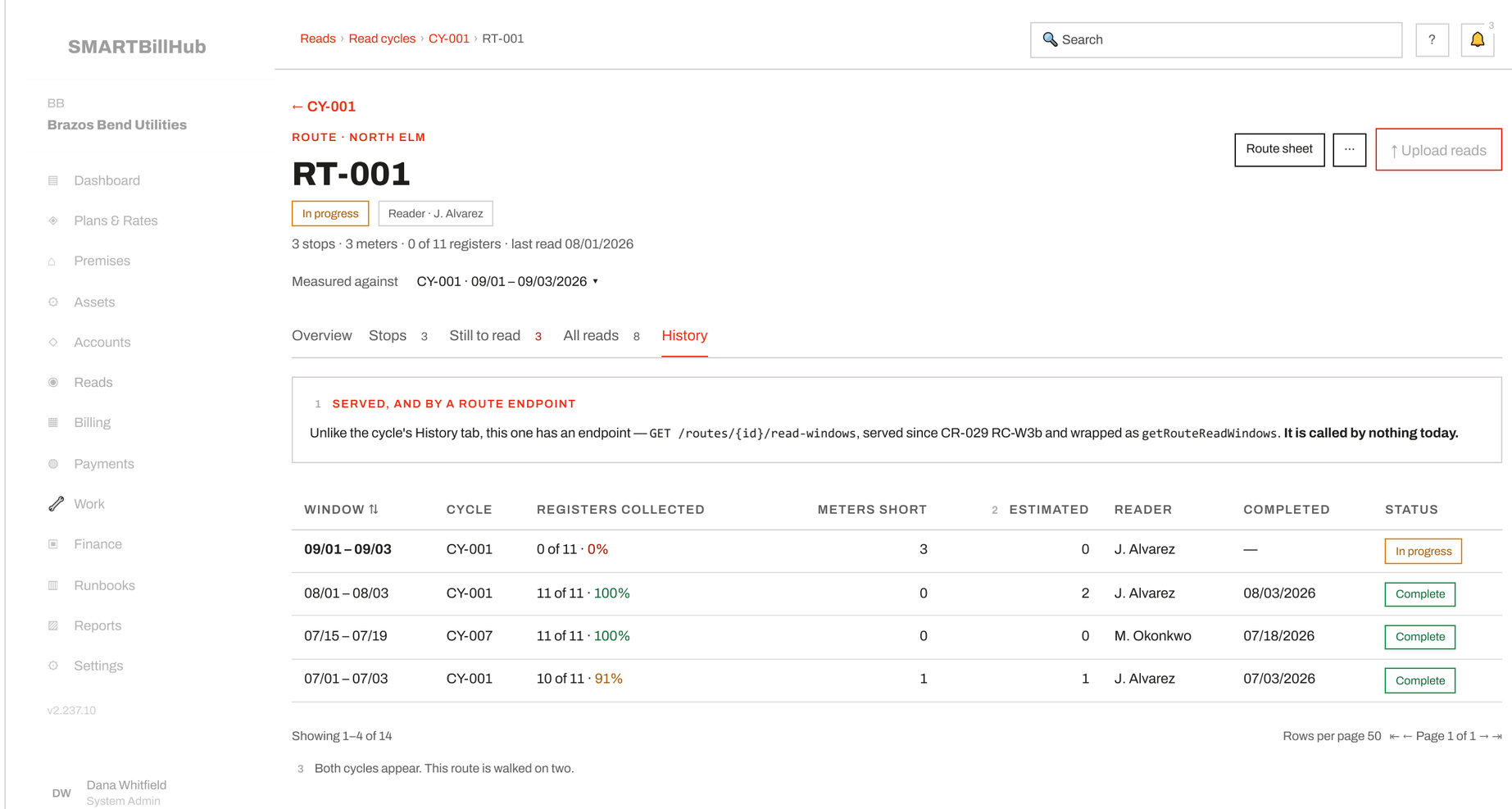
All reads



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **All reads · 1440.** The same table again, filtered by `f.route` — the sixth of the six filters `GET /reads` declares. **Three tabs of this screen say what is missing; this is the only one that says what arrived**, which is where a supervisor checks a reader's work rather than chasing them. The chip is removable, so one click widens it back to the whole module and the clerk keeps their place.

1 The route chip · `f.route`, the sixth declared filter · **NEW**
2 Which window · the same note the cycle's tab carries

History



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT** — **History · 1440. The route's windows across BOTH its cycles**, which is the view that only exists at this level: the cycle page can never show a July quarterly window beside an August monthly one for the same reader. **The Estimated column is the one a supervisor reads first** — 11 of 11 collected with 2 of them estimated is a different month from 11 of 11 walked. **And this tab is served where the cycle's is not**, by an endpoint wrapped two weeks ago and called by nothing since.

1 **Served, and called by nothing** · `getRouteReadWindows` wrapped 2026-08-23 · **NEW** — unlike the cycle's History, which needs building

2 **Estimated** · 11 of 11 with 2 estimated is a different month from 11 of 11 walked · **NEW READ**

3 **Both cycles in one list** · the view that exists only at this level

For	The supervisor who has to answer for one reader's walk.
How you get here	A route row on entry 20's Routes tab, and a pasteable address carrying its window.
Primary action	↑ **Upload reads**, pre-scoped to this route and window.
Tabs	**Five**, each with its own address — Overview · Stops 3 · Still to read 3 · All reads 8 · History.
Table controls	Server pager on all four lists. Search on Stops and All reads; register chips on Still to read.
Write surfaces	**Update** estimate and field visit, on the shortfall. **Detail** a stop's meter, a reading. **Add / edit NOTHING** — and the walk order is drawn read-only pending a ruling.

Walked	**No — it does not exist.** Every figure is read from an endpoint that answers today.
States	**Loading** per panel. **Empty** a route with no stops says *Add a premises in Settings* →. **Failed** per section. **Domain states:** before the window opens · open · closed and short · complete · walked on two cycles.
Rules	BR-MET-010 · 011 · 026 · 049 · 057 · CR-029 D11 · B14 17 · NFR-3
Audience	Staff. Drawn desktop.

Screen health: SERVED — every figure on all five tabs. **This is the only new screen in this document with nothing unserved behind it**, and one of its endpoints has been waiting three weeks for a caller.

---

16 Meters still to read

EXISTS BEHIND · NEW READ · BLOCKED · MR-10 · MR-51

`was a panel behind a table cell on entry 15 — no address of its own; now the` **`Still to read`** `section of entry 20`

The question: *“Name them. Which meters, where, and how long have I got?”*

Revised

Drawn in place, as section ⑥ of entry 20 above — this entry is not drawn twice. What changes, and why, is the table below.

As-built element	Disposition	Why
Meter · Premises · Unit · Registers still needed · Days left	**UNCHANGED**	All five columns survive intact
The panel itself	**MOVED**	Out from behind a table cell, onto the cycle page. It was two levels of unreachable; now it is on the screen the ruling creates
“Meters still to read on CY-002” heading	**REMOVED**	The cycle is the page — the heading was naming the thing the breadcrumb now names
The 17-word standfirst	**REMOVED**	B14 rule 33: rationale is zero words on screen. What it explained is what the column headings already say
Close	**REMOVED**	Nothing to close — it is a section, not an overlay
Meter (as text)	**RELABELLED**	Becomes a link to the asset. A list of meters you cannot open is a list you have to re-find

```
As-built fields    5
  unchanged        5
  removed          0
Remedy fields      9   = 5 + 4 new (tick column, Route, Estimate these, Send someone out)
                       + a server pager, which is a control rather than a field
```

> **The pager is not a nicety here**
>
> The chase list is **unpaginated and sorted in Python** (`completeness.py:894`, `:913`), and the only bound stated anywhere is a code comment: *"Bounded by the cycle's own membership (<3,000 connections)"*. **That ceiling was retired on 2026-09-06** — the ICP now reaches 40,000 accounts — so the comment is not a bound, it is a note about a world that no longer exists. `NFR-3` and `NFR-5` both apply, and the drawing carries the pager rather than leaving it to be noticed later.

For	Whoever has to get three readings before Friday.
How you get here	**Entry 20, in the open.** Today: the Completeness cell on entry 15 — **and entry 15 has no door**, so in practice this panel is two levels of unreachable.
Primary action	**Estimate these** on the ticked rows — which is the gap this closes: as built **there is no "estimate these" or "raise a visit" control here**, though step 3 of the upload wizard has exactly those two for the same shortfall.
Tabs	None.
Table controls	**As built: all four NONE.** No search, no filter, no columns, no sort, no pager. **Revised: a server pager**, and no search — the list is bounded by the cycle.
Write surfaces	**As built NOTHING** — detail, add, edit and update all absent; a meter name is text, not a link. **Revised: detail** (meter → asset) and **update** (estimate, visit) on the ticked rows.
Walked	**No.** The *Chase 5 meters* control was located on entry 15 but the panel did not open under the driver, and the failure was not diagnosed. Graded UNVERIFIED rather than guessed at.
States	**Loading** *"Counting…"*. **Empty** before the window opens the list is genuinely empty and the section says when it opens instead. **Failed** the section carries its own error. **Domain state:** nothing expected in this window.
Words	*Registers still needed* — the plan's requirement, named per meter.
Components	`ChaseListPanel` · `PlainTable`
Rules	BR-MET-049 · BR-MET-057 · BR-MET-132 · NFR-3 · NFR-5
Audience	Staff.

```
the plan-required register set  –  committed reads in this window
  ↳ completeness.chase_list(cycle, period=…)   completeness.py:894   unpaginated
     ↳ rows.sort(days_left, unit, meter)       completeness.py:913   in Python
        ↳ GET /api/{tenant}/cycles/{id}/chase-list   api.py:965
           ↳ <ChaseListPanel/>                        BREAKS AT PAGINATION
```

#	Field	Comes from	Owned by	Health
①	Meter	`assets.asset_no`	metering ⇄ assets	SERVED
②	Premises	walked to `premises`	metering	SERVED
③	Unit	`units.label`	metering	SERVED
④	Registers still needed	the plan-required set minus what arrived	metering ⇄ rating	SERVED

	⑤ Days left	computed from the window this row was asked about	metering	SERVED
	⑥ Route NEW	`routes.route_no` via cycle membership	metering	SERVED

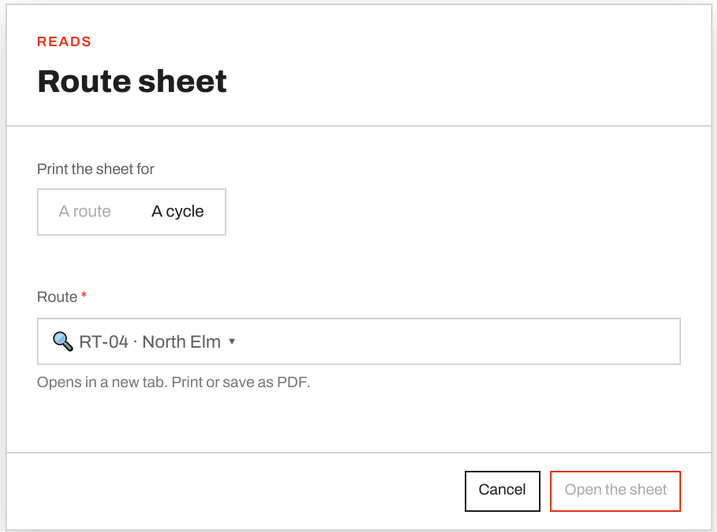
Screen health: SERVED — 6 of 6. Flow BREAKS AT PAGINATION.

17 Route sheet

EXISTS BEHIND · NEW BUILD · MR-06 · MR-31

`a dialog on entry 02; the sheet itself opens at /api/{tenant}/{routes|cycles}/{id}/route-sheet in a new tab`

The question: *"Print me the walk for tomorrow."*



REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — As built, 1440.** The scope is a **segmented choice rather than two buttons**, because it is one question with two answers — *what are you holding?* — and the picker below changes with the answer. Choosing a scope clears the other one's selection, so the schema can never hold a route id and a cycle id at once.

Why the cycle half exists, and what it says about testing

Until 2026-08-27 this dialog offered a single required *Choose a route…* — while the generated sheet **already printed its cycle in the header**. So the cycle was a first-class fact of the artefact with no way to ask for one, and a clerk who works a monthly walk by cycle had to know which routes were on it and print them one at a time.

Nothing caught it because the feature worked perfectly for the case it was built for. Every test asked for a route sheet by route and got one. **A missing entry point is invisible to any test that only exercises the entry point that exists.**

For	A clerk sending someone out with paper.

How you get here	Entry 02's *Route sheet PDF* secondary action (`ReadsListPage.tsx:830`), enabled.
Primary action	**Open the sheet** — `window.open` on a same-origin API path with `noopener`, so the session cookie rides along and the sheet needs no handle back into the app.
Tabs	None — a segmented control, which is a different thing and is the right one.
Table controls	None.
Write surfaces	**NOTHING.** The sheet is a read: *"a clerk PRINTS the sheet; nothing writes."*
Walked	**Yes.** Opened from entry 02. The segment offered *A route* / *A cycle* and the picker listed all six seeded routes by code and name. The sheet itself was not opened (it targets a new tab).
Cognitive load	The help line says what will happen *before* the tab opens — print-ready HTML, and your browser makes the PDF. The label keeps r10's own words ("Route sheet PDF") even though no PDF library is a backend dependency, and the screen reconciles the two rather than pretending.
States	**Loading** the route and cycle lists arrive with the page behind. **Empty** — a tenant with no routes gets an empty picker, which is the one state this dialog cannot distinguish from "still loading". **Failed** the required-field refusal. **Domain state:** cycle scope, which changes the picker and its help line.
Words	*A route* / *A cycle* — articles, because the segment answers "what are you holding?".
Components	`FormModal` · `SegmentedField` · `SelectField`
Rules	BR-MET-162 (a sheet is printed for a route *or* a cycle) · DEF-532's tenantPath discipline · B14 8, 9, 10
Audience	Staff.

```
GET /routes · GET /cycles          → the two pickers    unpaginated
  ↳ window.open("/api/{tenant}/{routes|cycles}/{id}/route-sheet")   api.py:788 / :814
     ↳ route_sheet.render_sheet / render_cycle_sheet
        ↳ print-ready HTML (text/html), the WHOLE sheet, built from the D11 walk order
                                                    WHOLE — and unpaginated by nature
```

#	Field	Comes from	Owned by	Health
①	Scope	*a two-value segment*	—	SERVED
②	Route — value	*chosen*	metering	SERVED
②b	Route — **options**	`GET /routes`, every page, fetched whole	metering	SERVED, UNBOUNDED
③	Cycle — options	`GET /cycles`, every page, fetched whole	metering	SERVED, UNBOUNDED
④	The sheet	`route_sheet.py` — premises in walk order, each meter's previous reading pre-filled	metering ⇄ assets	SERVED

Screen health: SERVED, two option lists unbounded — 5 fields, 3 SERVED, 2 unbounded. 0 ungraded. Flow WHOLE.

“Can it handle every kind of register?” — yes, and it is the best-handled screen in the module

The sheet does not print one line per meter. It asks the completeness resolver what that meter's plan expects and prints **one line per expected register** (`reading_sheet.py:349`, `for channel in expected.channels`):

Plan	Lines printed for one meter	Unit on each
Flat · tiered	1 — *Total register*	kWh
Peak / off-peak	3 — *On-peak, Off-peak, Total register*	kWh
Time of use	N+1 — the plan's **own** windows, plus total	kWh
Demand	2 — *Total register* and *Demand*	**kWh and kW** — per register, `BR-MET-143`
Net-metered	+1 — *Export*	kWh
Unresolvable	1, with **no register named** — it prints the absence rather than guessing `total`	—

Two details worth copying elsewhere. The unit is taken per register (`register_units.register_unit`), so a demand line says **kW** and the energy line beside it says **kWh** — the distinction whose absence cost `DEF-797` $4,378.00. And each line carries a `row_key` of `<cycle>|<channel>|<meter>`, so **a filled-in sheet types back with its register already identified** — which is the one real defence in this module against the wrong-register trap at entry 04, because the clerk never has to choose the column again.

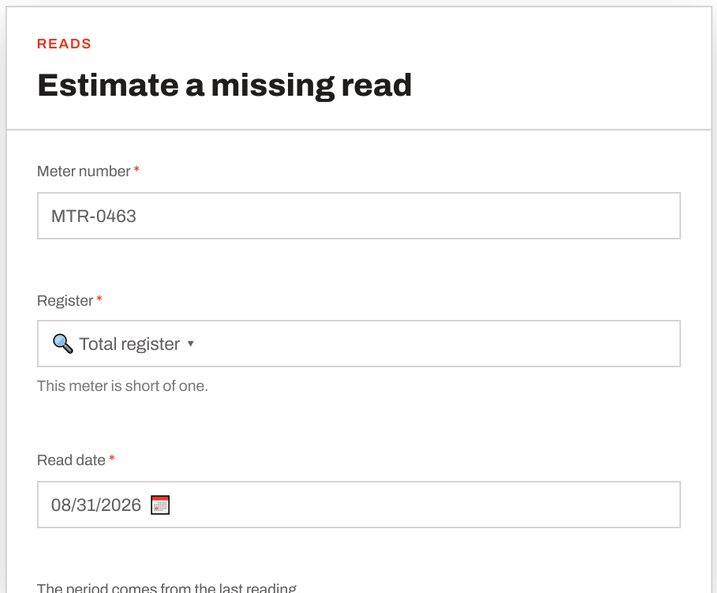
---

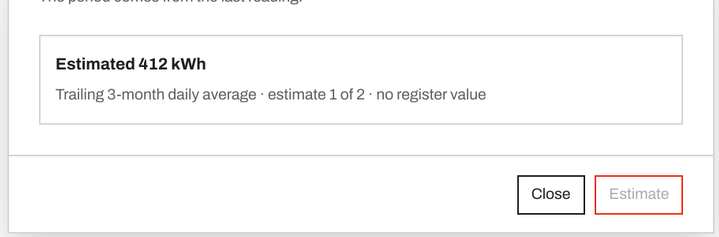
18 Estimate a missing read

EXISTS BEHIND · NEW BUILD · BLOCKED · MR-06 · MR-14 · MR-31 · MR-42

`a dialog on entry 02 — it had its own route, /reads/estimate, and the PO took it away`

The question: *“The meter could not be read and the bill run is Friday. Put a lawful number in.”*





REDESIGN · published 2026-09-12/13, drawn to the PO rulings of those days · **NEVER BUILT — 1440, trimmed and re-scoped. The answer is rendered as a child of the form**, so the clerk reads it over the inputs that produced it — and a *refusal* is rendered the same way, as an answer rather than as an error: *"No estimate was made — the earlier readings on this meter cover no days between them, so a daily average cannot be calculated."* The absent *days* input has a sentence explaining why it is absent, which is one of the few places in this module where rationale is deliberately on screen. **Two changes.** The 25-word lede is gone — rule 33 caps rationale at zero on screen, and the sentence explaining that the period cannot be nudged was addressed to a reader who is not the clerk. And the **Register list offers this meter's MISSING set**, not the flat seven: as built it lists every register in the product, including ones the meter does not have.

For	A clerk with an unreadable meter and a bill run coming.
How you get here	Entry 02's *Estimate a missing read* secondary action, enabled. **The page it used to be is gone** — the PO, on the live app at v2.61.1: *"this can be modal - instead of page"*, and its three siblings on Reads were already dialogs.
Primary action	**Estimate** — `POST /reads/estimate`, which creates a `meter_reads` row and an audit row.
Tabs	None.
Table controls	None.
Write surfaces	**Add** — the whole dialog. **Update / delete** — NOTHING. An estimate has **no register value** and is a lawful bill input that travels flagged (BR-MET-028/029); the cap is two consecutive.
Walked	**Yes.** Opened from entry 02. Its lede reads *"When a meter could not be read, this works out the consumption from that meter's own recent actual readings. It never guesses a rate and it never rounds."* — 25 words of rationale on screen, which B14 rule 33 caps at zero. The Register select offered five options. Not submitted.
Cognitive load	It removes the arithmetic and refuses to let anyone nudge it.
States	**Loading** working on submit. **Empty** n/a. **Failed** a **refusal-as-data** — a 200 carrying a named reason, of which there are seven, each with its own sentence. **Domain state:** *no history — field visit*, which is the honest answer rather than a number.
Words	*Estimate* is used for the act and *est.* for the badge; a demand peak is **never** estimated and the product says so where that is asked.
Components	`FormModal` · `TextField` · `SelectField` · `DatePickerField` · `EstimateOutcome`
Rules	BR-MET-027 (two consecutive) · 028 / 029 · B14 10, 11, 32
Audience	Staff.

```
the form → POST /api/{tenant}/reads/estimate      api.py:1553
  └→ estimation.estimate()   reads read_validation_settings.est_method  (never a constant)
      ├→ ok        → meter_reads.create + audit_log         services.py:9553
      └→ refused   → 200 with a NAMED reason, rendered as the answer    WHOLE
```

#	Field	Comes from	Owned by	Health

#	Field	Comes from	Owned by	Health
①	Meter number	*typed*	metering ⇄ assets	SERVED
②	Register — options	`READ_CHANNELS`, *a literal array*	—	HARDCODED
③	Read date	*picked*	metering	SERVED
④	The estimate, with its unit	`EstimateFigure` over the server's answer; a figure with no unit is **not shown** and says why	metering	SERVED
⑤	Method · consecutive number	`estimation_method`, `consecutive_estimate_no`	metering	SERVED
⑥	Refusal reason	one of seven named reasons from the engine	metering	SERVED

Screen health: HARDCODED on ② — 6 fields, 5 SERVED, 1 hardcoded. 0 ungraded. Flow WHOLE.

“All registers? And in bulk?” — one at a time, and the bulk endpoint is built, served, and has no door

Registers: the dialog takes exactly one. `estimateRead({meter_no, channel, read_at})` — **one (meter, register, date) per open**. A TOU meter short of three windows is three trips through this dialog, and the register list it offers is the flat seven, not that meter's missing set, so the clerk picks from a list that includes registers the meter does not have.

Bulk: `POST /reads/bulk-estimate` **exists and works** (`api.py:2137` → `services.bulk_estimate_reads:9840`). It takes a set of targets and one reason, runs **every target through the same single-row pipeline** — so `BR-MET-026`'s trailing-3-cycle average, `BR-MET-027`'s two-consecutive-estimates limit and `BR-MET-058`'s per-channel history all apply unchanged (*“a bulk button is a different surface, never a different pipeline”*) — and a target the rules refuse **comes back with its reason** rather than being silently skipped. One audit row each.

Nothing in the front end calls it. This is the module's third *served-with-no-caller* (with the billed correction and the two doorless screens), and this one **says why, in the code, and names its home**: *“bulk ESTIMATE is served by the backend and has NO control on this screen yet: the missing set it operates on is the DERIVED `f.status=pending` list, whose rows carry `id: null` and are a different selection model from the stored rows this bar owns. Building one selection that means two different things is how a clerk estimates the wrong meter — REPORTED and left to W1's completeness surface, which owns that list.”*

That surface is entry 20. The chase list *is* the derived pending list, it is the only place in the product where those rows are selectable, and the **Estimate these** control drawn on its Still-to-read table is therefore not a new idea — **it is the delivery of a deferral the code itself wrote down.** The selection-model objection does not apply there, because on that screen there is only one kind of row.

So R2 is now two things, and the smaller one is the bigger win: put the bulk control on entry 20 (drawn), and default this dialog's register list to **that meter's missing set** rather than all seven. Neither needs a new endpoint.

---

19 The retired drawer address

EXISTS BEHIND NO COMMITMENT

`/:tenant/reads?read=<id>  →  forwards to entry 03`

The question: *“I have a ticket from August with this link in it.”*

Not a screen — an address. It is listed as its own entry so that nobody later finds a URL nothing in this document describes. S29 used to be a drawer opened by this parameter, and

the linkability was the parameter's whole point, so those links live in tickets. `ReadsListPage.tsx:697-702` forwards them with `navigate(..., {replace: true})` rather than silently opening nothing.

How you get here: from outside the product — a ticket, an email, a bookmark. Nothing inside the product writes this shape any more, which is correct.

Counts: 20 entries · **17 drawn in full**, in 21 drawings · three are undrawn and each says why in its own entry: **01** (retired by ruling), **11** (the walk never rendered the receipt) and **this one** (a second address).

05

What broke when the product was driven

One row per failure **seen in the running app** — not per journey and not per screen. This is the only section whose rows cannot be re-derived from the source, which is exactly why an invented one would be undetectable. Everything here was produced by a browser against the deployed join on **2026-09-12**.

> **The stack that produced these rows**
>
> PostgreSQL 16 provisioned by the repository's own `10-scripts/13-provision-database.sh` — **338 of 338 migrations applied, one per process, in dependency order**, with the app role verified `rolsuper=f rolbypassrls=f` and **RLS ENABLE+FORCE on 281 of 282 tenant-scoped tables**. Then `seed_demo` (tenant `brazosbend`), `vite build`, and `07-frontend/server.mjs` serving `dist` on :8080 with `/api` passing through to Django on :8000 — **the deployed join, not the dev server**, because the vite dev server has no `/api` proxy and cannot serve this app same-origin. Chromium at 1440×1000, signed in as the seeded owner.
>
> **Journeys walked: 3 of 3** processes in this module — capture & exceptions, getting the readings in, planning and chasing the walk. **Not walked:** the repository's own `J2-read-to-cash`, `J3-exception-handling` and `J11-bulk-correction` end to end, because each leaves this module at its third or fourth step and this audit is scoped to Meter Reads. **One of nineteen entries was never rendered** — entry 11, the upload receipt.

#	Journey · step	Screen	Severity
J-01	Capture & exceptions · 3 of 5 — open a reading	03 · `/reads/:readId`	Blocks the journey

DID Clicked the meter `M-7210` in the first row of *All reads*. EXPECTED The reading detail: the four figures, the math line, the flag reason with its thresholds, the compliance stamp. HAPPENED **The page is replaced by React Router's error boundary.** `Unexpected Application Error! t.history.map is not a function · TypeError: t.history.map is not a function.` Reproduced on a second reading. Console carries the same error twice, once raw and once from the router. **`getAssetReads()` is typed `AssetReadRow[]` and the endpoint answers `{count,next,previous,results}`** — measured directly: `GET /assets/{id}/reads → 200 {"count":2,…}`.

J-02	Plan & chase the walk · 1 of 4 — get to the screen	14 · `/reads/routes` 15 · `/reads/cycles`	Blocks the journey

DID Signed in, opened Reads, and enumerated every `<a href>` the screen renders. EXPECTED A link to Routes and a link to Cycles — they are declared as children of the Reads nav item. HAPPENED **Fourteen links on the page and neither is there.** The full set is the thirteen sidebar destinations plus one external docs link. Both screens open perfectly **when the URL is typed**, and by no other means. **The journey cannot be started by a user.**

J-03	Getting the readings in · 4 of 5 — post the batch	10 · stage 4 · Load	The journey completes, with a

completes, with a wasted commit

DID Staged a real 2-row CSV, ran the rules, reached *Load*, and pressed **Commit 1 rows ✓** without ticking the certification. EXPECTED Either the commit is disabled with its reason — this product's posture everywhere else — or it posts. HAPPENED **The button was enabled, the press was accepted, and the answer came back afterwards:** *"Tick the certification before posting the batch."* Nothing posted. The click is spent and the clerk learns the rule by breaking it.

J-04 Plan & chase · 2 of 4 — read the route progress — 14 · Routes — The journey completes, wrongly

DID Read the table. EXPECTED Two comparable numbers: how many meters, how many collected. HAPPENED **RT-001 reads *Meters 3* and *Collected 8*; RT-003 reads *2* and *12*.** Collected counts *readings* and Meters counts *meters*, and they sit in adjacent columns with no label saying so. This is the shape a defect already renamed once on the Cycles screen; the Routes screen kept it.

J-05 Plan & chase · 2 of 4 — when was this route last walked? — 14 · Routes — The journey completes, wrongly

DID Read the *Last completion* column. EXPECTED A date: `08/01/2026`. HAPPENED **`07/31/2026 19:00`** — a date column rendering a clock, and the clock has moved the date **back one day**. That instant is 08/01 00:00 UTC shown in `America/Chicago`. It is the same defect that was fixed on the reads list's own Read date column, still live here.

J-06 Plan & chase · 3 of 4 — can I bill CY-001? — 15 · Cycles — The journey completes, wrongly

DID Read the CY-001 row. EXPECTED One coherent answer. HAPPENED **Four figures that do not agree, on one row:** *Meters 5 · 0 of 18 · 0% · Chase 5 meters*. Three different denominators for one cycle — meters, plan-required registers, and meters again — and the completeness quotient is over registers while the column beside it counts meters. It also reads ***0 days left*** **on 09/12 for a window that closed on 09/03**, which reads as "just in time" rather than "nine days late".

J-07 Getting the readings in · 3 and 4 of 5 — 09 · Validate, 10 · Load — Cosmetic

DID Read the split statement and the commit button on a one-row post. EXPECTED English. HAPPENED ***"1 rows will post. 1 will not"*** and ***"Commit 1 rows ✓"***. An interpolated count with no singular form, on the button a clerk presses to write to a financial record.

J-08 Getting the readings in · 4 of 5 — the compliance stamp — 10 · Load — Cosmetic here; it is evidence elsewhere

DID Read the stamp on a real staged batch. EXPECTED Six facts, all present: this is the evidence the stamp exists for. HAPPENED Five are there and correct — batch `RB-2609-0006`, source, the full SHA-256, *Dana Whitfield* as a name rather than a key, and `09/12/2026 08:31` correctly converted from 13:31 UTC. ***Session · IP*** **came back – · –.** The row is drawn either way, which is right; the evidence is simply absent.

J-09 Plan & chase · 4 of 4 — name the meters I am short — 16 · Meters still to read — Unresolved

DID Pressed *Chase 5 meters* on the CY-001 row. EXPECTED The panel listing five meters with their premises and days left. HAPPENED **The panel did not open under the driver, and I did not diagnose why.** It may be the driver's selector rather than the product. Recorded as unresolved rather than filed as a defect; entry 16 is graded UNVERIFIED for it.

J-10 Getting the readings in · 5 of 5 — the receipt — 11 · Result — Not walked

DID Attempted to tick the certification and commit. EXPECTED The receipt: posted count, held rows, the rejects file. HAPPENED **Never rendered.** The certification control was not the page's last checkbox and the driver's tick missed it, so the commit kept refusing. The stage exists in the rail and was named; the screen behind it was not seen. **Entry 11 is therefore the one entry in this document with no wireframe**, deliberately.

And four things that were checked and were right

A register of failures alone is a misleading picture, so: **the worked arithmetic reconciles** — row 1 shows previous 44,000 kWh, current 46,754 kWh and usage 27,540 kWh, which is `(46,754 − 44,000) × 10` on a ×10 meter, and the demand row's 412 kW is `41.2 × 10` the same way. **Every quantity on every screen carried its unit**; nothing printed a bare number and nothing printed `1008.000000`. **The page title measured 32px at weight 800 in Archivo** — B14 rule 40, ratified the same day, is live in the build. And **`scrollWidth === clientWidth` at 1440** on the reads list: no horizontal overflow.

06

Every stated absence, and its one decision

These are not an auditor's impressions of what is missing. Every row is a string the product itself shows a person when it cannot do something, read out of the eleven locale catalogues and the components that render them. That is a far better source than an opinion: it is the product testifying against itself.

READS KEYS, ENGLISH CATALOGUE	OF THOSE, REFUSAL OR ABSENCE	HOLDING A CONTROL THE USER CAN SEE AND CANNOT USE	DEAD — RENDERED BY NOTHING	EXPIRED HOLDS
741	181	28	3	2

The contrast worth stating: the wireframes in `05-prototype/r10` state **zero** absences. The product's own copy states **181** — 103 server-raised refusals and 78 front-end holds — and **28 of them sit on a control a person can see and cannot use**. Roughly one reads string in four is the product explaining why it will not do something.

⚠ The rule was applied without exception, and three times to the wrong kind of evidence

PO instruction 2026-09-12: *"check section 06 absences for missed commitments"*. Thirteen rows read **OUT OF SCOPE**. **Six of the thirteen cited no ruling at all** — they cited a code comment, a design note, or an open question, and the rule below admits **only a ruling**.

Three were genuinely misfiled and are now specifications:

Was out of scope on	Why that is not a ruling	Now
"a deliberate split — the counter case is one meter, one date, one value"	The code's own design note. Nobody ruled that a clerk at a demand meter should be sent into a five-stage wizard for two numbers.	MR-25
"a refusal to guess … will not guess them from a join made in the browser"	**The reason does not even apply.** `meter_reads.account_id` is a stored foreign key — serving it is not a guess and needs no join.	MR-39
"open questions, both registered"	**An open question is the ABSENCE of a ruling**, which is the exact condition the rule says makes something a specification.	MR-42

The other three stand, with sharper reasons. One is **not absent at all** — Override serves it. One is **correctly absent**, and its on-screen explanation goes anyway under rule 33. One is **another module's to specify** — rating's credit math — which is a different thing from ruled-to-wait and now says so.

What this is really about. A comment in the code and a ruling in the decision log read the same on the page and carry completely different weight: one is an engineer's note, the other is the Product Owner having decided. **Treating the first as the second quietly converts our design choices into the customer's requirements** — which is the same fault CLAUDE.md's change process guards with *"when it is ambiguous it is a DEFECT, not a change"*. Three rows in this table had made that conversion.

change process guards with *"when it is ambiguous it is a DEFECT, not a change"*. Three rows in this table had made that conversion.

The judgement rule, stated once and applied without exception

A **recorded ruling** that a thing waits makes it **out of scope**, with the ruling quoted. **No ruling** makes it a **specification** that names what it waits on.

The borderline case, and how the rule decided it. The Missing chip on entry 01 does not drill in, while Flagged and Disputed both do. There is no ruling that it should not — so by the rule it is a specification. But the *reason* it cannot is a fact, not a preference: `READ_FILTERS` declares no `missing` filter, and an undeclared `f.` param is *ignored*, so the button would have returned the whole list while looking like a slice. So it is a specification (**MR-03**) naming the server work it waits on — not out of scope, and not a design decision.

The two expired holds — a sentence that stopped being true and nothing noticed

⛔ **1 · The primary button on a billed reading refuses over a door that is built**

On screen: *"This reading is already billed. Its correction has to travel with the money fix, **and that path is not built**."* (`metering.read.action.billedLocked`, rendered at `ReadDetailPage.tsx:189` through `overrideBlockedReason`.)

What is actually built: `POST /reads/{id}/billed-correction` — handler `api.py:1479`, routed `api.py:3058`, whose own docstring calls it *"the ONLY door a BILLED read's correction has"*. The money screen exists too: S30's Fix-a-bill tab, routed at `App.tsx:952`. The server even contradicts the button in its own copy — *"This reading is billed. Correct it through the billing correction path instead."*

And the front end never calls it. There is no occurrence of `billed-correction` in any `.ts` or `.tsx` outside comments.

So the control renames itself for a door it will not open. On a billed reading the primary button's label changes to *Correct billed read* — and `overrideBlockedReason` returns the *not built* sentence for that same state, unconditionally. The button announces the capability and refuses it in the same breath, with a sentence that is false. **This is hard-won law 2's mirror exactly: a live capability behind a dead control**, and the mechanism is the one CLAUDE.md names — a rule with a clock in it, correct the day it was written, rotted on the day the endpoint landed, because landing the endpoint and releasing the control that pointed at it were two acts and only the first felt like the work.

⛔ **2 · Two screens say the reads module is not wired. It shipped a month ago.**

On screen (a tooltip on the *Not collected yet* cell, `RoutesCyclesPage.tsx:528`): *"Collected counts come from committed reads. **The reads module is not wired yet**, so this is unknown rather than zero."*

It is wired. `seams.collected_read_count` returns `None` only when the implementation cannot be resolved, and the implementation lives in the *same app* (`services.py:10243`) — so it always resolves and `collected` is never null. The reads screens have existed since August. **The branch is unreachable and its stated reason is false**, and the walk confirms it: every route on screen printed a number.

The register

28 rows that hold a control, and the rulings that decide them. The 50 prose absences and the 103 server refusals are counted above and summarised below rather than listed one by one — the point of the register is the decision, and the decision is the same for a family.

What is absent	Screens	Decision	Reason, and who ruled it
Correcting a billed reading from the reading *"that path is not built"*	03	**MR-15**	**Specification.** The endpoint and the money

— *"that path is not built"*			screen both exist, nobody ruled that the control stays shut. The sentence is false today.
"The reads module is not wired yet" on the Collected cell	14	**MR-16**	**Specification.** An expired hold. The branch cannot be reached and the sentence cannot be true.
Estimate, on a reading that exists — *"estimating here would append a second one"*	03	OUT OF SCOPE	**Not absent — served elsewhere.** `POST /reads/estimate` fills a *missing* record and has no guard against a second head for a date that already has one. The right control for an existing reading is Override, which is the button beside it.
Clearing a dispute, for a Billing Clerk	03 · 06	OUT OF SCOPE	**A ruled permission.** Gated at the route *and* in the service; the reason names the role and what clearing does.
Bulk verbs other than Accept — *"Accept only — a correction is done one reading at a time."*	13	OUT OF SCOPE	**BR-MET-023.** Every other verb needs a per-row value or a per-row reason; the screen says so rather than leaving a clerk hunting for Override.
Bulk estimate — served at `POST /reads/bulk-estimate` and on no screen but stage 3	13	**MR-10**	**Specification.** The bar's own trailer records it: the missing set it operates on is the *derived* pending list, whose rows carry `id: null` and are a different selection model. Left to the completeness surface, which does not have it either.
A second register on the quick-entry form — *"This meter has 3 registers. Use Upload reads to enter all of them."*	12	**MR-25**	**RE-DECIDED 2026-09-12.** It was out of scope on the strength of *"a deliberate split — the counter case is one meter, one date, one value"*. **That is the code's own design note, not a recorded ruling**, and the judgement rule above admits only a ruling. It is a specification, and the modal already holds what it needs: it has fetched `channels` by name before it decides to hand off.
AMI replay as a transport	07	OUT OF SCOPE	**BR-MET-123 clause 3, quoted:** *"a transport card with nothing behind it is not offered"*. The server's refusal keeps a name on this side and nothing renders it.
Two estimation methods — *"Not available yet: a rate plan holds no average for us to read."* and *"nothing here says which accounts count as similar."*	Settings · S64	**MR-42**	**RE-DECIDED 2026-09-12. Two open questions are not a ruling** — they are the absence of one, which by the rule above makes this a specification that names what it waits on rather than a thing ruled to wait. **Blocked**, and the options stay disabled with their reason

			meanwhile.
Turning off an integrity check — *"These protect the data itself."* · *"Always runs"*	Settings · S64	OUT OF SCOPE	**BR-MET-136 clause 4.** Refused 422 with the rule named.
The days input on the estimate form	18	OUT OF SCOPE	**Deliberate, and said:** *"You do not enter it, so the calculation cannot be nudged."*
Co-sign holding the post — *"it does not hold the post"*	10	**MR-17**	**Specification.** Half of OQ-042 is built — the writer — and the blocking half needs a tenant toggle nobody has ruled on.
The account on a reading — *"open it through the unit"*	03	**MR-39**	**RE-DECIDED 2026-09-12.** It was out of scope as *"a refusal to guess"* — *"this screen will not guess them from a join made in the browser"*. **Not a ruling, and the reason does not apply:** `meter_reads.account_id` is a stored foreign key in the reviewed cross-module set, so serving it is not a guess and needs no browser join. A column the database holds and the payload declines to send is a specification.
The net-metering credit — *"the credit calculation is not built, so nothing here is subtracted from anything."*	03	OUT OF SCOPE	**A module boundary with an open question on it.** Subtracting export from import is rating's credit math; nothing in rating or billing computes it.
A drill-in on the Missing chip	01	**MR-03**	**Specification** — see the borderline case above.
A batch *number* on a reading — *"the batch number is not served with a reading"*	02	**MR-04**	**Specification.** The consequence is a raw uuid in a column, which B14 rule 31 forbids; the fix is a served `batch_no`.
The other 12 control-holding refusals — role gates, superseded-version guards, already-disputed guards, unresolvable registers, unavailable read-method and service lists	03 · 07 · 12 · 13 · 14 · 15	OUT OF SCOPE	**Each is a true statement about state**, rendered per BR-SHL-025 as a disabled control *with* its reason rather than a missing one. They are the rule working, not gaps.
The 50 prose absences — stated gaps carrying no control (no prior register pair, no threshold stored, nothing expected in this window, no rows in this view, no peak recorded, seven named estimate refusals, seven named "rule could not run" reasons)	all	OUT OF SCOPE	**Each names a fact about the data rather than a missing feature**, and several are the product being scrupulous: *"A demand peak is never estimated. A missing one needs a field visit."*
The 103 server-raised refusals	all	OUT OF SCOPE	**Business rules refusing, which is what they are for.** All reach the screen through one renderer, so none can be silent.
Three dead strings — `metering.read.device_unit_required` ·	none	**MR-18**	**Specification, and a small one.** Present in all eleven catalogues, rendered by no source file.

`metering.read.generation_requires_device`
· `metering.reads.deviceReadNote`

Already parked in the i18n gap register. A fourth, `metering.read_batch.errors_need_resolution`, is dead in substance — its raise was deleted and only comments quote it.

Nothing is left unresolved. Every absence above is a specification or out of scope: **seven specifications** — MR-03, MR-04, MR-10, MR-15, MR-16, MR-17 and MR-18 — and thirteen rows out of scope, each with its reason.

07

Where the screens depart from the rules

One row per violation, never per screen. The authority is **B14** (`04-docs/02-decisions/14-ui-conventions.md`), with `05-prototype/r10-2026-08-16/UI-RULES.md` as its source and the build file as how the two are obeyed in this tree. A violation is only a violation against a stated authority, so every row quotes one.

> **Every row here is closed by a commitment, and now says which**
>
> **All eighteen.** Each finding's prose row ends with **Closed by** and the commitment that answers it, linked to its register row. **Until 2026-09-12 this section cited no commitment at all** — the violations were on one page, the undertakings on another, and nothing joined them, so a reader could not tell a violation somebody had signed up to fix from one nobody had. Four of the eighteen are closed by commitments minted *because* of that check: MR-33 to MR-36, the four that change no screen.

> **Read the last column before the others**
>
> **3 of the 18 rows below could ever have been caught by a machine.** Most of B14 is RATIFIED and UNENFORCED, because no gate in this repository reads rendered output — so a green suite over this module is evidence of nothing at all, and 15 of these rows were found by reading the source or by clicking. That column is the point of the table.

#	Screen	Where	Enforced?
7-01	02	`ReadsListPage.tsx`, the `batch` cell	RATIFIED G28 reads strings, not rendered values; the two render-checks that do cover S30 and S15 only

RULE **B14 31** — *"Nothing internal is ever visible to a utility user — not a primary key... Where the reference is genuinely what a user needs it has a HUMAN-FACING number and that is what is shown."* THE SCREEN The **Batch** column renders `import_batch_id`, a raw uuid. The code says so itself: *"the id is what renders — reported rather than dressed up as a number."* Hidden by default, so the blast radius is small; the breach is not. CLOSED BY **MR-04**

7-02	10	stage 4, `UploadReadsWizard.tsx` post panel	RATIFIED

RULE **BR-SHL-025** — a control this role or release cannot offer renders **disabled with its reason**, never live onto nothing. **B14 11** — a consequence is shown *before* the commit. THE SCREEN **Commit 1 rows** ✓ is enabled while the certification is unticked. The press is accepted and the refusal arrives afterwards. **Walked.** CLOSED BY **MR-09**

7-03 14 · 15 `RoutesCyclesPage.tsx` — the file imports no toolbar — RATIFIED

RULE **B14 12** — *"Table toolbar: [Search…] [All | filter chips ×] [+ Filter ▾] … [▦ Columns]"*. **B14 13** — headers show ⇅. **B14 14** — ▦ opens show/hide with reorder and Reset. THE SCREEN **No toolbar at all.** No search, no filter chips, no `+ Filter ▾`, no ▦, no sortable header. The pager is the only toolbar element present on either screen. **Walked.** CLOSED BY **MR-12**

7-04 14 · 15 `ListPager.tsx:107` · `endpoints.ts:2160` — RATIFIED no gate

RULE **NFR-3** — universal server pagination, and it is load-bearing rather than tidy: it is the flexibility T4 and T3 depend on. THE SCREEN Both page **in the browser**. `pageSlice()` is `rows.slice()`, over `getRoutes()`/`getCycles()` = `listAll()`, a loop that walks **every page at 1,000 rows a time**. The server-paginated route exists and `getRoutesPage`/`getCyclesPage` exist to ask it; these two screens use neither. CLOSED BY **MR-11**

7-05 08 `UploadReadsWizard.tsx` ~1495–1510 — PARTLY the chevron register exists; it does not cover this screen

RULE **B14 8** — dropdowns share one chevron style; the CR-016 conformance register asserts `nativeSelects: 0`. **B14 36** — *"compose the kit. Where the kit lacks a piece, ADD IT TO THE KIT."* THE SCREEN The manual-entry grid hand-rolls a raw `<select className="input">` per row for the register, rather than the kit's `SelectField`. CLOSED BY **MR-08**

7-06 08 `UploadReadsWizard.tsx` ~1478–1490 — PARTIAL rule 7 is half-enforced and only on four asset screens

RULE **B14 7** — date fields always use the date input, with keyed entry and a Today shortcut. **B14 36** — compose the kit. THE SCREEN The same grid hand-rolls a raw `<input type="date" className="input">` per row rather than `DatePickerField`. It is a date input, so the rule's first half holds; it is not the kit's, so bounds, the stated format and any future Today shortcut pass it by. CLOSED BY **MR-08**

7-07 07 · 17 · 18 `UploadReadsWizard.tsx:1140,1149` · `ReadsListPage.tsx:1098,1112` · `EstimateReadModal.tsx:373` — RATIFIED the searchable half is explicitly unchecked

RULE **B14 8** — *"5+ options are searchable (type-to-filter)"*. **BR-SHL-033** mandates `SearchSelectField` past `LONG_LIST_THRESHOLD = 10`. THE SCREEN Every long-list select in this module is the plain `SelectField`: the Cycle and Route pickers on stage 1, both pickers in the route-sheet dialog, and the Register select on the estimate dialog. At the seeded tenant the lists are 6 long, so nothing bites *yet* — and **NFR-5 forbids a cap**, so the list length is the tenant's, not ours. CLOSED BY **MR-06**

7-08 18 `EstimateReadModal.tsx` lede — PARTLY G33 caps refusals at six words per sentence; it does not read a lede

RULE **B14 33** — *"RATIONALE — why the product works this way… is **ZERO WORDS ON SCREEN**. It belongs in the code and the register."* THE SCREEN The estimate dialog opens with 25 words of rationale: *"When a meter could not be read, this works out the consumption from that meter's own recent actual readings. It never guesses a rate and it never rounds."* **Walked.** The second sentence is the clearest case — it is an argument about the engine's character, to a clerk who has already decided to estimate. CLOSED BY **MR-14** · **MR-31**

7-09 09 · 10 stage 3 split statement · stage 4 commit label — RATIFIED

RULE **B14 33** — a FACT is ≤ 6 words and is *a fact*. A count that reads as a grammatical error is not one. THE SCREEN *"1 rows will post. 1 will not"* and ***"Commit 1 rows ✓"***. No singular form. **Walked**, on the button that writes to a financial record. CLOSED BY **MR-20**

7-10 14 `RoutesCyclesPage.tsx`, the `lastCompletion` cell — RATIFIED

RULE **BR-SHL-034** — one date formatter. A *date* column shows a date. THE SCREEN *Last completion* prints **07/31/2026 19:00** — a clock in a date column, and the conversion has moved the date back a day. **Walked.** The identical defect was fixed on entry 02's Read date column and never swept here. CLOSED BY **MR-19**

7-11 14 `RoutesCyclesPage.tsx:460`; the same shape on entry 15 — RATIFIED

RULE **B14 17** — an empty state names *"what is missing, and the action that fills it"*. THE SCREEN *"No routes are configured yet."* — the first half only. No action, and no link to the Settings screen where a route is created, which the header offers two lines above. CLOSED BY **MR-21**

7-12 16 `RoutesCyclesPage.tsx` `ChaseListPanel` — G37 covers row-action icons where actions exist; it cannot ask for a missing one

RULE **B14 15** — row actions are icons with tooltips. **B14 16** — a pagination footer on every list table. THE SCREEN The chase panel is a list of meters with **no row action of any kind** — the meter is text, not a link — and **no pager**. Stage 3 of the wizard offers *Estimate all* and *Send someone out* for the same shortfall; this panel, which is where an operator actually reads it, offers neither.
CLOSED BY **MR-10**

7-13 03 `ReadDetailPage.tsx:188-190` — G103 exists for rotting holds; it does not cover this string

RULE **Hard-won law 2's mirror** — *"when you write a hold, write the test that fires when the hold expires, in the same commit."* THE SCREEN The primary button on a billed reading refuses over a built endpoint with a sentence that is false. Full case in section 06. CLOSED BY **MR-15**

7-14 14 `RoutesCyclesPage.tsx:528` —

RULE **Hard-won law 1** — a comment claiming a guarantee is not the guarantee. THE SCREEN The tooltip *"the reads module is not wired yet"* over an unreachable branch. Full case in section 06.
CLOSED BY **MR-16**

7-15 02 `ReadsListPage.tsx`, trailer — — no gate reads a trailer for truth

RULE **Hard-won law 1**, again, in the place a maintainer looks first. THE SCREEN `ReadsListPage.tsx`'s Lifecycle `exceptions:` trailer is **stale in five clauses**: it says the service filter is not rendered (it is), that there is no export control (there is), that both header actions are disabled (neither is), that S29 is a drawer (it is a page), and that *"the server IGNORES `dir`... REPORTED as a defect"* — which was fixed, as `selectors.py`'s own comment now records. CLOSED BY **MR-33**

7-16 14 · 15 `RoutesCyclesPage.tsx:853` · `SettingsRoutesCyclesTab.tsx:938` —

RULE **Hard-won law 1** — and this one is two files disagreeing about one fact. THE SCREEN `RoutesCyclesPage.tsx` and `SettingsRoutesCyclesTab.tsx` both state that `GET /routes` and `GET /cycles` do not paginate server-side. `api/endpoints.ts:2206` states the opposite, dates it, and calls the old claim *"hard-won law 1 pointed backwards"*. **Measured: the endpoint answers a paged envelope.** Two of the three are wrong. CLOSED BY **MR-34**

7-17 15 · 16 `completeness.py:894` · `RoutesCyclesPage.tsx` `chaseQueries` —

RULE **The ICP ruling of 2026-09-06** — the ceiling moved 3,000 → 40,000 and **NFR-5** forbids a cap anywhere. THE SCREEN Two live bounds are argued from the retired ceiling. The chase-list's own comment reads *"Bounded by the cycle's own membership (<3,000 connections)"*; entry 15 fires **one chase request per visible cycle** and justifies it as *"this product is <3,000 connections with a handful of cycles"*. Neither is a cap in code — which is why NFR-5 is not breached — but both are unpaginated reads whose safety rests on a number that is no longer the product's. CLOSED BY **MR-35** · **MR-11** · **MR-30**

7-18 — `07-frontend/src/styles/route-b-migrated.txt:149-150` — PARTLY the vitest check reads the list; neither reader refuses a duplicate

RULE **B14 21** — a utility is emitted only for a file listed in `route-b-migrated.txt`, and the ledger is what the two readers agree on. THE SCREEN `features/metering/readsShared.tsx` is listed **twice**, at lines 149 and 150. Harmless today — both readers key on the path — and it is a duplicate row in a file whose whole job is to be an exact list. CLOSED BY **MR-36**

Two product rules, checked separately because both fail invisibly

Rule	Verdict on this module
R14 new entries	**PASS, on the one list it can apply to.** Entry 02's default is `read_at` descending, honoured in SQL with a stable secondary sort on `id` (`selectors.py:111`, `READ_DEFAULT_SORT = "-read_at"`) — and the walk confirmed it, the URL reading `?sort=read_at&dir=desc`. **A reading's business date is the right key here**, not `created_at`: a clerk asking "what came in last" means the newest *reading*. Entries 14, 15 and 16 have **no sort control at all**, which R14 cannot bite on because there is nothing to default. **And the defect the rule was written from is**

sort to the top	**absent here:** nothing in Meter Reads sorts in Python after materialising every row — except the chase list, which does (`completeness.py:913`), on a list with no pager.
R15 money never reaches JSX unformatted	**NOT APPLICABLE, and deliberately so.** `BR-MET-078` puts no money on any of these screens, and the walk confirmed it: not one currency figure on nineteen entries. **The quantity half is the live version of the same rule and it passes:** Previous, Current and Usage are composed server-side through `quantity_display.quantity_with_unit`, the history table goes through `formatQuantity`, and a reading whose unit never arrived renders a **dash** rather than a bare number. Nothing printed `1008.000000` anywhere in the walk — which is what this rule looked like when it was broken.

The copy budget — every sentence on every screen, against rule 33

B14 rule 33 caps on-screen text by what it is FOR: a label ≤ 3 words and no sentence · a fact ≤ 6 · an instruction ≤ 6 *per sentence*, and two sentences of six beat one of twelve · a consequence ≤ 6 per sentence and at most two · and **rationale — why the product works this way — is ZERO WORDS on screen**. Every helper line and floor note in the module is measured below. **Five passed. Twelve were over, three of those pure rationale, and all twelve are cut.**

Screen	On screen today	Words	What it is	What it becomes
02	Showing the last two years. Older readings need the archive search.	5 + 6	instruction	PASSES — two sentences, both inside six
02	A corrected reading appears once: the list shows the current version of each chain.	14	**rationale**	**Cut.** It explains why the list looks the way it does; that belongs in the code and the register. **Applied — the revised drawing no longer carries it.**
04	In kgal — consumption and every rule are recomputed.	8	consequence	*"In kgal."* + *"Usage and flags recompute."* — 2 and 4
04	Stored on the correction — "collector re-read".	6	fact	PASSES
05	A System Admin decides this later from exactly this sentence.	10	**rationale**	*"A System Admin reads this."* — 5. The *why* goes.
06	Stored in the audit trail, not on the reading.	9	fact	*"Stored in the audit trail."* — 5
07	Any day in the month; the batch stores the first of it.	12	instruction	*"Any day in the month."* + *"We store the first."* — 5 and 4
07	The cycle's read window is what out-of-window compares against.	9	**rationale**	**Cut.** A clerk choosing a cycle does not need the rule engine's join explained.
07	Leave empty when the file covers the whole cycle.	9	instruction	*"Empty means the whole cycle."* — 5
07	A row naming its own method overrides this one.	9	fact	*"A row's own method wins."* — 5
07	Which services this file covers. Each row states its own service.	5 + 6	instruction	PASSES
08	A reading already on file for that meter, date and register is not written over. Those rows are held back for you to look at before anything posts.	16 + 13	consequence	*"Existing readings are not overwritten."* + *"Those rows are held back."* — 5 and 5. **The worst line in the module, and it sits on the disposition that decides what a file does to live data.**

12	We look the meter up when you leave the box.	10	instruction	*"We look it up."* — 4
12	The number on the dial, not the usage.	8	fact	*"The dial, not the usage."* — 5
13	Say what you checked. It is stored on each reading.	4 + 6	instruction	PASSES
14, 15	Completeness and progress derive from committed reads — nothing on this page is a stored counter, and nothing on it can be edited.	22	**rationale**	**Cut.** It is `BR-MET-011` addressed to a reader who is not the one on this screen. **Applied — the revised drawing no longer carries it.**
17	Opens print-ready in a new tab; your browser's Print / Save as PDF makes the PDF.	15	instruction	*"Opens in a new tab."* + *"Print or save as PDF."* — 5 and 5
20	*The new screen, measured as it was drawn.*			PASSES — longest line is *"5 meters are short."* + *"Estimate or read them first."*, 4 and 5

> **All seventeen are applied**
>
> **Every row above is now APPLIED in the drawing** — PO ruling of 2026-09-12, *"run all 18"*. The table is kept as the record of what each line was, what it is, and which cap decided it, because a reader asking *"why is this sentence so short?"* should find the answer beside the sentence. **The seventeen lines were 202 words and are 102.** Half the helper copy in Meter Reads is gone, and three explanations a clerk was never the audience for went with it.

Where Meter Reads meets the other modules

This pass is run once, from above, because it finds the failures that belong to nobody — the ones that fall between two owners. It starts from the ⇄ **rows** in the field tables above rather than from the import graph, so the boundary it describes is the one the screens actually cross. Every row names **both** modules, because a finding filed against one is a finding the other's owner never reads.

Metering has **ten** live boundaries in this module's screens. Eight are inbound reads through a declared seam, one is a write-side lock applied by billing, and one is a column that exists and is served to nobody.

1 · Does the handoff land?

From → To	The contract	Verdict
metering → billing	`J2` step 5→7: *"committed `meter_reads`, one row per channel, plan-required channel set complete (BR-MET-049), status ok/estimated"*.	**The shape lands; the completeness clause is not a thing this module can promise.** On the walked tenant, CY-001 reported **0 of 18 registers** and 5 meters short. The handoff is satisfiable only when the chase list is empty, and the one screen that says so is entry 15 — **which has no door**. So the contract's precondition is checked on a screen nobody can reach.
billing → metering	`BR-MET-053`: at issue, billing marks each priced read BILLED,	**Lands, and is the tightest seam here.**

	and from then the read may be corrected only through the J6 path.	`apps/billing/metering_seam.py` runs *inside* `issue_invoice`'s transaction, resolves `apps.metering.services.mark_read_billed` **by name**, and is on the human-owned list. **But the door it unlocks is shut on the screen** — see section 06's expired hold. Billing does its half; metering's screen refuses to do the other.
metering → rating	The plan-required register set — what *should* have arrived for each meter this window.	**Lands.** It drives four separate surfaces — the Missing tile, the wizard's shortfall table, the cycle completeness cell and the chase list — and all four agree, which is the test that matters.
metering → work	*Send someone out* on a shortfall group raises field work.	**Lands on stage 3 and nowhere else.** Entry 16 — the chase panel, which is where an operator reads the same shortfall — has neither *Estimate* nor *Send someone out*.

2 · Does any module write another's tables?

No — and the repository's own gates say so, not just this document. `G84` passes (every ORM write in a file carrying a write allowance is one the allowance names); `G65` passes with 13 baselined cross-module imports, a number that may fall and never rise; `G85` passes with four ruled address homes and no unruled one. Metering's own seam file states the rule it is built to — *"metering NEVER touches another module's ORM"* — and resolves every neighbour call at call time from the owning module's services.

What does exist is a set of cross-boundary READS that bypass a declared seam. They are LEAKY rather than WRITING, and one of them says so about itself.

3 · Does one fact have two homes?

The fact	Modules	Verdict
The service address on a reading	metering ⇄ metering	ONE HOME Entries 02, 03 and 16 all show `premise` + `unit_label` **walked to `premises`**, composed server-side. No reading, route, cycle or batch declares an address column. `G85` confirms it.
The account a reading belongs to	metering ⇄ customers	A HOME WITH NO READER `meter_reads.account_id` is a real FK to `customers.Account`, in the reviewed cross-module set (`models.py:1847`) — and `read_dict` does not serve it, so entry 03 states the absence instead: *"The account is not served on a reading; open it through the unit."* **The column is a stored fact the product declines to show**, which is the opposite of two homes and is its own kind of drift: the day somebody serves it, two paths to the account exist — the column, and the walk through the unit — and nothing says which wins.
"How many were collected"	metering ⇄ metering	TWO FIGURES, ADJACENT Entry 14's *Collected* counts readings; its *Meters* counts meters. Entry 15 carries a third, plan-required registers. One row on the walked tenant read **5 · 0**

		of 18 · 5 meters short. Journey issues J-04 and J-06; remedy R3.
The current read window	metering ⇄ metering	FORKED, AND KNOWN Two derivations disagree — `cycle_dict`'s window and `completeness.current_window` answered different months on one measured day. Entry 15 works around it by passing the window *the row is itself printing* into the chase call, so both halves of one line describe one period. **The workaround is right and the fork is filed, not fixed.**

4 · Does the same thing have two names?

Concept	Words in use	Verdict
A meter's register	**Channel** (API, filter, column header, S29 tag) · **Register** (the wizard's manual grid, its mapping table, the quick-entry form) · **Total register** (the value)	TWO NAMES, SPLIT BY AUDIENCE — and the split is defensible: *Channel* is the schema's word and *Register* is the clerk's. It is not defensible that **both appear on screen**: entry 02's column header says CHANNEL while entry 08's grid says Register, for one thing.
A reading's state	**Staged** → **Validated** → **Posted** (batch) · **Committed** → **Billed** (reading) · **OK / Flagged / Estimated** (status)	THREE VOCABULARIES, THREE SUBJECTS — a batch, a reading's lifecycle and a reading's validation verdict are genuinely different facts, and no screen uses one word for two of them.
B14 rule 34's pair — **Recorded** / **In the books**	neither appears in this module	NOT APPLICABLE The rule settles *held here* versus *in the books*, which is a money distinction. `BR-MET-078` puts no money on any screen here, so the pair has nothing to name.

Every boundary crossing, graded

From	To	What crosses	Through	Grade
metering	assets	the meter number; the read history table; meters on a route	`apps/metering/seams.py`, resolved at call time	HEALTHY
metering	rating	the plan-required register set	the same seam	HEALTHY
metering	tenancy	the enabled service catalogue and its names	the same seam	HEALTHY
metering	audit	the uploader's **name** for the compliance stamp	audit's `actor_names` seam	HEALTHY — and it is the fix for a sixteen-day defect that printed the key instead
metering	identity	a route's *assigned to*	the same seam, **read-only by construction** — the ops screens import no mutation wrapper at all	HEALTHY

metering	work	*Send someone out* on a shortfall group	the work module's own service	HEALTHY
billing	metering	the BILLED lock, at issue	`apps/billing/metering_seam.py` — inside the issue transaction, resolving `mark_read_billed` by name; 🔒 human-owned	HEALTHY
metering	billing	`read_corrections` — references and a reason, **never a money figure**	the J6 linkage table	HEALTHY
rating	**metering**	the latest consumption per unit per service	`apps/rating/seams.py:841` — `django_apps.get_model("metering","MeterRead")` then `.objects.filter(...)`	LEAKY **MR-37** — read-only and tenant-filtered, and **it declares itself**: *"`apps/metering/selectors.py` has no such read… A published `metering.selectors.recent_consumption_by_unit` would be better and is named in this lane's write-back."*
portal	**metering**	a bill's readings, premises, service points	`apps/portal/seams/metering_seam.py:317` reaches the ORM directly then passes each row through `metering_selectors.read_dict`; `portal/selectors.py:859,891` and `portal/credentials.py:256,323` reach metering's models from **outside** portal's own seam file	LEAKY **MR-37**
integrations	**metering**	the route vocabulary for a value mapping	`apps/integrations/mappings.py:164` — `from apps.metering.models import Route`, inline, in a function body	LEAKY **MR-37**
customers · portal	**metering**	service interruptions	four files import `apps.metering.interruptions` models directly (`customers/service_request_api.py:96`, `service_request_service.py:458`, `portal/services.py:73`, `portal/guest_api.py:58`)	LEAKY **MR-37** — reads only, and the shape is consistent across all four, which usually means a seam function is owed rather than four callers being careless

Grades: 8 HEALTHY · 4 LEAKY · 0 WRITING · 0 ABSENT. The four LEAKY are MR-37; the blind gate is MR-22. No module writes another's tables anywhere in this surface, which is the grade that would have mattered most.

> ⛔ **And one gate is blind to exactly one file, because of an import alias · MR-22**
>
> `G96` exists to catch a portal read over another module's table that names only the account set and leans on RLS `FORCE` for the tenant. `BR-CWP-010` requires **both halves in the query**, and hard-

won law 14 is why: the suite runs on in-memory SQLite, which has no row-level security, so a read that omits the tenant **passes every test in this repository, forever, indistinguishably from one that does not**.

It reports `PASS: 30 portal file(s) read; every queryset over another module's model names the tenant`, and that sentence is true of everything it can see. **It cannot see** `apps/portal/seams/metering_seam.py`**.** The gate matches chains by **import alias** against a hardcoded list of eleven — `metering_models`, `customers_models`, `assets_models` and so on — and that file writes `from apps.metering import models as reads_models`. The alias is not on the list, so the chain is classified as *"over the portal's own tables and not this gate's business"*, which is its own comment's words for the case.

What is behind the blind spot: `reads_models.MeterRead.objects.filter(id__in=wanted)` — **no tenant keyword**. Verified by running the gate's own `findings()` over that file with its own scoped-model set: `MeterRead` *is* in the tenant-scoped set, and the result is `[]`.

How bad is it, honestly? The read is still tenant-safe in production — RLS is `ENABLE+FORCE` on 281 of 282 tenant-scoped tables, measured on the stack that produced section 05, and the app role holds neither `SUPERUSER` nor `BYPASSRLS`. **The defect is in the check, not (today) in the data.** But the check is the only thing standing between this file and law 14's trap, and it has been reporting a clean zero over it.

Two other aliases are also off the list — `identity_models` and `asset_models` (the list has `assets_models`) — and both are in `sample_data.py`, which the gate exempts by name anyway. So the live blind spot is one file.

The cure is the one CLAUDE.md already prescribes for this shape: *"a check that cannot cover its whole subject must print what it skipped beside what it checked."* A gate that resolved aliases from the import statement instead of matching a list would have no blind spot to print — and a gate that counted *chains classified as portal-own* beside its zero would have made this visible the day the alias was written.

09

The specification register

One row per commitment, no exceptions, and every commitment appears both here and as a card in the section that owns it.

Read this before the table

If a thing is not in a commitment card, nobody has undertaken to build it. Everything else in this document — every finding, every drawing, every honest statement that something is missing — is a **description**. A description commits nobody to anything. This section is the only place where the word "will" appears, and even here every row starts at **Unverified**.

The ID series is this document's own — `MR-`, minted here and allocated nowhere. The repository's live series are `US-`, `BR-<MOD>-`, `OQ-<MOD>-`, `DEF-<MOD>-` and `CR-###-BRC##`, and borrowing one would make this look like scheduled work when it is an analysis. **An analysis may propose and never allocate.** If any of these is taken on, it gets a real id at that point, through `10-scripts/allocate.py`, from the register's own maximum.

Description — grey rail. A statement about what the code does today, including an honest statement that something is missing. Commits nobody.

Commitment — accent card. A named piece of work, what it waits on, the screens it changes, and what a person could do and see once it is done.

Read these five first — ordered by what their absence costs a person

The table below is in number order, which is the order they were minted and means nothing. **This is the order to do them in**, and the measure is not effort — it is what a person cannot do today.

#	Do	What a person cannot do today
1	MR-01	**Open a reading. At all.** Every click on a meter replaces the page with a type error, on four screens. Everything else on this list is invisible while this stands
2	MR-46	**Tell a peak from a total.** A demand register read as a difference bills **412 kW as 14 kWh — $154.00 against $4,532.00**. It has happened once and the screen still cannot show the difference
3	MR-22	**Trust a clean gate.** A tenant-scoping guard reports zero over a file it never reads. Hard rule 1 is the one rule whose breach is not recoverable
4	MR-45	**See that a sibling register never arrived.** A clerk looking at one TOU window has no way to know the meter is short, and the register set is what decides whether it can be billed
5	MR-52	**See that a dispute was cleared.** A System Admin clears it, the product records who and when, and **the reading shows no trace** — so the next person reopens a question that was answered

Two of those five are not screens — MR-01 is one line of TypeScript with a trap in it, MR-22 is a gate. **The most expensive things in this module are not drawings**, and a document made of drawings is exactly the document that would let you forget it.

ID	What a person could do, and what it waits on	Screens	What done looks like on screen	Status
MR-01	**A clerk can open a reading again.** Waits on: nothing — `getAssetReads()` is typed `Promise<AssetReadRow[]>` over an endpoint that answers `{count,results}`.	03	Clicking `M-7210` on Reads opens *Reading · M-7210 · Total register* with its four figures, and *Read history (this meter)* lists rows rather than the page being replaced by *"Unexpected Application Error!"*.	Unverified
MR-02	**A clerk can get to Routes and Cycles without knowing a URL.** Waits on: nothing — both screens are built and translated.	01 · 14 · 15	The Reads header shows four tabs — *Needs attention · All reads · Routes · Cycles* — and `/reads/routes` still opens, redirected to `?tab=routes`.	Unverified
MR-03	**A clerk can see which readings are missing, not just how many.** Waits on: `READ_FILTERS` declares no `missing` filter, and an undeclared `f.` param is ignored — so a drill-in over it would return every read while looking like a slice.	01	The Missing tile reads *"18 · Show the cycles that are short →"* and the link lands on the Cycles tab with the short cycles at the top.	Unverified
MR-04	**A clerk can read a batch number in the Batch column.** Waits on: `read_dict` serves `import_batch_id` and not `batch_no`.	02	The Batch column reads **`RB-2609-0006`**, and the uuid rides on `title` as evidence — B14 rule 31's own sentence.	Unverified
MR-05	**The reading page survives a failed history read.** Waits on: nothing.	03	With the history endpoint down, the page still shows its figures, math line, flag reason and compliance stamp, and the history section alone reads *"The read history could not be loaded."* with a Retry.	Unverified
MR-06	**A clerk with hundreds of routes can find one by typing.** Waits on: nothing — `SearchSelectField` exists and `BR-SHL-033` already mandates it past ten options.	07 · 17 · 18	Typing *"riv"* into the Route picker narrows it to *RT-003 · Riverbend* and *RT-004 · Riverbend*.	Unverified
MR-07	**A supervisor sees a window that has closed, as closed.** Waits on: nothing — both dates are already served.	15	CY-001 shows ***Closed 9 days ago*** rather than *"0 days left"*, and CY-003 shows *Opens in 3 days*.	Unverified
MR-08	**The wizard's grids use the product's own controls.** Waits on: nothing — `DatePickerField` and `SelectField` are the kit.	08 · 09	The manual grid's date cell carries the same calendar affordance as every other date field in the product, and its register cell the same chevron.	Unverified
MR-09	**A clerk cannot spend a press on an uncertified commit.** Waits on: nothing.	10	*Commit 1 row ✓* is greyed with *"Tick the certification first"* on hover, and becomes live the moment the box is ticked.	Unverified

MR-10	**A clerk can estimate a group of missing readings from the list that names them.** Waits on: the pending set is derived and its rows carry `id: null`, which is a different selection model from the stored rows the bulk bar owns.	13 · 16 · 20	The chase panel carries *Estimate these 5* and *Send someone out* — the same two controls stage 3 of the wizard already has for the same shortfall.	Unverified
MR-11	**Routes and Cycles page on the server.** Waits on: nothing — the endpoints paginate, sort and filter server-side, and `getRoutesPage`/`getCyclesPage` exist.	14 · 15	With 300 routes seeded, the footer reads *Showing 1–50 of 300* and page 2 issues a request rather than slicing an array the browser already holds.	Unverified
MR-12	**Routes and Cycles get the toolbar every other list has.** Waits on: MR-11 — a search box over a client-side array is the defect, not the fix.	14 · 15	Both screens carry `[Search…] [All | chips ×] [+ Filter ▾] … [▦ Columns]`, and column headers show ⇅.	Unverified
MR-13	**One cycle row states one set of numbers.** Waits on: nothing — every figure is already served.	14 · 15	CY-001 reads *Meters 5* and *Registers collected 0 of 18 · 5 meters short*, and no two adjacent columns count different things without saying so.	Unverified
MR-14	**The estimate dialog stops arguing.** Waits on: nothing.	18	The dialog opens on its three fields. The 25-word lede is gone; its content lives in the code comment and the register, which is where B14 rule 33 puts rationale.	Unverified
MR-15	**A clerk can correct a billed reading from the reading.** Waits on: nothing on the server — `POST /reads/{id}/billed-correction` is served and S30's Fix-a-bill exists. The front end has never called it.	03	On a billed reading, *Correct billed read* is **live**, opens a dialog asking for the invoice it was priced on, and the sentence *"that path is not built"* is gone from all eleven catalogues.	Unverified
MR-16	**Two screens stop claiming the reads module is unwired.** Waits on: nothing — the branch is unreachable.	14 · 15	No cell on either screen carries *"The reads module is not wired yet"*; a route with no readings this window says *Not collected yet* and stops there.	Unverified
MR-17	**A utility that requires a second pair of eyes can enforce it.** Waits on: a tenant toggle nobody has ruled on — half of `OQ-042`, the writer, is built.	10	With the setting on, the commit is greyed until a second person co-signs, and its reason says who may.	Unverified
MR-18	**Three dead strings leave eleven catalogues.** Waits on: nothing.	—	Nothing visible changes, which is the point: the catalogues stop carrying copy no screen can render.	Unverified
MR-19	**Last completion shows a date.** Waits on: nothing.	14	RT-001 reads **`08/01/2026`**, not `07/31/2026 19:00`.	Unverified
MR-20	**The commit button counts in English.** Waits on: nothing.	09 · 10	*Commit 1 row ✓* and *"1 row will post. 1 will not."*	Unverified
MR-21	**An empty Routes or Cycles screen says what fills it.** Waits on: nothing — the Settings link is already in the header.	14 · 15	*"No routes are configured yet."* carries *Add a route in Settings* → beneath it.	Unverified
MR-22	**A gate that cannot see a file says so.** Waits on: nothing. `G96` matches import aliases against a hardcoded list of eleven and is blind to `reads_models`.	—	Not a screen. `G96`'s own line reads *"30 files read · 0 unresolved · **0 chains skipped as portal-own**"*, and the number it prints beside its zero is what makes a future blind spot visible on the day it is created.	Unverified
MR-23	**A client helper cannot claim an array over a paginated endpoint.** Waits on: nothing. This is MR-01's class rather than its instance.	—	Not a screen. A check reads every `client.get<T[]>` in `endpoints.ts` and refuses one whose path is served by a paginated handler — both halves already exist, in the client and in `core/pagination.py`.	Unverified
MR-24	**A reading filed on the wrong register can be put right.** Waits on: **a ruling** — migration `0007`'s trigger freezes `channel` with the rest of the measurement, and the narrowing is `OQ-209`, open. **Four shapes are on the table and this document picks none of them.**	04 · 08	Four options, one of which needs no ruling: *(a)* a correction carries a channel, reopening OQ-209 · *(b)* a re-register path that supersedes the wrong row and creates the right one — two rows, no update, hard rule 4 intact · *(c)* leave it and say so on screen, which no screen does today · ***(d)* refuse it at the door**: validate rejects a row whose register is not in that meter's expected set, where today it is a silent `unexpected`. **(d) is the only one this document recommends without a ruling**, and it prevents rather than repairs.	Unverified

MR-25	**A clerk can key a multi-register meter without opening the import wizard.** Waits on: nothing — `QuickReadModal` has already fetched `channels` by name when it decides to hand off.	12	A meter with four expected registers shows four labelled number boxes in the same dialog, named *On-peak* rather than `tou_on`, and all four post together. The hand-off stays for `channel_count === null`, the one case where the modal genuinely does not know what to ask for. **Today a demand meter with two numbers on a card sends a clerk into a five-stage file wizard.**	Unverified
MR-26	**A clerk lands on the readings instead of on three tiles.** Waits on: nothing. PO ruling 2026-09-12 — *"Need Attention — does not add any value"*.	01 · 02	The Needs attention tab and its three tiles are withdrawn; **All reads** is the landing tab. Flagged and Disputed become filters counting with the same predicate the tiles used; Missing is stated per cycle on 20. **The tenant-wide all-clear is lost and is recorded as a cost**, not argued against. Front end only — the endpoint keeps its Dashboard caller.	Unverified
MR-27	**A clerk can see which plan a reading belongs to, and find one by address.** Waits on: nothing — every field is already served.	02	The tab strip carries **All reads** and **Read cycles**; search widens to meter number *and* address; *Channel* reads **Register**; the batch shows its human number rather than a uuid; a **Plan** column, hidden by default, reads the stored `plan_registers` outcome. **Nineteen fields carry over and none is removed.**	Unverified
MR-28	**Routes stop being a screen with no door and become part of the cycle that measures them.** Waits on: MR-30. PO ruling 2026-09-12 — *"we should say Read Cycles and within that we should show Routes and their completion"*.	14 · 20	`/reads/routes` redirects to the Read cycles tab; all nine columns survive inside the cycle whose window measures them. **The all-routes view is lost** — a supervisor reads routes cycle by cycle — and that is recorded as a cost of the ruling. If it bites, the answer is a Routes tab beside Read cycles, not the old orphan screen.	Unverified
MR-29	**A person can reach the cycles at all.** Waits on: nothing — the endpoints paginate and the client helpers exist, unused. PO ruling 2026-09-12 — *"we will need a list view of read cycles"*.	15	A **Read cycles** tab beside All reads, with the toolbar the screen has never had and a **server** pager. Ten columns become nine: start day and days-to-complete fold into *Frequency*; *Completeness* reads **Registers collected**; the countdown becomes a **Window** pill that can say *Closed 9 days ago*. ⚠ **the Window pill and the collected figure both compute from a window whose two derivations disagree** — 3.14, unresolved.	Unverified
MR-30	**A person can answer for one cycle in one place.** Waits on: nothing for three of its four tabs; **History needs an endpoint that does not exist**. PO ruling 2026-09-12 — *"we will need a detail view of read cycles"*.	20	A cycle page at its own address, with four tabs each separately linkable: **Overview** (the facts, the schedule, the billing), **Routes**, **Still to read** — which carries the bulk-estimate control `POST /reads/bulk-estimate` has been served and uncalled for since CR-038 — and **History**. Four actions, all pre-scoped; **no cycle edit**, which stays in Settings by `BR-MET-011`. **History is the only unserved figure and is drawn in its own tab so it can be dropped from wave one without moving anything else.**	Unverified
MR-31	**Every helper line on every reads screen fits its purpose.** Waits on: nothing — the replacement string for each is written out in §07's copy budget.	02 · 04 · 05 · 06 · 07 · 08 · 09 · 10 · 12 · 17 · 18	**Blast radius: 17 lines across 11 screens.** Seventeen lines measured against `B14` rule 33 and rewritten: **202 words become 102**. Three are pure rationale and go to zero — a clerk is not the audience for why the list de-duplicates a chain, why a cycle's window is the comparison, or why the write mode is asked before the file is read. Nothing loses a fact; the longest survivor is two sentences of six.	Unverified
MR-32	**The reading's information groups become Panels.** Waits on: nothing. PO ruling 2026-09-12, pointing at the bill detail — *"in detail section use Panel to information groups"*.	03	**Blast radius: 8 groups on ONE screen.** The same top-ruled idiom appears in **6 files** across the front end and `<Panel>` is already used **548 times in 127 files** — **unifying the two product-wide is a separate and much larger decision, and this commitment does not take it.** Calculation · Flags · History · Location · Evidence each render as `components/Panel.tsx` — the bordered surface	Unverified

			used 149 times — instead of `ReadDetailPage.tsx:139`'s top-ruled section. **This is a change to the product, not a correction of a drawing**: both containers are real and the ruling chose between them. One screen, no data change.	
MR-33	**The reads list's own trailer stops lying to the next maintainer.** Waits on: nothing.	—	Not a screen. `ReadsListPage.tsx`'s exceptions trailer is **false in five clauses** — it still says the export does not exist and the detail is a drawer. A maintainer reads the trailer first, which is what makes a stale one expensive. **Hard-won law 1.**	Unverified
MR-34	**One answer about whether Routes and Cycles paginate.** Waits on: nothing — the endpoints have paginated since 2026-08-23 and one file already says so, dated.	—	Not a screen. Two files claim the opposite and are deleted of that claim. **This already cost this document a wrong finding**, corrected only by measuring the endpoint — which is the argument for fixing a comment nobody would call urgent.	Unverified
MR-35	**No live bound is argued from a ceiling this repository retired.** Waits on: MR-11 for one of the two.	—	Not a screen. Two comments justify an unbounded list with *"<3,000 connections"* — the ICP ceiling that moved to **40,000 on 2026-09-06**. The chase list's pager (MR-30) and the list pagers (MR-11) remove the dependency; these sentences remove the false reasoning, which is the half that would otherwise be copied into the next unbounded list.	Unverified
MR-36	**A file appears once in the migration list.** Waits on: nothing.	—	Not a screen. `features/metering/readsShared.tsx` is listed twice in `route-b-migrated.txt`. Harmless today — both lines say the same thing — and it is on the register because a file that appears twice in a build manifest is a question somebody has to answer later, at the worst moment.	Unverified
MR-37	**Four direct reads of metering's tables go through the seam.** Waits on: nothing — the seam file exists and serves these shapes.	—	Not a screen, and **reads only — nothing writes**, which is why it is here rather than in the first two. It matters because a direct read is what put `MeterRead.objects.filter(...)` where `G96` could not see it: the boundary and the blind spot are the same defect from two ends.	Unverified
MR-38	**Two derivations of "the current window" give one answer. Waits on: a ruling** — this is **3.14**, open, and it is a backend disagreement rather than a design question.	15 · 20	Measured on one day, **the cycle row printed August while the completeness engine answered September**. Three drawn figures compute from it — the **Window** pill, **Registers collected** and **Last read** — so **building them on the fork produces a screen that disagrees with itself** in a tidier typeface. The screens work around it today by passing the row's own window into the chase call, so one line describes one period. **This document does not choose between the two derivations**; the prime directive forbids resolving an open question by picking a value.	Unverified
MR-39	**A clerk can open the account a reading belongs to.** Waits on: nothing — `meter_reads.account_id` is a real foreign key to `customers.Account`, in the reviewed cross-module set (`models.py:1847`).	03	The account renders as a link beside the premises and unit. **Today `read_dict` declines to serve a column the database already holds**, and the screen states the absence instead — *"the account is not served on a reading; open it through the unit"*. **A stored fact the product refuses to show is its own kind of drift**: the day somebody serves it, two paths to the same account exist and neither is wrong, which is how a second source of truth starts.	Unverified
MR-40	**A staged batch is read a page at a time, and no row count is capped.** Waits on: nothing — `StandardPagination` is what every other list on this module uses.	08 · 09	`GET /read-batches/{id}` serves **every staged row with no pager, to a hardcoded 50,000-row cap**. Both halves are ruled against: `NFR-3` is universal server pagination and `NFR-5` is **no hard caps anywhere** — and CLAUDE.md names a cap that would have to be removed when T4 arrives as the specific	Unverified

			thing nobody may add. **A 40,000-account utility importing a monthly cycle can reach that ceiling on one file**, and the failure at the ceiling is silent truncation rather than a refusal.	
MR-41	**The receipt is walked rather than composed from its source.** Waits on: nothing.	11	A driver that can tick the certification, so stage 5 renders in front of a person. **This is the reason entry 11's drawing carries a disclaimer no other drawing needs** — the commit refuses until the tick and the tick could not be driven, so the one screen in this module nobody has seen is also the one that reports what a write did. **A test that cannot reach the last stage of a five-stage wizard is not covering the wizard**, whatever its assertions say.	Unverified
MR-42	**A tenant can choose how a missing read is estimated. Waits on: two open questions**, both registered — a rate plan holds no average to read from, and no document says which method a tenant may store.	Settings · 18	Both options render and both are disabled with their reason, which is `BR-SHL-025` working. **It is on the register because two open questions are not a ruling** — they are the absence of one, and §06's judgement rule admits only a ruling as grounds for out-of-scope. A `PUT` may store only a runnable method, so the guard exists; what is missing is the answer, not the guard.	Unverified
MR-43	**A person can see what a cycle DID collect, not only what it is short of.** Waits on: nothing — `f.cycle` is one of the six filters `GET /reads` already declares.	20	A fifth tab on the cycle page: the reads list, pre-filtered to this cycle, with the chip removable so one click widens it back to the module. **Three of the other four tabs say what is missing; this is the only one that says what arrived** — which is where a supervisor checks a reader's work rather than chasing them. The floor note names the window the readings belong to, because five readings under a cycle that collected nothing this window would otherwise read as progress.	Unverified
MR-44	**A route has a page, and it is measured against a window a person chose.** Waits on: nothing — `GET /routes/{id}`, `GET /routes/{id}/read-windows` and `f.route` are all served. PO ruling 2026-09-13, option A: *"this view should be similar to details view of Cycle"*, with its own address.	20 · 21	Five tabs at `/reads/routes/:id?window=…`: **Overview** (collection, the walk, the cycles it is on), **Stops** (`premises_order` — one row per premises, the ruled walk, exclusions kept), **Still to read** (the chase list with stop numbers), **All reads**, **History** (both cycles in one list). **A route can be walked on more than one cycle, so the header carries the window every figure is measured against** — without it *"0 of 11"* has no denominator. **The walk order is drawn read-only**: `PUT /routes/{id}/premises-order` is served, and whether Reads may call it or only Settings is not ruled. **One of these endpoints has been wrapped and uncalled since 2026-08-23.**	Unverified
MR-45	**A clerk looking at one reading can see the whole set its plan expects.** Waits on: nothing — the batch already stores the `plan_registers` outcome, with `expected`, `received` and `missing` resolved.	03	A line in **Calculation**: *Time of use · On-peak ✓ · Off-peak ✓ · Mid-peak missing · Total ✓ · 3 of 4*, the siblings linked. **Today this page says nothing at all about the other registers**, so a clerk holding one TOU window cannot see that a sibling never arrived — and the register set is what decides whether the meter can be billed.	Unverified
MR-46	**A demand reading is read as a maximum, never as a difference.** Waits on: nothing — `register_units.MAXIMUM_REGISTERS` already names which registers behave this way.	03	On a demand register the math line states the peak itself in **kW** and says the delta does not apply. **This is `DEF-797`'s defence on the screen where a person would catch it**: running `BR-MET-019`'s delta over two peaks billed **412 kW as 14 kWh — $154.00 against $4,532.00**. The unit is per register (`BR-MET-143`), so the screen has what it needs to tell them apart.	Unverified
MR-47	**Service and register sit with the reading's identity.** Waits on: nothing — both are on the row today. PO ruling 2026-09-12: *"on this screen can you show utility services chip in header"*.	03	The header carries **the service badge and the register** beside the status pill. **They decide how the number is read** — the unit, the plan's expected set and the rate all follow from them — and today the service is buried four groups	Unverified

			down in *Location* while the register appears only in the kicker. The *Location* group loses the service rather than showing it twice.	
MR-48	**A meter with years of history is read a page at a time.** Waits on: MR-01, whose fix is what makes the envelope's `count` reachable at all.	03	The history table gets the product's server pager. **The count is already in the payload** — it is the same envelope whose shape causes the crash — so this is the second half of one fix rather than new work. ⚠ **Its `count` is READ DATES and its rows are REGISTER ROWS**, so the pager must page dates and label rows, which is the trap MR-01 records.	Unverified
MR-49	**Wherever a shortfall is counted, the missing registers are named.** Waits on: nothing — the resolver already answers `missing` by name, and stage 3 already renders it.	09 · 10 · 20 · 21	**Blast radius: 4 screens.** *"tou_mid on MTR-0471 · demand on MTR-0488"* beside every count of what is short — on the wizard's Load stage, on the cycle page, on the route page. **This is the rate-type answer made visible:** a TOU meter short of one window and a flat meter short of its only register are the same number and completely different work, and only the register names tell them apart. Filter chips count by register so a clerk can clear one kind in one pass.	Unverified
MR-50	**A completeness figure states what its denominator is made of.** Waits on: nothing — the resolver computes the set it is summing.	20 · 21	*"0 of 18"* gains *"5 meters × their plan's registers"* under it. **Eighteen is not a number anybody can check without that line**, and the module has already shipped one screen where three different denominators sat on one row — meters, registers and meters again. Where the set is unresolvable the figure says so instead of printing a total nobody can reproduce.	Unverified
MR-51	**A meter that cannot be estimated says so before it is ticked.** Waits on: nothing — `BR-MET-026` and `BR-MET-027` already refuse these by name.	16 · 20 · 21	A row with **no actual reading on file** — nothing for the trailing average to average — and a **demand** register, which is a peak and is never estimated, are both marked on the chase list **before** selection. Today a clerk ticks five, presses *Estimate these*, and learns which two refused afterwards; the endpoint returns each refusal with its reason, so the information exists and arrives one step too late.	Unverified
MR-52	**A reading shows who cleared its dispute, and who filed it.** Waits on: nothing — `read_dict` already serves all nine fields.	03	**Measured: `read_dict` serves 41 fields, 32 reach the screen, 9 do not** — `dispute_cleared_at` · `dispute_cleared_by` · `imported_by` · `disposition` · `disposition_reason` · `event_context` · `trues_up_read_ids` · `read_type` · `period_end`. **The first two are the sharp ones:** entry 06 restricts clearing a dispute to a System Admin, writes who did it and when — and the reading then shows **no trace that it ever happened**. The audit fields join **Evidence**; the rest join **Flags** and the period line.	Unverified
MR-53	**The thirteen PO rulings of 12–13 September exist as text in the repository.** Waits on: **the PO lifting *"write it down in document and not in repo"* for the rulings** — the instruction was made about a code patch and took the rulings with it.	—	Not a screen. Each ruling enters `12-governance/02-decision-log.md` verbatim with its date, and the business rules it changes are stamped in the same commit. **Until then this document is the single point of failure for thirteen decisions**: they cannot be grepped, no lane building another module will meet them, and **nobody can count what depends on them**. Every August ruling quoted in this document is in the decision log; not one from this session is.	Unverified

53 commitments. Every one starts at Unverified, and that is not a hedge. Engineering fills the column in from the code, one row at a time, and several routinely come back already done — saying so on the register is what makes the column honest rather than decorative. **Nine of the 53 change no screen at all** (MR-18, MR-22, MR-23, MR-33 to MR-37) and are

Nine of the 53 change no screen at all (MR-18, MR-22, MR-23, MR-33 to MR-37) and are the ones most likely to prevent the next occurrence rather than this one. **Four of the 53 wait on a ruling and names its own four options rather than choosing** (MR-24, MR-38, MR-42 and MR-53) — the prime directive forbids resolving an open question by picking a value, and the freeze it runs into is `OQ-209`.

This table fills a gap rather than replacing one. This document had no such index before, because this document did not exist before. The repository's own indexes remain what they were — `12-governance/01-open-questions.md`, `10-defect-log.csv` and `11-change-requests.md` — and nothing here supersedes them.

10

What this pass changed, line by line

This is the first pass over Meter Reads, so there is no previous version to diff against. What follows is therefore a changelog of the **analysis** — what moved while it was being written, what was corrected after being wrong, and what was deliberately left as it is. It exists because a reader of the next version will need to know both.

The revision pass of 2026-09-12, and the four rulings that drove it

The PO read the as-built document and ruled. Each ruling is quoted at the entry it changes; this is the index of them, and of the one structural change the set forced.

The ruling, verbatim	What changed	Where
"Need Attention — does not add any value"	The landing tab and its three tiles are withdrawn. **All reads becomes the landing tab.** Flagged and Disputed become filters that count with the same predicate; Missing moves to the cycle, where it is stated against that cycle's expected register set. **The tenant-wide all-clear is lost** and is recorded as a cost rather than argued against.	Entries 01, 02
"All reads — all good"	Nineteen fields carry over unchanged. The tab strip, a widened search, a relabelled *Channel → Register* column and the batch number are the only movements.	Entry 02
"we should say Read Cycles and within that we should show Routes and their completion"	The two doorless addresses are replaced by **one Read cycles tab**. Routes stops being a screen; its nine columns move inside the cycle whose window measures them. **The all-routes view is lost**, recorded the same way.	Entries 14, 15
"we will need a list view of read cycles and we will need a detail view of read cycles"	**Entry 20 is new** — a cycle page carrying the routes, the shortfall, the schedule and the billing facts, with four actions and no cycle edit (that stays in Settings, `BR-MET-011`). Entry 16's chase list moves into it and gains a pager and two actions it never had.	Entries 16, 20

The structural change: the old section 08, *Drawn answers*, is **deleted**. Its three remedy

drawings are gone as a separate place, and every revised drawing now sits inside its own screen entry, beside the as-built it replaces — the PO's instruction, verbatim: *"Write all change in 02, 03 and 04 section only lets not create two places for revised screens."* Sections 09 to 12 are renumbered 08 to 11 and every cross-reference was moved with them. **A reader who knows a screen by number still finds it**: entry numbers were deliberately NOT resequenced, because sections 05 to 09 and the whole of appendix C cite screens by number, which is why the new cycle page is entry 20 and is read in its place rather than at the end.

Corrections applied during the pass, and what caused each

- **All five upload-wizard screens were redrawn.** The first draft named and drew them from `UPLOAD_STEP_IDS` — *Source & context · Upload / enter · Validation · Review & fix · Compliance & post*. **The running app draws Extract · Transform · Validate · Load · Result**, from the shared `WorkflowRunner`, and the panels do not map one to one: `validate` and `review` share stage 3 and stage 5 has no panel id at all. Found by clicking, by nothing else. The drawing rules say the running app wins and the disagreement is a finding; both halves were applied.
- **The pagination claim about Routes and Cycles was reversed.** The first draft repeated `RoutesCyclesPage.tsx`'s own trailer — *"unpaginated plain arrays"*. Measured: `GET /routes` answers `{"count":6,"next":null,…}`. The endpoints have paginated since 2026-08-23 and `api/endpoints.ts` says so; two other files still say the opposite. The finding changed from *"the server cannot"* to the sharper *"the server can, a client helper for it exists, and these two screens use neither"*.
- **`READ_CHANNELS` was cited to the wrong file.** Written as *"a literal array in the wizard"*; it is in `api/endpoints.ts:11657`. Corrected in place.
- **The page title was drawn at 32px, against the drawing rules this document was written under**, which pin it at 40px from an earlier measurement. B14 **rule 40** was ratified on 2026-09-12 — the day of this read — and `getComputedStyle` on the running app returned `32px / 800 / Archivo`. The rule is newer than the reference; the app agrees with the rule; the drawings follow both.
- **The column selector was very nearly reported missing.** A first probe searched for a `<button>` and found none. The control is a `<summary>` inside a `<details>`; opening it shows show/hide, ↑↓ reorder, *Reset table* and all three hidden columns. **The check was wrong, not the product**, and it is recorded here because the next person will write the same selector.
- **The first walkthrough never signed in** and produced a page of *"We couldn't load your regional settings"* — which would have read as a product defect. The sign-in email field is `type="text"` with id `signin-email`, and the selector looked for `input[type="email"]`. Every finding in section 05 comes from the second run.

Two more faults in this document, both found by a reader looking at it

> **The drawings were not in the product's typeface**
>
> Every drawing declared `--font-body:"Archivo"` and **the font was never loaded**, so all twenty-one rendered in `system-ui`. The product loads it at `tokens.css` line 22 — `fonts.googleapis.com/css2?family=Archivo:wght@400;600;800` — and this document simply copied the token and not the `@import`.
>
> **It is the same defect this repository already records against the product itself.** `tokens.css` opens with the words *"THE FONT WAS NEVER LOADED"*, filed after the PO reported on 2026-08-10 that *"font in Prototype and font in application is different"*. The document auditing that defect reproduced it. **Fixed:** the same family, at the same three weights, linked the way the product links it — and the four places a drawing used weight 700, which the product does not load, now use 600.

> **The top bar was wrong on all nine shells, and it omitted a real control**
>
> It was drawn as `[screen name] … [user · role]`. **The product's top bar is breadcrumb · global search · help ? · notification bell** (`AppShell.tsx:832–1007`), and the **identity is in the SIDEBAR FOOTER** — an initials avatar with name and current-tenant role, opening the S48 menu upward (`BR-SHL-008`, `AppShell.tsx:804`).
>
> So two things were wrong at once: a real control was in the wrong half of the screen, and **the product's only global search was missing from every drawing** — a bar the PO widened by ruling (*"if you have to increase the length of search bar please increase it"*) because it searches nine groups. Both corrected on all nine. **Found by a reader comparing this document with the Billing blueprint**, which draws the same shell — which is the argument for the two being drawn from one component rather than from two readings of it.

Does the revision close the findings? — sections 05 to 08, one row each

Asked by the PO before sign-off, and it is the right question: a drawing can close a finding about a **screen** and can never close one about **code**. The honest answer is **18 of 32 closed, 5 partly, 9 that no wireframe could ever close** — and the nine are named rather than implied.

#	The finding	Closed?	By what, or what is still needed
Section 05 · what broke when the product was driven			
J-01	The reading detail **throws on every reading**	OPEN	**No drawing can close this.** `getAssetReads()` is typed as an array over an endpoint answering a paged envelope — one line of TypeScript. It is the module's worst defect and the revision does not touch it
J-02	Routes and Cycles have **no door**	CLOSED	The Read cycles tab (15) and the cycle page it opens (20). This is the ruling's whole point
J-03	Commit live while the certification is unticked	CLOSED	Entry 10 — disabled until the tick, reason underneath, consequence above
J-04	Route progress: a Collected column counting readings beside a Meters column counting meters	CLOSED	Entry 20's Routes section — one denominator per row, scoped to the cycle whose window measures it
J-05	*"When was this route last walked?"* — a clock in a date column, a day early	PARTLY	The drawing prints a date and no clock. **The timezone conversion is code**, and it is `7-10`
J-06	*"Can I bill CY-001?"* — three denominators and a countdown reading 0 nine days after the window shut	CLOSED	Entry 15 — one denominator, and a Window pill that can say *Closed 9 days ago.* ⚠ **the two derivations of "the current window" still disagree** (3.14), so the figure is right and its input is not
J-07	*"1 rows will post"* · *"Commit 1 rows ✓"*	CLOSED	Entries 09 and 10 — *"1 row will post. 1 will not."* and a button that carries no count
J-08	Session · IP came back empty on a real post	PARTLY	The row now reads **"Not captured"** rather than an em dash, so the absence is visible. **Capturing it is backend work**
J-09	The chase panel did not open under the driver	PARTLY	The failing shape — a panel behind a table cell on a doorless screen — is gone; it is a section of entry 20. **Whether it renders is still untested**
J-10	The receipt was never rendered	PARTLY	Entry 11 is drawn, **and labelled as drawn from code**. The fix is a test that can tick the certification
Section 07 · where the screens depart from the rules			
7-01	The Batch column renders a raw uuid	CLOSED	Entry 02 prints `RB-2608-0044`, the human-facing number rule 31 asks for
7-02	Commit enabled while unticked	CLOSED	= J-03
7-03	No toolbar at all on either screen	CLOSED	Entry 15 carries all four, and the pager
7-04	Both page in the browser over a fetch-everything read	CLOSED	Server pagers drawn on 15 and on both of 20's tables. **The endpoints already paginate and the client helpers already exist, unused** — so this is wiring, not

			building
7-05	A raw `<select>` in the manual grid	CLOSED	Entry 08 drawn to the kit
7-06	A raw `<input type="date">` in the same grid	CLOSED	Entry 08
7-07	No long-list select is type-to-filter	CLOSED	Entries 07, 17 and 18 all draw the search affordance; `BR-SHL-033`'s threshold is stated in section 01
7-08	25 words of rationale on the estimate dialog	CLOSED	Cut to zero
7-09	No singular form, on the button that writes	CLOSED	= J-07
7-10	A clock in a date column, a day early	PARTLY	The drawing shows a date. **The conversion is code**
7-11	An empty state that names what is missing and no action	PARTLY	Entry 15's States row carries *"Add a cycle in Settings →"*. **Stated in the spec, not drawn** — empty states have no figure in this document
7-12	The chase list has no row action of any kind	CLOSED	Entry 20 — the meter is a link, the rows tick, and two actions sit above them
7-13	A primary button refusing over a **built** endpoint, with a false sentence	OPEN	**No drawing closes this.** `POST /reads/{id}/billed-correction` is served, S30 exists, the front end has never called it. It is wire-it or change the sentence, and both are decisions
7-14	*"the reads module is not wired yet"* over an unreachable branch	OPEN	An expired hold in a message catalogue
7-15	A file trailer false in five clauses	OPEN	A code comment. **4.8**
7-16	Two files say the endpoints do not paginate; a third says they do	OPEN	Code comments. **4.9**
7-17	Two live bounds argued from the retired 3,000 ceiling	CLOSED	Both get a pager in the drawings, so neither depends on that bound any more. **The comments still name it**
7-18	`readsShared.tsx` listed twice in the migration file	OPEN	A repository file. No screen
Sections 06 and 08 · absences and seams			
06	**Two expired holds** — a refusal that outlived the thing it was waiting for	OPEN	The same two as 7-13 and 7-14. This is hard-won law 2's mirror, and the cure named there is **a test that fires when the hold expires**, not a drawing
06	The other 179 stated absences	CLOSED	Each already carries its one decision. The revision does not disturb them
08	**G96 reports a clean zero over a file it cannot see**	OPEN	A gate matching import aliases against a hardcoded list of eleven. **4.10** — and it is the most dangerous item on this page, because it guards tenant scoping
08	Four places read metering's tables directly instead of through the seam	OPEN	Reads only, nothing writes. **4.11**

What this means before anybody builds from these drawings

The drawings close every finding that is about a screen. Eighteen of the thirty-two, including both doorless screens, every toolbar and pager gap, every rule-33 breach and both hand-rolled controls.

They close none of the nine that are about code, and two of those are worse than anything the revision fixes: J-01, which means the reading detail cannot be opened at all today, and **the G96 blind spot**, which is a tenant-scoping guard reporting a truthful zero over a file it never looked at. **Building the revised screens would leave both exactly where they are**, and a

reader who took this document as a work order would ship a prettier product with the same crash in it.

And three of the five PARTLY rows share one cause worth naming: the drawing is right and its input is not. **3.14** — two derivations of *"the current window"* disagree — feeds **J-05**, **J-06** and **7-10**. Draw them, build them, and the screens will disagree with each other in a new and tidier typeface. It is a filed backend question, not a design one.

⚠ The nine a wireframe cannot fix — what each one actually needs

PO instruction, 2026-09-12: *"fix J-01 — that crash is the priority"* and *"highlight these as important items"*. Each of the nine below is specified here to the point where it can be applied without re-deriving anything. **They are listed in the order they should be done**, and the first is worth more than the whole revision.

1 · J-01 — the reading detail throws, and it takes THREE other screens with it

First, a correction to this document's own finding. Section 05 called it *"one line of TypeScript"*. **That is wrong, and the reason it is wrong is the whole defect.**

What is broken. `GET /assets/{id}/reads` answers `paginator.get_paginated_response(…)` — `{count, next, previous, results}` (`apps/assets/api.py:282`). `getAssetReads()` declares `Promise<AssetReadRow[]>` and returns `requireData(...)`, which hands back the whole envelope typed as an array (`api/endpoints.ts:8674–8681`). Every caller then does `.map` on an object. **`tsc` passes**, because the lie is in the type argument, and **every unit test passes**, because each supplies its own array fixture — the fixture is right and the product is wrong.

It is not one screen. It is four, all broken identically, and only one of them was walked:

Caller	What the user sees
`ReadDetailPage.tsx:323`	**Walked.** The whole page is replaced by `TypeError: t.history.map is not a function`
`AssetDetailPage.tsx:1526`	The asset's own Reads tab
`AssetDetailPage.tsx:1588`	The linked meter's reads on an exporting inverter
`accountDetailTabs.tsx:163`	An account's meter reads

Why the obvious fix is also wrong, and this is the part that matters. The module already has a helper for exactly this shape — `listAll()` — and it terminates on `rows.length >= body.count`. **On this endpoint that comparison is between two different units.** The server says so in its own docstring: *"THE PAGE IS FIFTY READ DATES, NOT FIFTY ROWS … a three-register meter serves three rows for one date, and `count` here is the number of READ DATES."* So on a TOU meter `listAll` reaches `count` after about seventeen dates of a fifty-date page and **returns early, silently, with two thirds of the history missing** — a crash traded for a quiet wrong answer, which is worse.

The fix, and its termination condition is the fix: follow `next`, never count rows against `count`.

```
export async function getAssetReads(
  assetId: string,
  client: ApiClient = api,
): Promise<AssetReadRow[]> {
  // ⚠ NOT `listAll` — it stops on `rows.length >= count`, and this endpoint's
  // `count` is READ DATES while `results` is REGISTER ROWS (assets/api.py:271).
  // A three-register meter would truncate at a third of its history, silently.
  const rows: AssetReadRow[] = [];
  for (let page = 1; ; page += 1) {
    const body = requireData(
      await client.get<Paginated<AssetReadRow>>(
```

```
        `/assets/${encodeURIComponent(assetId)}/reads`,
        { query: { page, per: LIST_ALL_PER } },
      ),
    );
    rows.push(...body.results);
    if (body.next === null || body.results.length === 0) return rows;
  }
}
```

Blast radius: four call sites, and **none of them changes** — all four already expect an array and all four already get one afterwards. `Paginated` is already imported in this file. Nothing else calls the endpoint.

How to prove it, and a green suite is not the proof. Every existing test passes today, over the crash. The regression that catches this is a fixture **in the served shape**: a client stub answering `{count, next, previous, results}` rather than a bare array — and a second case with **count 2 and four rows**, which is what a two-date three-register meter looks like and what `listAll` would truncate. **Then open one reading in a browser**, because that is the only instrument that saw this at all (hard-won law 6).

This is a beta defect, not a change — the product does not do what it is expected to — so it is filed with `file_defect.py`, fixed, version-bumped on the fix digit, and closed by telling the reporter which build to look in.

⚠ **THE PATCH IS WRITTEN DOWN HERE AND HAS DELIBERATELY NOT BEEN APPLIED.** PO instruction, 2026-09-12: *"write it down in document and not in repo"*. This document is the deliverable; the branch is untouched. Whoever applies it gets the patch, the trap, the four call sites and the two fixtures without re-deriving any of it — which is the point of writing it down rather than landing it quietly.

2 · G96 — a tenant-scoping guard reporting a truthful zero

Why this is second and not fifth: every other item is a wrong pixel or a stale sentence. This one is **a gate that guards CLAUDE.md's hard rule 1** — never touch a tenant table without the scope set — and it is reporting **0 unresolved** over a file it has never looked at.

The mechanism: it matches import *aliases* against a hardcoded list of eleven. `apps/portal/seams/metering_seam.py` imports as `reads_models`, which is not on the list, so `MeterRead.objects.filter(id__in=wanted)` — **no tenant predicate** — is invisible to it. Verified by running the gate's own `findings()` against that file.

The fix is not "add the twelfth alias." That is the same defect with a longer list, and the next seam file re-opens it. **Resolve the module by what the import RESOLVES TO**, not by the name it was bound to. And per hard-won law 7's fifth shape, **a check that cannot cover its whole subject must print what it skipped beside what it checked** — a declared blind spot with no number on it is indistinguishable from an empty one, which is exactly how this survived.

3–9 · The rest, each with what it needs

#	What it is	What it needs
7-13	The primary button on a billed reading refuses with *"that path is not built"* — over `POST /reads/{id}/billed-correction`, which is **served**, with S30 built beside it and the server's own copy telling clerks to use it	**A decision, then one of two edits.** Wire the front end to the endpoint, or change the sentence to say what is actually true. **What must not happen is a third year of a button that lies.** This is hard-won law 2's mirror — a live screen behind a dead control — and the cure named there is **a test that fires when the hold expires**, written in the same commit as the hold
7-14	`metering.ops.notCollectedWhy` — *"the reads module is not wired yet"*, over a branch that cannot be reached	Delete the key and the branch. A message nothing renders is **the next lane's false lead** — the same reason `shell.search.unavailable` was retired from all eleven catalogues the day its branch went
7-15	`ReadsListPage.tsx`'s trailer is false in **five clauses** — it still says the export does not exist	Rewrite it. **Hard-won law 1:** a comment claiming a guarantee is not the guarantee — and

	and the detail is a drawer	a maintainer reads the trailer first
7-16	Two files say `/routes` and `/cycles` do not paginate. A third says they do, and dates it. **The third is right**	Delete the two false claims. This one already cost this document a wrong finding, which was corrected only by measuring the endpoint
7-18	`readsShared.tsx` listed twice in `route-b-migrated.txt`	Delete the duplicate line. Harmless today; it is on the list because a file that appears twice in a build manifest is a question somebody will have to answer later
06	Two **expired holds** — refusals that outlived the thing they were waiting for (the same two as 7-13 and 7-14)	Covered above. The general cure is the one law 2 states: **when you write a hold, write the test that fires when it expires**, and when you land a screen, grep for the control that was holding it
08	Four places read metering's tables directly instead of through the seam. **Reads only; nothing writes**	Route them through the seam file. Lower urgency precisely because nothing writes — but it is how G96's blind spot became reachable in the first place

The commitment audit of 2026-09-13 — and the family it found undrawn on the register

Fifty-five components are proposed in the drawings. Seven had no commitment behind them, and they were the ones the PO asked for by name.

The audit ran the whole document rather than the headings: **every legend item in every drawing is a build component** — a column, a filter, a pager, a line of derived text — and each one either has a commitment card or is a finding. Fifty-five are proposed across six entries.

The gap was not random. It was the rate-type work — drawn on PO instruction (*"do all these changes"*), applied to five screens, and **never written down as anything anybody had undertaken**:

Minted	What had been drawn and not undertaken
MR-45	**The register set a reading's plan expects**, ticked off with its siblings linked. The reading page says nothing about the other registers today
MR-46	**A demand reading read as a maximum, never a delta** — `DEF-797`'s defence on the one screen a person would catch it: **412 kW billed as 14 kWh, $154.00 against $4,532.00**
MR-47	**Service and register in the reading's header**, where they belong because they decide how the number is read
MR-48	**A pager on the reading's history** — the second half of MR-01's fix, and it inherits MR-01's trap: the envelope's count is READ DATES, not rows
MR-49	**The missing registers NAMED wherever a shortfall is counted** — four screens. A TOU meter short of one window and a flat meter short of its only register are the same number and completely different work
MR-50	**A denominator that states what it is made of** — *"5 meters × their plan's registers"*. This module has already shipped a row carrying three different denominators
MR-51	**A meter that cannot be estimated marked BEFORE it is ticked** — today a clerk ticks five, presses Estimate, and learns which two refused afterwards

Why a heading-level check could not have found these. Every one of those seven sits on a screen that already carried commitments — entry 03 had five, entry 20 had four — so the headings were consistent, the register was consistent, and the two agreed with each other **about the wrong set of components**. **The check that finds it reads the LEGENDS, not the chips:** a drawing that proposes a column nobody has undertaken to build looks exactly like a drawing that does not, until you list what each one is asking for.

51 commitments. Eight change no screen; three wait on a ruling.

A route gets a page — six new screens

PO ruling 2026-09-13, and the design question it forced

Verbatim: *"in this Route Tab — what if User click on Route link — we will have to give a details view of Route. this view should be similar to details view of Cycle. in Cycle you will need to add a tab — all reads and same in route details also."*

Six drawings. One new entry. Entry 20 goes from four tabs to five with **All reads** (MR-43); **entry 21** is the route page and carries five — Overview · Stops · Still to read · All reads · History (MR-44).

The question the ruling forced, and the PO answered it: option A. A route can be walked on **more than one cycle** — the drawn one is on two — so *"0 of 11 collected"* has no denominator until a window is named. The route page therefore has **its own address**, `/reads/routes/:id?window=2026-09`, with the window in the header defaulting to the cycle you arrived from. The alternative, nesting it under the cycle, would have left a route with no address anybody could paste — **which is exactly how the old Routes screen became an orphan**.

Nothing new is served for any of the six. `GET /routes/{id}` answers the membership and the ruled walk; `f.cycle` and `f.route` are two of the six filters the reads list already declares; and the History tab runs on `GET /routes/{id}/read-windows` — **wrapped on 2026-08-23 and called by nothing since**. Entry 21 is the only new screen in this document with **nothing unserved behind it**.

And one thing it deliberately does not do. The Stops tab shows the ruled walk order, and `PUT /routes/{id}/premises-order` is served and writes it. **Whether a clerk may reorder a walk from Reads, or only from Settings, is not ruled** — `BR-MET-011` puts route configuration in Settings and `SET-06D` numbers these stops there. Drawn read-only, with the question named on the screen rather than answered by a drawing.

Section 06 re-decided, and one drawing was clipping its own primary action

Three absences were ruled out of scope on evidence that is not a ruling

The judgement rule in §06 is one sentence: **a recorded ruling that a thing waits makes it out of scope; no ruling makes it a specification.** Thirteen rows read OUT OF SCOPE and **six cited no ruling** — a code comment, a design note, or an open question.

Three are now specifications. MR-25 (a second register on the quick-entry form, out of scope on *"a deliberate split"* — the code's own note), MR-39 (the account on a reading, out of scope on *"a refusal to guess"* — **a reason that does not apply**, since the column is a stored foreign key), and MR-42 (two estimation methods, out of scope on *"open questions, both registered"* — **an open question is the absence of a ruling**, which is the exact condition that makes something a specification).

Why it matters more than three rows. A comment in the code and a ruling in the decision log read the same on the page and carry completely different weight. **Treating the first as the second quietly converts our own design choices into the customer's requirements** — the fault CLAUDE.md's change process guards with *"when it is ambiguous it is a DEFECT, not a change"*.

And the Plan column pushed entry 02's drawing 64px past its own window

Reported by the PO: the right-hand side was cut. **The primary action, the CSV download, the Status column and the pager were all clipped** — fifteen columns do not fit 1440.

The check could not see it, and the reason is the same one twice. It compared the drawing's own box with its container — and the box is fixed at 1440, so it stays inside while its CONTENTS spill. **Measure the widest DESCENDANT, not the frame**; the check does now, and it found exactly one offender in twenty-one drawings.

Fixed by drawing what a clerk sees: twelve visible columns, with the ▦ picker drawn **open** beside them — *Register*, *Batch*, *Plan* and *Days since prior* are `defaultVisible: false`, and all three of this screen's changes live inside that picker. It is a better drawing than the one that broke.

The commitment sweep — every change this document asks for, registered

From 23 commitments to 41, and the last eighteen were the ones nobody had undertaken

PO instruction: *"commitment chip — not just on screens but all changes that we need to ask Claude Code to do"*, then *"check all screens once more for missed commitments"*. Three sweeps, each finding a different kind of gap.

Sweep	What it found
Screens	**Five revisions had no card at all** — entries 01, 02, 14, 15 and the whole cycle page. `MR-26`–`MR-30`. And **13 existing commitments appeared on no heading** while two headings carried a commitment belonging to another screen; entry 15 showed one where **nine** apply.
Non-screen	Seven pieces of work described in full and undertaken by nobody: the copy budget across eleven screens, the Panel ruling, three false comments, a duplicated line in a build manifest, four seam reads, and **the window disagreement three drawn figures compute from**. `MR-31`–`MR-38`.
Re-sweep	Three more, found by reading every entry for a described absence rather than for a heading: **the account link the database already holds and `read_dict` declines to serve** (`MR-39`); **a staged batch served whole with a hardcoded 50,000-row cap**, against both `NFR-3` and `NFR-5` (`MR-40`); and **the test that cannot reach stage 5**, which is why one drawing in this document carries a disclaimer no other needs (`MR-41`).

The structural fix matters more than the count. Every state chip and every commitment chip is now **derived from the register at assembly** rather than typed on a heading — the commitments that name a screen decide its states. **A chip and a card can no longer disagree, because there is only one of them.**

And the shape of the answer changed. Before the sweep, five screens said *No commitment*. **Now one does** — entry 19, a second address rather than a screen. The four others were not uncovered; their work was described and simply never written down as an undertaking. **Seven screens now carry** BLOCKED, which is the honest number: four commitments wait on a question nobody has answered, and between them they touch a third of the module.

The chip vocabulary drifted, and nothing in the document could see it

Nine chips that were not in the vocabulary, and five revisions with no commitment behind them

The rule is two words. **A description** says what the code does today, including an honest statement that something is missing, and commits nobody. **A commitment** is a change specification with a named piece of work, what it waits on, the screens it changes, and what a person could do once it is done. **Four state chips carry it and only four** — Exists behind · New read · New build · Blocked — with **No commitment** where no specification covers a screen, because that absence is itself the finding.

The revision pass grew nine chips that are in neither list: *Revised*, *New screen*, *Drawn from code*, *Not walked*, *Throws on open*, *Retired*, *Retired as a screen*, *Behind a doorless screen*, *Second address*. Every one was true. **Every one was a DESCRIPTION wearing a commitment chip's clothes** — which is exactly the confusion the two words exist to prevent, and it went unnoticed for nine versions because **nothing in this document validated its own vocabulary.**

Worse, five screens carried a revision with no commitment card behind it — entries 01, 02, 14, 15 and 20 — and by the register's own opening sentence that means *nobody had undertaken to build them*. The drawings were there; the undertaking was not.

Both closed. Every heading is back to the four states, each displaced fact moved into the entry's own text where a description belongs, and `MR-26` to `MR-30` give the five revisions their cards. **30 commitments, and every chip on every heading now has a row behind it** — checked mechanically at assembly rather than by reading, which is what would have caught this on the first version.

The finding tables were unreadable, and the cause was arithmetic

Seven columns of prose in an 890px measure

Reported by the PO looking at **J-06**: one finding running **twenty lines** down a column about 154px wide. The cause is not the writing — it is `table-layout:fixed` with seven columns, five of them unsized, splitting one text measure between them. Every long finding was a ribbon.

Both finding tables are restructured, and no word changed. Each finding is now **two rows**: a compact metadata line — id, screen, where, severity — and a full-width row beneath it carrying the prose, each part introduced by its own inline label. **Measured: J-06 went from roughly 800px tall to 171px**, and the worst row in the rule table from 321 to under 200. Same facts, a third of the height, and the severity is now readable beside the id instead of at the end of a long scroll.

Entry 20's other three tabs, drawn

PO instruction: *"please create screens for Routes 2, Still to read 5, History — I need to see what the details in it"*. Each tab is a separate address under `BR-SHL-011`, so each is a separate screen and gets its own drawing. **Four drawings for entry 20, and no duplication** — the header is identical on all four and only the body changes.

Tab	What it shows, and what is new in it
Routes	Entry 14's retired table, scoped and with a toolbar it never had. **The column that made it unreadable is gone** — no *Cycle* column, because the cycle is the page — and *Collected* splits into two honest numbers: **registers collected** and **meters short**. Row actions open the route, print its sheet, or reassign it
Still to read	Entry 16's panel, actionable. **Filter chips count by REGISTER** — *Total register (3) · On/off-peak (1) · Demand (1)* — which is what lets a clerk work one rate type at a time, and the product's bulk bar drives `POST /reads/bulk-estimate`, served since CR-038 and called by nothing. **A meter with no actual reading says so before it is ticked**, because BR-MET-026 refuses it by name and finding that out after the press is a wasted round
History	Twelve windows of this cycle, one row each — and **the only tab with no endpoint behind it**, drawn with the absence stated as a warning Panel rather than hidden. **Its Estimated column is why it earns its place:** a cycle reading 100% on estimates is not a cycle read by a person, and no other screen in the module puts those two numbers side by side

The Panel ruling — every information group on every detail screen

> **PO ruling 2026-09-12, pointing at the bill detail**
>
> Verbatim: *"use Panel — a box, 1px border on all four sides, surface fill, 16px padding, heading inside … in detail section use Panel to information groups"*. **Ten groups changed container**: the read detail's five (Calculation · Flags · History · Location · Evidence) and the upload stamp's five (Evidence · Composition · Coverage · Co-sign · Certify). Entry 20 and entry 11 were already drawn that way.
>
> **This is a change to the product, not a correction of a drawing, and it is recorded as one.** The read detail today renders the OTHER container — `ReadDetailPage.tsx:139`, `"border-t border-divider pt-3.5"`: a rule across the top, no border at the sides, no fill. **Both are real components and the PO chose between them.** The specimen in section 01 draws the two side by side so nobody has to guess which is which when comparing a drawing with the running app.
>
> **The heading follows the ruling too.** `Panel.tsx` declares its heading at 16px (`text-h5`); the screen the ruling points at shows the red small-caps label. Where the component and the named screen disagree, the drawing follows what the PO can see.

The screen-by-screen pass of 2026-09-12 — 51 changes across 14 screens

The PO read every entry and ruled on it. This is the ledger.

#	Screen	Changes	What changed
01	Needs attention	1	Retired. The tab and its three tiles are withdrawn

01	Needs attention	1	Retired. The tab and its three tiles are withdrawn
02	All reads	6	Landing tab · widened search · *Channel → Register* · batch number · one rationale line cut · **Plan column**, hidden by default (R3)
03	One reading	3	Two-column information architecture with named groups · **the register set this plan expects, ticked off** (R1) · **a demand register reads as a maximum, never a delta** (R4)
04	Correct the reading	3	Two help lines cut to the cap · **MR-24**: the register of a filed reading can never be changed, and the database enforces it
05	Dispute	2	A 10-word rationale becomes a 5-word fact · what a dispute does and does not fix
06	Clear the dispute	1	9 words → 5
07	Extract	3	**~190 words of copy → ~50** · pickers type-to-filter · the Route picker scoped to the chosen cycle
08	Transform	2	Copy cut · the hand-rolled date box and register select drawn as the kit renders them
09	Validate	2	Four headings and three sentences cut · the missing registers already named per plan model
10	Load	6	**Commit disabled until the tick**, reason underneath, consequence above · the count off the button label · *Session · IP* reads **"Not captured"**, never blank · **Evidence** group · 62 words cut · short registers named
11	Result	1	**Drawn** — from the source, and labelled as such. It is the one drawing here that is a picture of the code rather than of the product
12	Add a single reading	4	Two help lines cut · the off-cycle strip 24 words → 12 · registers by **name**, not code · **MR-25**: the modal grows the rows it has already fetched instead of handing off to a five-stage wizard
13	Accept flags in bulk	0	Its copy already passes
14	Routes	2	Retired as a screen · its drawing dropped, its columns kept inside entry 20
15	Read cycles list	2	The list the ruling asked for · one rationale line cut
16	Still to read	1	Folded into entry 20, with a pager and two actions it never had
17	Route sheet	3	15 words → 10 · type-to-filter picker · **confirmed fully register-aware** — one printed line per expected register, with the right unit on each
18	Estimate	4	**25 words of rationale → 0** · the Register list offers this meter's missing set, not all seven · type-to-filter · **the bulk endpoint is served and has no door**, and its home is entry 20
19	Retired address	0	A second address, not a screen
20	A cycle — the detail	5	**New screen**, then redrawn to the shared `PageHeader` · the denominator states its basis (R5) · short registers by name · the bulk-estimate control the code deferred to this surface
Total		**51**	All applied.

A defect in this document, found by measuring the thing instead of the proxy

> **Every wireframe was clipped by 550px, and the acceptance check said zero**
>
> The drawings are one 1440 desktop window meant to be scaled down as a unit. The scale factor was `var(--pw-s,1)` — declared with a fallback and **set by nobody** — so each drawing rendered at its full 1440 inside an **890px** column whose container said `overflow:hidden`. **The right-hand third of every one of the 21 drawings was cut off and unreachable**, including a table's last three columns and the whole right-hand side of the read detail.

The acceptance check reported `scrollWidth − clientWidth === 0` at every width, and was telling the truth. `overflow:hidden` guarantees that answer whether the drawing fits or not — the check measured the page and the fault was inside a box. That is CLAUDE.md law 7's *structurally incapable of failing*, in this document rather than in the product it audits.

Fixed with a container query and `zoom` rather than `transform:scale`: a transform leaves the pre-scale box in the layout, so the figure would keep a 1440-wide hole and be clipped again. Measured after: the drawing's right edge and its container's right edge are the same pixel at 1920, 1440 and 1280. **The check now compares those two edges**, not the page's.

What the construction and copy pass changed

- **Every picker and date in the module now states where its options come from**, in one table in section 01 rather than repeated per screen. It is what turned up two things worth knowing: the Route picker on the upload wizard is **not narrowed by the cycle chosen above it**, and the register list is a client constant mirroring the server's enum, so a new register is a string in two files.
- **Spacing, label order, two-fields-per-row, dropdown and date-field rules are stated once** and every drawing is held to them — B14 rule 38, the UI rules win over a drawing on how a form is BUILT.
- **Every helper line in the module was measured against rule 33's caps** — 17 lines, 202 words, 5 passing. The two cuts that fall on a revised drawing were **made**; the other fifteen carry their replacement string.

Structural faults reported and deliberately not repaired

- **Entry 11 has no wireframe.** The receipt was never rendered, because the certification control could not be ticked from the driver. Drawing it from `Receipt.tsx` would be a picture of the source rather than of the product. **Seventeen of twenty entries are drawn, and each of the three that is not says why in its own entry.**
- **Entry 19 is a second address, not a screen**, and is numbered anyway so that nobody later finds a URL nothing in this document describes.
- **This document itself scrolls sideways by 12px at exactly 400px wide**, and that is recorded rather than hidden. R10's acceptance test asks for `scrollWidth − clientWidth === 0` from 1920 down to 400; measured in Chromium at eight widths it is **0 at 1920, 1600, 1440, 1280, 1024, 820 and 600, and 12 at 400**. It began at 1,305px — a grid item defaults to `min-width:auto`, so `minmax(0,1fr)` constrains the track and not the item, and one wide table made the whole page scroll. Four fixes closed it: `min-width:0` on `main` and `section`, wrapping the verdict pills (one of them is 316px on a single line), `overflow-wrap:anywhere` on `code`, and less side padding below 520px. **The last 12px was not chased**, on a desktop-only document, and saying so is better than a claim of zero that a reader could check.
- **The three processes were audited in parallel rather than depth-first.** The process prescribes closing each before opening the next, precisely because breadth-first produces a thin last group. That was not followed here: steps 1 to 4 were run across all three together. The counts are recomputed from the markup rather than accumulated by hand, which is the mitigation — but it is a mitigation, not compliance, and a reader should weigh the third group accordingly.

Figures added, and what they are

Every worked number in every drawing is listed in appendix B, verbatim and unrounded. **They are test cases, not served values**, and B14 rule 37 is explicit about it: *“Treat every number drawn in a wireframe as an illustration, never a fact to build to.”* A figure taken from the walk is marked as such and is a reading of the seeded demo tenant at one moment — not a signed figure, and none of them is.

Counts checked against what they count

The masthead's screen, drawing, door, field and violation counts were **recomputed from this document's own markup** at assembly and again at the end, not carried forward from the draft. The recomputation is what catches an entry added late that never made it into a total.

And nothing was deleted

No finding was removed once written. Where one turned out to be wrong it was **corrected in place and the correction says what it replaced** — the wizard names, the pagination claim, the citation, the column selector. A deleted row loses the history and invites the same finding again next quarter.

11

Appendix

Eleven tables, each built from a section-by-section walk of this document rather than from memory, and each exhaustive over the document.

I · Every field, and where its data comes from

The union of every screen's field table, **sorted by source rather than by screen** — the point is to put every consumer of one column on adjacent rows, because the question this table gets asked after delivery is *“if I change this column, what breaks?”*, and a per-screen layout cannot answer it.

100 of 121 fields are plainly SERVED. 21 are qualified or broken. 0 were never traced. An ungraded field is not a healthy one, and a table that quietly grades untraced fields as SERVED is decoration. **28 rows cross a module boundary** and carry ⇄.

Screen	Field	Comes from	Owned by	Health
07 · Upload reads · stage 1 · Extract	Reading month	*a native `<input type="date">`; truncated to the first of the month server-side*	metering	SERVED, BROWSER-FORMATTED
03 · One reading	Kicker — meter · channel	code>read_dict.meter + `.channel`	metering ⇄ assets	SERVED
15 · Cycles	Bill date · due date	code>cycles.bill_date, `.due_date`	metering ⇄ billing	SERVED
14 · Routes	Route code · name	code>metering.routes.code / `.name`	metering	SERVED
02 · Reads — All reads	Search box	posts `?q=`; matched in SQL on the meter number only	metering	SERVED, NARROWER THAN r10 DRAWS
08 · Upload reads · stage 2 · Transform	Manual grid — Register options	code>READ_CHANNELS, *a literal array at `api/endpoints.ts:11657`*	—	HARDCODED
02 · Reads — All reads	Select checkbox	*nothing — page state; `bulkSelectable(row)` decides enabled*	—	SERVED
10 · Upload reads · stage	Certify checkbox	posted as `certified`; stored as `certified_by_uploader`	metering	SERVED

4 · Load				
10 · Upload reads · stage 4 · Load	Source · file name	code>source_kind + `compliance.file_name`	metering	SERVED
18 · Estimate a missing read	Method · consecutive number	code>estimation_method, `consecutive_estimate_no`	metering	SERVED
12 · Add a single reading	Reading, with its unit in the label	*typed*; the unit is `context.meter.unit`	metering	SERVED
03 · One reading	Previous / Current	code>prior_value_display / `current_value_display`, composed with the unit server-side	metering	SERVED
03 · One reading	Usage · days	code>usage_display, `days_since_prior`	metering	SERVED
01 · Reads — Needs attention	The two drill links	*nothing — they write `f.status=flagged` / `f.disputed=true` into this screen's own URL*	—	SERVED
02 · Reads — All reads	Route / Cycle filter options	code>GET /routes, `GET /cycles` through `listAll()` — every page, fetched whole	metering	SERVED, UNBOUNDED
10 · Upload reads · stage 4 · Load	Session · IP	code>session_ref, `ip` — **both empty on the walked post**	metering	SERVED, EMPTY IN PRACTICE
07 · Upload reads · stage 1 · Extract	Cycle — options	code>GET /cycles through `listAll()`, which walks **every page** at 1000 a time	metering	SERVED, FETCH-EVERYTHING
03 · One reading	Premise · premise no · unit	walked to `premises`; links to the premises detail	metering	SERVED
16 · Meters still to read	Premises	walked to `premises`	metering	SERVED
03 · One reading	Math line	assembled from `prior_value`, `current_value`, `multiplier` and the server's own `usage_display`	metering	SERVED
02 · Reads — All reads	Batch hidden	code>meter_reads.import_batch_id — **the raw uuid**. `read_dict` serves the id and not `batch_no`.	metering	SERVED, BUT RULE 31
02 · Reads — All reads	Meter	code>assets.assets.asset_no via `read_dict.meter`	metering ⇄ assets	SERVED
08 · Upload reads · stage 2 · Transform	Four starter buttons	code>GET /reads/template?shape=, validated against `READ_SHAPES`	metering	SERVED
09 · Upload reads · stage 3 · Validate	“Cannot run” + its reason	he outcome's own `reason` — here, *no signed threshold exists*	metering	SERVED
02 · Reads — All reads	Source	code>read_method + `source`, two catalogue lookups	metering	SERVED
07 · Upload reads · stage 1 · Extract	Transport	*a literal pair, `SOURCE_KINDS`; the third value exists only to name the server's refusal*	—	SERVED
03 · One reading	Supersession chain	he server's walk of `superseded_by_id`	metering	SERVED
03 · One reading	Uploaded by	code>compliance.uploaded_by_name; the key rides on `title` as evidence	metering ⇄ audit	SERVED
03 · One reading	Peak demand (a	code>consumption + `unit`, the register's own value at the cycle read	metering	SERVED

	Peak demand (a demand register draws this instead of Usage)	code>consumption + unit, the register's own value at the cycle read	metering	SERVED
02 · Reads — All reads	Premise · unit	code>premises.address_line1 + `units.label`, walked	metering (premises is metering's own table)	SERVED
09 · Upload reads · stage 3 · Validate	Row disposition — options	*a literal disposition list in* `workflowKit`	—	HARDCODED
16 · Meters still to read	Meter	code>assets.asset_no	metering ⇄ assets	SERVED
10 · Upload reads · stage 4 · Load	Batch number	code>batch_no — human-facing	metering	SERVED
13 · Accept flags in bulk	Why a row cannot be selected	code>bulkSelectable(row) in the browser, from the row the list already holds	—	SERVED
09 · Upload reads · stage 3 · Validate	Registers each plan expects	code>completeness against the plan-required set	metering ⇄ rating	SERVED
01 · Reads — Needs attention	Missing count	code>completeness.missing_records_count(), the plan-required set minus committed reads	metering ⇄ rating (the plan declares the required registers)	SERVED
03 · One reading	File hash	code>compliance.file_hash (SHA-256)	metering	SERVED
10 · Upload reads · stage 4 · Load	File hash	code>compliance.file_hash — the same hash stage 2 showed	metering	SERVED
10 · Upload reads · stage 4 · Load	Uploaded at	code>compliance.uploaded_at, rendered in the tenant's zone — measured 09/12/2026 08:31 against 13:31 UTC, which is correct for America/Chicago	metering	SERVED
10 · Upload reads · stage 4 · Load	Uploaded by	code>compliance.uploaded_by_name. This printed a raw primary key to a clerk for sixteen days and then *explained* it.	metering ⇄ audit	SERVED
02 · Reads — All reads	Total row count	code>count on the paged envelope	metering	SERVED
02 · Reads — All reads	Current	code>current_value_display	metering	SERVED
15 · Cycles	Meters	code>cycle.expected	metering ⇄ assets	SERVED
10 · Upload reads · stage 4 · Load	Cycle shortfall	code>cycle_coverage, or its absent-reason when the batch carries no cycle	metering ⇄ rating	SERVED
15 · Cycles	Read window	code>cycle_dict's dated window when one exists, otherwise the day-of-month shape the three fields derive	metering	SERVED, TWO DERIVATIONS DISAGREE
15 · Cycles	Cycle code	code>cycles.code	metering	SERVED
14 · Routes	Cycle	code>cycles.code — one cycle at a time, BR-MET-010	metering	SERVED
15 · Cycles	Status	code>cycles.status	metering	SERVED

18 · Estimate a missing read	The estimate, with its unit	code>EstimateFigure over the server's answer; a figure with no unit is **not shown** and says why	metering	SERVED
03 · One reading	Read history rows	code>GET /assets/{id}/reads, the assets read seam	metering ⇄ assets	SERVED, UNPAGINATED
17 · Route sheet	Cycle — options	code>GET /cycles, every page, fetched whole	metering	SERVED, UNBOUNDED
15 · Cycles	Completeness · days left · chase count	code>GET /cycles/{id}/chase-list, one call per visible row	metering ⇄ rating	SERVED
08 · Upload reads · stage 2 · Transform	Reading sheet · row count	code>GET /cycles/{id}/reading-sheet/summary — the button names its own number	metering ⇄ rating	SERVED
08 · Upload reads · stage 2 · Transform	Staged rows behind the panel	code>GET /read-batches/{id} — the whole batch, to a 50,000-row cap	metering	SERVED, UNPAGINATED
07 · Upload reads · stage 1 · Extract	Read method — options	code>GET /reads/read-methods, the **tenant's** list, not a constant	metering	SERVED
17 · Route sheet	Route — **options**	code>GET /routes, every page, fetched whole	metering	SERVED, UNBOUNDED
07 · Upload reads · stage 1 · Extract	Route — options	code>GET /routes, same helper	metering	SERVED, FETCH-EVERYTHING
07 · Upload reads · stage 1 · Extract	Service chips	code>GET /settings/services, the enabled set	metering ⇄ tenancy	SERVED
02 · Reads — All reads	Service filter options	code>GET /settings/services, the tenant's enabled set	metering ⇄ tenancy	SERVED
01 · Reads — Needs attention	Flagged count	code>meter_reads — “neither ok nor estimated”, the same predicate the Flagged filter uses	metering	SERVED
02 · Reads — All reads	Channel hidden	code>meter_reads.channel	metering	SERVED
02 · Reads — All reads	Days since prior hidden	code>meter_reads.days_since_prior	metering	SERVED
01 · Reads — Needs attention	Disputed count	code>meter_reads.disputed_at not null	metering	SERVED
02 · Reads — All reads	Read date	code>meter_reads.read_at — a **calendar date**, stamped 00:00Z of the read day	metering	SERVED
02 · Reads — All reads	Status	code>meter_reads.status	metering	SERVED
02 · Reads — All reads	Cycle	code>metering.cycles.code joined server-side	metering	SERVED
02 · Reads — All reads	Route	code>metering.routes.code	metering	SERVED
01 · Reads — Needs attention	Meters that could not be checked	code>missing_unresolved_meters on the same payload	metering ⇄ rating	SERVED

03 · One reading	Net meter pair	code>net_pair, served on the read to save a round trip; renders for a net meter and nothing else, and **never nets the two**	metering	SERVED
14 · Routes	Download CSV	code>POST /exports {resource:"routes", state:null} — **posts an empty state**, so it exports the whole list rather than the page	metering ⇄ tenancy	SERVED, UNSCOPED
10 · Upload reads · stage 4 · Load	Co-sign	code>POST /read-batches/{id}/cosign — and the screen says plainly that it **does not hold the post**	metering	SERVED, HALF THE RULE
09 · Upload reads · stage 3 · Validate	Estimate all / Send someone out	code>POST /reads/bulk-estimate and the work-order path	metering ⇄ work	SERVED
10 · Upload reads · stage 4 · Load	The seven composition figures	code>postSplit()	metering	SERVED
09 · Upload reads · stage 3 · Validate	What will post / will not	code>postSplit() — the same computation stage 4 names on its button	metering	SERVED
02 · Reads — All reads	Previous	code>prior_value_display — the server composes value+unit	metering	SERVED
12 · Add a single reading	Context: address · unit · register · multiplier	code>quick_read.quick_entry_context	metering	SERVED
18 · Estimate a missing read	Register — options	code>READ_CHANNELS, *a literal array*	—	HARDCODED
03 · One reading	Billed corrections	code>read_corrections — references and a reason; **no money figure**, the credit note is billing's	metering ⇄ billing	SERVED
03 · One reading	Tags: service · cycle · route · channel · transport · method	code>read_dict since DEF-045	metering ⇄ tenancy (service name)	SERVED
08 · Upload reads · stage 2 · Transform	Column mapping + “not in your file”	code>read_import.detect_mapping, re-detected on every upload; nothing is stored	metering	SERVED
08 · Upload reads · stage 2 · Transform	Batch number	code>read_import_batches.batch_no — **a human-facing number, not the key**	metering	SERVED
03 · One reading	Batch number	code>read_import_batches.batch_no, falling back to the id when absent	metering	SERVED
15 · Cycles	Frequency · Start day · Days to complete	code>read_schedules, the ruled CR-029 D2 vocabulary	metering	SERVED
14 · Routes	Last completion	code>read_windows	metering	SERVED
13 · Accept flags in bulk	Per-row refusals	code>refused[] with a named reason each	metering	SERVED
17 · Route sheet	The sheet	code>route_sheet.py — premises in walk order, each meter's previous reading pre-filled	metering ⇄ assets	SERVED
14 · Routes	Assigned to	code>routes.assigned_to — **rendered, never written**; reader assignment is out of scope by ruling	metering ⇄ identity	SERVED
14 · Routes	Type	code>routes.route_type	metering	SERVED

14 · Routes	Status	code>routes.status	metering	SERVED
15 · Cycles	Pager total	code>rows.length in the browser	—	DERIVED
14 · Routes	Pager total	code>rows.length in the browser — **not a server count**	—	DERIVED
03 · One reading	Rules passed / evaluated	code>rule_outcomes JSON, tallied in the browser from stored outcomes	metering	SERVED
09 · Upload reads · stage 3 · Validate	Rule · severity · hits · action	code>rule_outcomes rolled up per rule	metering	SERVED
14 · Routes	Collected	code>seams.collected_read_count, derived from committed reads	metering	SERVED
02 · Reads — All reads	Service	code>tenancy.services.code through the service point	metering ⇄ tenancy	SERVED
16 · Meters still to read	Unit	code>units.label	metering	SERVED
02 · Reads — All reads	Usage	code>usage_display	metering	SERVED
16 · Meters still to read	Days left	computed from the window this row was asked about	metering	SERVED
15 · Cycles	Routes	erived from membership — never a second stored list (BR-MET-010)	metering	SERVED
13 · Accept flags in bulk	Result — applied of selected	he endpoint's own answer	metering	SERVED
16 · Meters still to read	Registers still needed	he plan-required set minus what arrived	metering ⇄ rating	SERVED
14 · Routes	Service(s)	he route's service set (CR-029 D10)	metering ⇄ tenancy	SERVED
12 · Add a single reading	Derived cycle and period	he same payload; absent when the meter is on no route	metering	SERVED
12 · Add a single reading	Register name	he served **code** — no signed vocabulary maps a channel to a display word on this surface	metering	SERVED AS A CODE
08 · Upload reads · stage 2 · Transform	The five “what is in this file” figures	he staged batch's counts	metering	SERVED
08 · Upload reads · stage 2 · Transform	File name · rows · size · SHA-256	he staged batch's own compliance block	metering	SERVED
03 · One reading	Flag reason + thresholds	he stored outcome bag — nothing is compared again here	metering	SERVED
09 · Upload reads · stage 3 · Validate	Rows · Passed · Warnings · Errors · Missing records	he validate answer's own counts	metering	SERVED
07 · Upload reads · stage 1 · Extract	Write mode	*a two-value choice; posted with the batch*	metering	SERVED
17 · Route sheet	Scope	*a two-value segment*	—	SERVED

17 · Route sheet	Route — value	*chosen*	metering	SERVED
03 · One reading	**Account**	*nothing. A reading does not carry its account, and the screen refuses to join it in the browser*	—	NOT SERVED — stated
13 · Accept flags in bulk	Selected count	*page state*	—	SERVED
12 · Add a single reading	Read date	*picked*	metering	SERVED
18 · Estimate a missing read	Read date	*picked*	metering	SERVED
18 · Estimate a missing read	Meter number	*typed*	metering ⇄ assets	SERVED
12 · Add a single reading	Note	*typed, optional*	metering	SERVED
13 · Accept flags in bulk	Reason	*typed; required; stored on every row*	metering	SERVED
12 · Add a single reading	Meter number	*typed; resolved by exact lookup on blur*	metering ⇄ assets	SERVED
14 · Routes	Meters	membership count	metering ⇄ assets	SERVED
18 · Estimate a missing read	Refusal reason	one of seven named reasons from the engine	metering	SERVED

J · Health, and what the grade rests on

A screen's health is the worst of its fields, never an average. An average turns one broken field among nine good ones into “90% healthy”, and the screen still does not work. Three tables rather than one, because the three things are graded by different tests and mixing them produces a number that means nothing.

J1 · Fields

Verdict	Count
SERVED	100
SERVED, UNBOUNDED	3
HARDCODED	3
SERVED, UNPAGINATED	2
SERVED, FETCH-EVERYTHING	2
DERIVED	2
SERVED, BUT RULE 31	1
SERVED, NARROWER THAN r10 DRAWS	1

SERVED, NARROWER THAN r10 DRAWS	1
NOT SERVED — stated	1
SERVED, BROWSER-FORMATTED	1
SERVED, EMPTY IN PRACTICE	1
SERVED, HALF THE RULE	1
SERVED AS A CODE	1
SERVED, UNSCOPED	1
SERVED, TWO DERIVATIONS DISAGREE	1
ungraded	0

Every row that is not plainly SERVED, in full:

Screen	Field	Verdict
02 · Reads — All reads	Batch hidden	SERVED, BUT RULE 31
02 · Reads — All reads	Search box	SERVED, NARROWER THAN r10 DRAWS
02 · Reads — All reads	Route / Cycle filter options	SERVED, UNBOUNDED
03 · One reading	**Account**	NOT SERVED — stated
03 · One reading	Read history rows	SERVED, UNPAGINATED
07 · Upload reads · stage 1 · Extract	Reading month	SERVED, BROWSER-FORMATTED
07 · Upload reads · stage 1 · Extract	Cycle — options	SERVED, FETCH-EVERYTHING
07 · Upload reads · stage 1 · Extract	Route — options	SERVED, FETCH-EVERYTHING
08 · Upload reads · stage 2 · Transform	Manual grid — Register options	HARDCODED
08 · Upload reads · stage 2 · Transform	Staged rows behind the panel	SERVED, UNPAGINATED
09 · Upload reads · stage 3 · Validate	Row disposition — options	HARDCODED
10 · Upload reads · stage 4 · Load	Session · IP	SERVED, EMPTY IN PRACTICE
10 · Upload reads · stage 4 · Load	Co-sign	SERVED, HALF THE RULE

Load		
12 · Add a single reading	Register name	SERVED AS A CODE
14 · Routes	Download CSV	SERVED, UNSCOPED
14 · Routes	Pager total	DERIVED
15 · Cycles	Read window	SERVED, TWO DERIVATIONS DISAGREE
15 · Cycles	Pager total	DERIVED
17 · Route sheet	Route — **options**	SERVED, UNBOUNDED
17 · Route sheet	Cycle — options	SERVED, UNBOUNDED
18 · Estimate a missing read	Register — options	HARDCODED

J2 · Flows — one row per screen

A flow is never graded WHOLE on a read alone. UNVERIFIED is the honest verdict for a screen Step 5 did not walk.

Screen	Verdict
01 · Reads — Needs attention	WHOLE — walked
02 · Reads — All reads	WHOLE — walked; 24 rows, the toolbar, the column panel and the pager all rendered
03 · One reading	BREAKS AT THE HISTORY FETCH — and it takes the whole page with it
04 · Correct the reading	UNVERIFIED — unreachable behind 03
05 · Dispute the reading	UNVERIFIED — unreachable behind 03
06 · Clear the dispute	UNVERIFIED — unreachable behind 03
07 · Upload · Extract	WHOLE — walked
08 · Upload · Transform	BREAKS AT PAGINATION — walked; the staged-row fetch is unpaginated to a 50,000 cap
09 · Upload · Validate	BREAKS AT PAGINATION — walked; same fetch behind the segments
10 · Upload · Load	WHOLE — walked to the refusal
11 · Upload · Result	UNVERIFIED — never rendered
12 · Add a single reading	WHOLE — walked, not posted
13 · Accept flags in bulk	WHOLE — walked, not submitted
14 · Routes	BREAKS AT PAGINATION — walked by typed URL; the browser pages a fetch-everything read
15 · Cycles	BREAKS AT PAGINATION — walked by typed URL; same, plus one chase request per visible row

16 · Meters still to read	UNVERIFIED — the panel did not open under the driver and the reason was not diagnosed
17 · Route sheet	WHOLE — walked; the sheet itself was not opened
18 · Estimate a missing read	WHOLE — walked, not submitted
19 · The retired drawer address	WHOLE — a redirect, read in the source

8 WHOLE · 4 BREAKS AT PAGINATION · 1 BREAKS AT THE HISTORY FETCH · 5 UNVERIFIED · 1 redirect.

J3 · Seams — one row per boundary crossing

From → To	The file that crosses	Grade
metering → assets	`apps/metering/seams.py`	HEALTHY
metering → rating	`apps/metering/seams.py`	HEALTHY
metering → tenancy	`apps/metering/seams.py`	HEALTHY
metering → audit	audit's `actor_names` seam	HEALTHY
metering → identity	`apps/metering/seams.py`	HEALTHY
metering → work	the work module's own service	HEALTHY
metering → billing	`read_corrections`, the J6 linkage	HEALTHY
billing → metering	`apps/billing/metering_seam.py` 🔒 human-owned	HEALTHY
rating → metering	`apps/rating/seams.py:841` — `get_model("metering","MeterRead")`	LEAKY
portal → metering	`apps/portal/seams/metering_seam.py:317` · `portal/selectors.py:859,891` · `portal/credentials.py:256,323`	LEAKY
integrations → metering	`apps/integrations/mappings.py:164`	LEAKY
customers · portal → metering	four files importing `apps.metering.interruptions` models directly	LEAKY

8 HEALTHY · 4 LEAKY · 0 WRITING · 0 ABSENT. A seam that hands back a raw queryset is LEAKY even though it only reads — the caller can then filter, join and write through it, and the boundary the seam exists to draw is gone. The grade is on what the seam *returns*, not on what its docstring says it does.

A · Every rule this document states

The **home** column is the point. `INHERITED` — it exists elsewhere and this document follows it. `EXTENDS` — it adapts an existing rule to this context, and says what changed. `NO HOME` — stated only here. **Count the NO HOMEs: a rule stated only in a document nobody reads is a rule that is about to be broken.**

id	The rule, as stated	Where	Home	What it extends or restricts
A-01	The page-head title is 32px on every screen, set on the element rather than inherited from its tag.	§01, every drawing	INHERITED	B14 rule 40, ratified 2026-09-12. **Supersedes the drawing rules this document was written under**, which pin 40px from an earlier measurement.
A-02	Take the tokens from the shipped app, the layout from the prototype, and never present a prototype class as though the product had it.	§01	INHERITED	B14's own precedence, plus CLAUDE.md's ban on porting the prototype's runtime.
A-03	A door is a link a user can press from another screen. A registered route is not a door, and a config entry naming a path is not a door.	§02, §04, §05	EXTENDS	Extends G100's question — which covers customer-portal screens only — to staff screens, where no gate asks it.
A-04	Draw the as-built screen as it is, not as the rules say it should be. A tidied as-built wireframe destroys the audit.	every drawing	INHERITED	The skill's own rule; stated here because it decides what four drawings look like.
A-05	A remedy is additive: every field and every piece of information architecture survives unless a ruling says otherwise, and the carry-over arithmetic must reconcile.	§02–04	INHERITED	The general form is CLAUDE.md's *"before retiring a value, grep for its CONSUMERS"*.
A-06	A walk you did not take is not a walk. Where the app was not driven, the row says so and the flow is UNVERIFIED.	§05, J2	INHERITED	Hard-won law 6, applied to this document's own claims.
A-07	An ungraded field is not a healthy one, and the count of ungraded fields goes on the masthead.	§11 I	EXTENDS	Extends the health vocabulary with an explicit zero rather than an implied one.
A-08	Every figure in every drawing is an illustration until somebody verifies it, and a figure taken from the walk says so.	§11 B	INHERITED	B14 rule 37, verbatim.
A-09	A check that cannot cover its whole subject must print what it skipped beside what it checked.	§08	INHERITED	CLAUDE.md hard-won law 7, fifth shape. Quoted, not invented — and it is what MR-22 asks for.
A-10	If a thing is not in a commitment card, nobody has undertaken to build it.	§09	INHERITED	The skill's own rule, stated on the document's face.
A-11	An analysis may propose and never allocate: this document mints its own `MR-` series and borrows no live id.	§09	NO HOME	**Stated only here.** It protects the repository's id registers from an analysis that looks like scheduled work — CLAUDE.md's id grammar governs the live families and says nothing about an analysis's own.
A-12	A second address is numbered as its own entry, drawn as nothing, and says which screen it opens.	entry 19	NO HOME	**Stated only here.** Prevents the case where somebody later finds a working URL that nothing in the document describes.

Two NO HOMEs, both about how an analysis behaves rather than how the product behaves — which is the comfortable kind. Ten of the twelve are inherited or extend a rule that already has an owner.

B · Every worked figure, numbers intact

None of these is a signed figure and the drawings are illustrations. The rules that govern them are B4 (money math, ❄ frozen) and CLAUDE.md hard rule 3 (every quantity with its unit) — a verifier checks against those, not against this table. Figures marked **WALKED** were read off the running app on 2026-09-12 against the seeded `brazosbend` tenant and are a reading of demo data at one moment.

Figure	The sentence around it	Screen	Status
18	Missing — plan-required readings not yet collected	01	WALKED
2	meter(s) could not be checked — no reading is expected of them yet	01	WALKED
24	Flagged	01	WALKED
0	Disputed	01	WALKED
44000 kWh · 46754 kWh · 27540 kWh	Previous · Current · Usage on M-7210. **Reconciles**: (46,754 – 44,000) × 10 on a ×10 meter	02	WALKED
39.8 kW · 41.2 kW · 412 kW PEAK	the demand row on the same meter; 41.2 × 10 = 412	02	WALKED
4110 kWh · 4320 kWh · 210 kWh	M-7450, on-peak register	02	WALKED
24	rows returned by `GET /reads` on the seeded tenant	02	WALKED
1,008 kgal · 1,016 kgal · 8 kgal · 30 days	the illustrated water reading on entry 03	03	illustration
5.2 kgal · 154% · 150% · 3	trailing average · percent of it · threshold · readings in the average	03	illustration
11 of 12	rules passed	03	illustration
8,634 kWh · 424 kWh	the read-history row, from `GET /assets/{id}/reads`	03 · R2	WALKED (API)
RB-2609-0005 · 2 rows · 1 KB	the staged batch	08	WALKED
808e48f7176c84	File hash (SHA-256), unrounded and unabbreviated	08 · 10	WALKED

e1fb2a237e8939 5b452aa0d05fd2 8c9ae7fe55ae7d c0f4e804			
2 · 0 · 1 · 1 · 2	Rows in batch · Passed · Warnings · Errors · Missing records. **Reconciles**: 0 + 1 + 1 = 2	09	WALKED
2 · 8 · 2 · 6	Time-of-use plan: meters · expects · arrived · short	09	WALKED
1 · 1 · 1 · 0 · 0 · 0 · 0	the seven composition figures at stage 4	10	WALKED
RB-2609-0006 · 09/12/2026 08:31	batch and upload stamp; 08:31 local against 13:31 UTC, correct for America/Chicago	10	WALKED
3 of 5 · 2	meters covered · meters still unread	10	WALKED
RT-001 · 3 meters · 8 collected · 07/31/2026 19:00	the first Routes row. **Does not reconcile** — J-04 and J-05	14	WALKED
RT-003 · 2 meters · 12 collected	the same shape again	14	WALKED
CY-001 · 5 meters · 0 of 18 · 0% · 0 days left · Chase 5 meters	the cycle row. **Three denominators** — J-06	15	WALKED
09/01/2026 – 09/03/2026 · 11/05/2026 due 11/25/2026	CY-001's window and bill date; the window closed nine days before the read	15	WALKED
6	routes, and cycles, on the seeded tenant — the option lists on entries 07, 17 and 18	07 · 17	WALKED
MTR-0451 · MTR-0452 · 45,210 → 45,910 = 700 kWh	the `J2-read-to-cash` cohort, used where a drawing needed a second illustration	02 · 04 · 05	fixture
32px · 800 · Archivo	`getComputedStyle` on the page title	all	MEASURED
240px · 13 · 1440 = 1440	rail width · nav items · `scrollWidth` vs `clientWidth`	all	MEASURED
#f3f2f2 · #ec3013 · 0px	`--color-bg` · `--color-accent` · `--radius-md`, read off the running document	all	MEASURED
338 of 338 · 281 of 282	migrations applied · tables with RLS ENABLE+FORCE	§05	MEASURED

C · Every screen, with its exact contents

#	Route	Tabs	Table controls	Write surfaces	How you get here
01 R ETI RE D	`/reads`	attention · all	none — three tiles	detail: two drill links · add/edit/update: NOTHING	the sidebar's *Reads* item
02	`/reads?tab=all`	attention · all	search SERVER (meter no only) · 7 filters SERVER · columns ▦ show/hide/reorder/ reset · sort SQL · pagination SERVER	detail: the meter cell · add: 12, 07–11 · edit: NOTHING · update: 13	the sidebar, plus both drill links on 01
03	`/reads/:readId`	none — ten sections	one table, **no controls at all**	detail: itself · add: NOTHING · edit: 04, 05, 06 · update: 4 endpoints	the meter cell on 02; the legacy `?read=`; itself after an override
04	dialog	none	none	update: `POST /reads/{id}/supersede`	03's primary action
05	dialog	none	none	update: `POST /reads/{id}/dispute`	03's secondary action
06	dialog	none	none	update: `.../dispute/clear`, System Admin only	03's secondary action, once a dispute is open
07	`/reads/upload` 1/5	none — a 5-stage rail	none	**NOTHING**	02's primary action; the Dashboard's *Import reads*
08	2/5	none	none (wizard tables, exempt)	add: *+ Add a row* · update: `POST /read-batches` · delete: discard	stage 1
09	3/5	4 segments, client-side	filter only, over rows already fetched	update: validate · PATCH a row · DELETE a row · bulk demand-reset	stage 2
10	4/5	none	none	update: `POST .../post` — **the widest write in the**	stage 3, once the rules have run

				module · co-sign	
11	5/5	none	*not walked*	**NOTHING** — the write happened at stage 4	stage 4's commit, once certified
12	dialog	none	none	add: stage → validate → post, the same pipeline a file walks	02's *+ Add reading*
13	bar + dialog	none	none	update: `POST /reads/bulk-fix`, accept only, cap 500	ticking a row checkbox on 02
14 RETIRED	`/reads/routes`	**none**	**search NONE · filter NONE · columns NONE · sort NONE** · pagination CLIENT-SIDE	detail: the route code → Settings · add/edit/update: **NOTHING, by rule**	**NOTHING**
15	`/reads/cycles`	**none**	**all four NONE** · pagination CLIENT-SIDE	detail: the cycle code → Settings; the completeness cell → 16	**NOTHING**
16	panel	none	**all four NONE**, no pager	**NOTHING** — the meter is text, not a link	15's completeness cell — behind a doorless screen
17	dialog	none	none	**NOTHING** — the sheet is a read	02's *Route sheet PDF*
18	dialog	none	none	add: `POST /reads/estimate`	02's *Estimate a missing read*
19	`/reads?read=<id>`	—	—	—	from outside the product — a ticket, an email, a bookmark
20 NEW	`/reads/cycles/:cycleId`	none — two sections and a band	server pager on both tables · no search, no filter (bounded by the cycle)	detail: a route row, a meter row · update: estimate, field visit · add/edit: **NOTHING, by BR-MET-011**	a row on 15; and a pasteable address

As built: 2 of 19 name no door, and a third sits behind one of them. **Six screens have a table; two of the six have a toolbar.**

Revised: 01 and 14 are retired by ruling, 20 is new, and the two doorless addresses become the *Read cycles* tab and the cycle page it opens — so **0 of 18 live entries name no door**, and the count of screens with a toolbar goes from two to three.

D · Decisions recorded here and nowhere else

D1 · Rules that live only in this document

Two — A-11 and A-12 in table A, both about how an analysis behaves. Nothing about the product is stated only here.

D2 · Rulings already recorded elsewhere — do not re-ask

Ruling	Where it is recorded
The read detail is a **page**, not a drawer — *"i want page to be open for consitency - we dont open side panel"*, and *"no read drawer required"*	PO 2026-08-26 · BR-MET-131 · decision log
The estimate form is a **dialog**, not a page — *"this can be modal - instead of page"*	PO 2026-08-27 at v2.61.1 · DEF-897
A route sheet prints for a **route or a cycle** — *"allow route of Cycle - both"*	PO 2026-08-27 · BR-MET-162 · DEF-898
The source cell carries **no bracket** — *"you dont have to provide sub text in bracket"*	PO 2026-08-27 · BR-MET-160 · DEF-895
The toolbar is **one row** — *"as per our stadard first raw of tookkit is more than suficient"*	PO 2026-08-27 · DEF-1700
A meter on no cycle **posts with a warning** — *"(b) Accept and warn"*	PO 2026-08-27 · OQ-422 · BR-MET-140
The sidebar renders **no submenus** — *"In plan Nav bar - no sub menu required"*, *"same is the case with premises UI; no Sub menu"*	PO 2026-08-10 ×2 · BR-SHL-001
The page-head title is **32px** — *"use 32 pixels — for every ordinary screen"*	PO 2026-09-12 · B14 rule 40
The ICP ceiling is **40,000 accounts**, not 3,000	PO 2026-09-06 · CLAUDE.md, decision log *"THE ICP, CORRECTED THE SAME HOUR"*
Reader **assignment** is out of scope; the ops tab's read-only reader display is untouched	CR-029 D6 · B17 §3

A search technique that gave a false negative, recorded because the next person will hit it too: grepping the decision log for `OQ-\d{3}` returns `OQ-119` for a document containing `OQ-1190` — a different, real question. The pattern that works is `(?:OQ|DEF)-(?:[A-Z]{3}-)?\d{3,5}(?!\d)`, and the lookahead is load-bearing.

E · Every guarantee, and what enforces it

Only a constraint refuses. A convention is a habit with a good reason, and a habit does not stop a caller. Sorting the guarantees this way is uncomfortable and always useful.

The guarantee	What enforces it	Level
A read batch cannot be posted twice	a DB unique index on file hash + tag fingerprint, plus two service guards that name who posted it and when	DATABASE CONSTRAINT
A committed reading is never updated — a correction appends	the supersede service; a billed read is refused by name and routed to the other door	SERVICE RULE

No tenant sees another's readings	RLS `ENABLE+FORCE` on 281 of 282 tenant-scoped tables, with the app role holding neither SUPERUSER nor BYPASSRLS — **measured**	DATABASE CONSTRAINT
Every quantity carries its unit	`quantity_with_unit` returns null for an absent unit and the cells dash it; G44's display-truth ratchet counts the rest	SERVICE RULE
The reads list shows one row per correction chain	`_read_heads()` filters `superseded_by_id__isnull=True` in every consumer's query	SERVICE RULE
An integrity check cannot be turned off	422 at the settings endpoint, with the rule named	SERVICE RULE
Routes and Cycles cannot be edited from the ops screens	**by construction** — the file imports no mutation wrapper at all	SERVICE RULE
A reading never carries an address of its own	G85, four ruled homes and no unruled one	SERVICE RULE
No screen in this module shows money	**nothing** — BR-MET-078 is a rule and no check reads a screen for a currency symbol	CONVENTION
Every list paginates on the server	**nothing** — NFR-3 has no gate, and 11 of the 12 reads list endpoints do not	CONVENTION
No internal id reaches a clerk's eye	G28 over strings, and two render-checks covering **two screens in the whole product**, neither of them here	CONVENTION for this module
Every screen has a door	**nothing, for staff screens.** G100 asks the question for the customer portal only	CONVENTION
A portal read names the tenant in the query	G96 — **which is blind to one file because of an import alias** (§08)	CONVENTION in practice, for that file

F · What this document assumes and never says

Gaps in the agreement, not in the product. Ordered by cost, most expensive last.

The assumption	Where it hides	What breaks if it is wrong
The seeded demo tenant's data is representative enough for a walk to be meaningful	every WALKED figure in table B	Findings that are artefacts of six routes and 24 readings. J-04's *3 meters, 8 collected* is the one most likely to look different on real data — though the *shape* of the mismatch would not.
A wizard panel's component maps to the stage the rail shows	§03's stage-to-panel mapping	The mapping was derived by walking four of five stages and reading the kind config. Stage 5 was never seen, so *"stage 5 has no panel id"* rests on the absence of a fifth id rather than on having watched it render.
The chase panel's failure to open was the driver, not the product	J-09, entry 16's UNVERIFIED grade	If it is the product, entry 16 is a second dead control behind a doorless screen, and MR-10 is a bug fix rather than a feature.
What a clerk needs from Routes and Cycles is what those screens already show	remedy R1, which moves them into tabs unchanged	R1 makes two unreachable screens reachable. **It does not ask whether they are the right screens** — nobody has used them, because nobody could.
That the reads engine's figures are right	every field graded SERVED	**The most expensive one.** This audit checks whether a number is *served, reached and displayed with its unit*. It does not check whether the number is *correct*: the billing-math contract is B4/B5's 61 golden tests, of which **14 are engine-verified**. A

field can be SERVED, WHOLE, rule-compliant and wrong, and nothing in this document would notice.

G · Every UI rule violation

Section 07 is that table in full — 18 rows plus the two product-rule verdicts, each with its rule, its `path:line` and whether a gate would ever have caught it. **3 of 18 are mechanically reachable today.** It is not repeated here.

H · Every cross-module finding

Section 09 carries these with both modules named in every row, split the way Step 10 asks the questions. **Of the four questions, one found nothing:** no module writes another's tables anywhere in this surface, and the repository's own G84, G65 and G85 agree. The other three found: one handoff whose precondition is checked on an unreachable screen, one fact with a column nobody reads and two figures that count different things, and one concept with two names split by audience but appearing on screen both ways.

K · Everything an upgrade would drop

> **Two things, and both were dropped by a ruling rather than by an upgrade.**
>
> **Field-level, the answer is still none.** Five screens are redrawn and the carry-over arithmetic reconciles at **19 → 19** (entry 02), **17 → 19** (entry 03), **11 → 13** (entry 15), **5 → 9** (entry 16) and **9 → 9** (entry 14's columns, inside entry 20): **zero fields REMOVED across all five**, and therefore zero rows with no quoted ruling, which is the row type this table exists to print first.
>
> **Two VIEWS are lost, and each is recorded at the entry that lost it rather than argued against:**

What is gone	Where	The ruling	What answers it now
The tenant-wide all-clear — one place saying “nothing needs attention” across the whole utility	Entry 01	*“Need Attention — does not add any value”*	Per cycle, on entry 20. Across the tenant, **nothing**.
The all-routes view — every route in the utility in one list, regardless of cycle	Entry 14	*“we should say Read Cycles and within that we should show Routes and their completion”*	Routes read cycle by cycle, on entry 20. If it bites, the answer is a Routes tab beside Read cycles — **not** the old orphan screen, which had no door and no window to measure against.

> Three further removals are chrome rather than content, all on entry 16: its own heading (the breadcrumb names the cycle now), its 17-word standfirst (B14 rule 33 caps rationale at zero words on screen) and its *Close* (a section has nothing to close). Listed because a reader counting rows would otherwise find the arithmetic short.
>
> Two dispositions are worth a reader's attention even though nothing is lost, because both cost somebody their muscle memory: on entry 15, **Completeness → Registers collected** and **“N days left” → the Window pill**. A user searching the screen for the old word will not find it. That is why a relabel is its own disposition and not a tidy-up.

—

Completeness, scoped — and what was left out on purpose

What this document is exhaustive over

Itself. Every rule it states is in appendix A with its home. Every worked figure is in appendix B, verbatim and unrounded. Every screen it describes is in appendix C with its tabs, its table controls, its write surfaces and its door. Every decision recorded only here is in D1, and every ruling it relies on that belongs to somebody else is in D2 so nobody re-asks it. Every guarantee is in E, sorted by whether anything actually refuses. Every field is in I, sorted by source, with **zero ungraded**.

It is not exhaustive over the product, which has behaviour these nineteen addresses never touch: the validation settings screen, the premises, unit and asset flows the same module owns, the customer portal's own reading surface, and everything the reads engine does that no screen displays.

And it is exactly as stale as the read it was made from: version **2.237.10**, commit **3706b459**, **2026-09-12**. Two of its findings — B14 rule 40 and the ICP ceiling — moved within the fortnight before that date, and one of them moved on the day itself.

What was deliberately left out, and why

- **The argument behind each screen.** It lives in that screen's own spec table, where a reader meets it beside the drawing it explains, rather than collected into a section nobody reads twice.
- **The source catalogues.** 741 reads keys across eleven languages. Section 06 states what they contain and counts what matters; reprinting them would bury the twenty-eight rows that hold a control.
- **Line numbers, except where a claim needs one.** They move. Files and values do not, so this document cites files by path and pins values as literals — and where a line number appears it is because the claim is about one specific line.
- **The planning of the pass.** Which files were read in which order, which searches gave false negatives, how long each step took. One exception, in section 11: the searches that gave false negatives *are* recorded, because the next person will write the same selector.
- **The demo tenant's data, as anything but illustration.** Six routes, six cycles, 24 readings. Table B marks every figure taken from it, and appendix F names the assumption that it is representative enough for a walk to mean anything.
- **Entry 11's wireframe.** Never rendered, so never drawn. The entry says so on its face.
- **A depth-first pass.** The process asks for one process closed before the next opens; this was run breadth-first across three. Section 11 says so, says why it matters, and names the mitigation.

It is fine for those to be out. It is not fine for them to be out silently — which is the reason this section exists.

> **And the honest closing count**
>
> **21 entries = 17 drawn + 4 undrawn, and each of the four says why on its own face: 01** and **14** retired by ruling — a retired screen has no final state — **16** drawn in place inside entry 20, and **19** a second address rather than a screen. **20 drawings, one per screen** — entry 08 has two because it draws both its branches, and entries 20 and 21 have five each, because every tab is a separate address under `BR-SHL-011` and therefore a separate screen. The as-built twins are gone by PO ruling of 2026-09-12: *“just keep the what is final screen”*, so every drawing here is the screen as it should be, and what moved to get there is the carry-over table under each one. **53 specifications + 10 out of scope** resolve every stated absence, with nothing left unruled. **121 fields = 100 SERVED + 21 qualified + 0 never traced. 2 of the 19 as-built entries name no door**, and a third sits behind one of them; the revision gives all three a door and adds none that lacks one.

> Every one of those numbers was recomputed from this document's own markup at assembly, not carried forward from a draft — which is what catches the entry added late that never made it into a total.